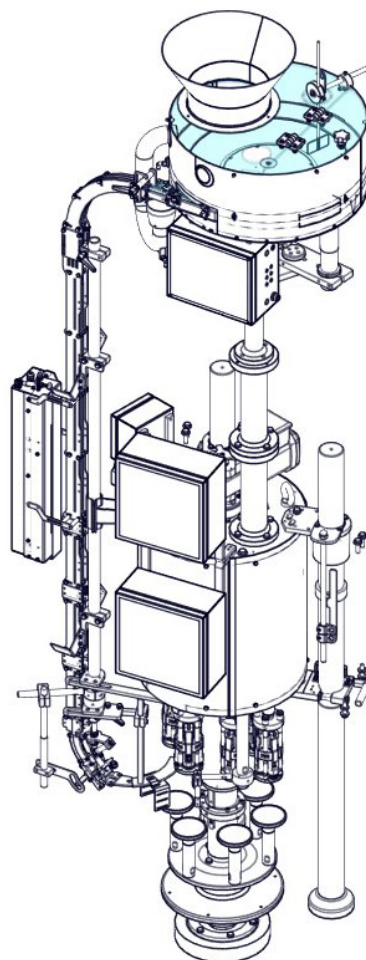




Model
TSI - EURO VA IES LIGHT

Serial number
17203

Pitch diameter
288 mm

Nr of heads
6

Manufacturing year
2018

USE AND MAINTENANCE

*TRANSLATION OF THE
ORIGINAL INSTRUCTIONS*

Code
Z17203ABSEN001

Revision
0



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BEFORE START UP THE PARTLY COMPLETED MACHINERY, READ THE TECHNICAL INSTRUCTIONS IN THIS MANUAL AND FOLLOW THE INDICATIONS SHOWN.

KEEP THE USE AND MAINTENANCE MANUAL IN PLACE KNOWN AND ACCESSIBLE TO OPERATORS AND THE SERVICE PERSONNEL.



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1 PURPOSE OF THE MANUAL - EDITION - UPDATING METHOD

The manual is integral to the partly completed machinery TSI - EURO VA IES LIGHT model, 17203 serial number; it should be kept all the life of partly completed machinery; it must follow the same machinery when it is moved to other premises or if it sold.

It is in Italian and translated in target language whenever required.

In case of doubts about the interpretation of the text, always refer to the original language and contact Customer Service in case of need:

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The text of the manual will refer to the "closing machine" as an abbreviation of the standard name **"Closing machine which uses a closure"** as stated in the UNI EN 415-1 standard (Terminology and classification of packaging machines).

The manual contains information and instruction about all the operating cycle of the closing machine:

- Installation.
- Start up.
- Use.
- Size change.
- Maintenance.
- Dismantling and disposal.

The manual should be kept so that:

- avoid any damage.
- Arrange for its availability to the professional operators working on the closing machine.
- Prevent it from being modified or damaged during the use.
- Make it available in case of need (training, changed ownership of the closing machine, etc.)



Ask for a replacing copy if damaged.

This manual was written with care; if some of the sections needs to be reviewed to add information or instructions, or simply for clarity reasons, AROL S.p.A. will proceed as follows:

- *should the modification involve only a few pages*: the replacement pages will be sent.
- *should the modifications involve more pages of different chapters*: all the chapters involving modification will be sent again.
- *should the modification involve many chapters* : the whole replacement manual will be sent.

Should the manual be supplied in electronic form, the supplied manual will always be the complete and updated one.

The consultation of this manual is made easier by the addition of the initial pages of the general table of contents, for an immediate tracking of the topic concerned.

The chapters are organised a hierarchy structure for a fast tracking of the info required.



The manual shows graphic symbols that are used to distinguish:

- Informations and prescriptions.
- Obligations.
- Warning about residual risks.

The following documents are attached to this manual:

- Pneumatic diagram.
- Closing machine installation diagram.
- Identification tables for size change.
- Closure head calibration tables.
- Instructions manuals for accessories and/or equipment installed on the closing machine and not integrated in this manual.



Before any operation with or on the closing machine read the procedures and warnings herein carefully.



IMAGES OF ANY GENERAL KIND ARE NOT USED IF THE DETECTION OF SPECIFIC STRUCTURE OF CLOSING MACHINE IS ESSENTIAL IN THE CONTEXT TO THE SUBJECT MATTER. The reported instructions and use modes for all described components anyway remain valid.



2 IDENTIFICATION OF THE CLOSING MACHINE

The upper part of the closing machine is provided with an identification plate. (See Figure).

On the plate are reported:

- Closing machine model (**MODEL**)
- Type of closing machine (**TYPE**)
- Pitch diameter (**P.D. in mm**)
- Number of closure elements (**HEADS**)
- Serial number (**SERIAL N.**)
- Manufacturing year (**YEAR**)

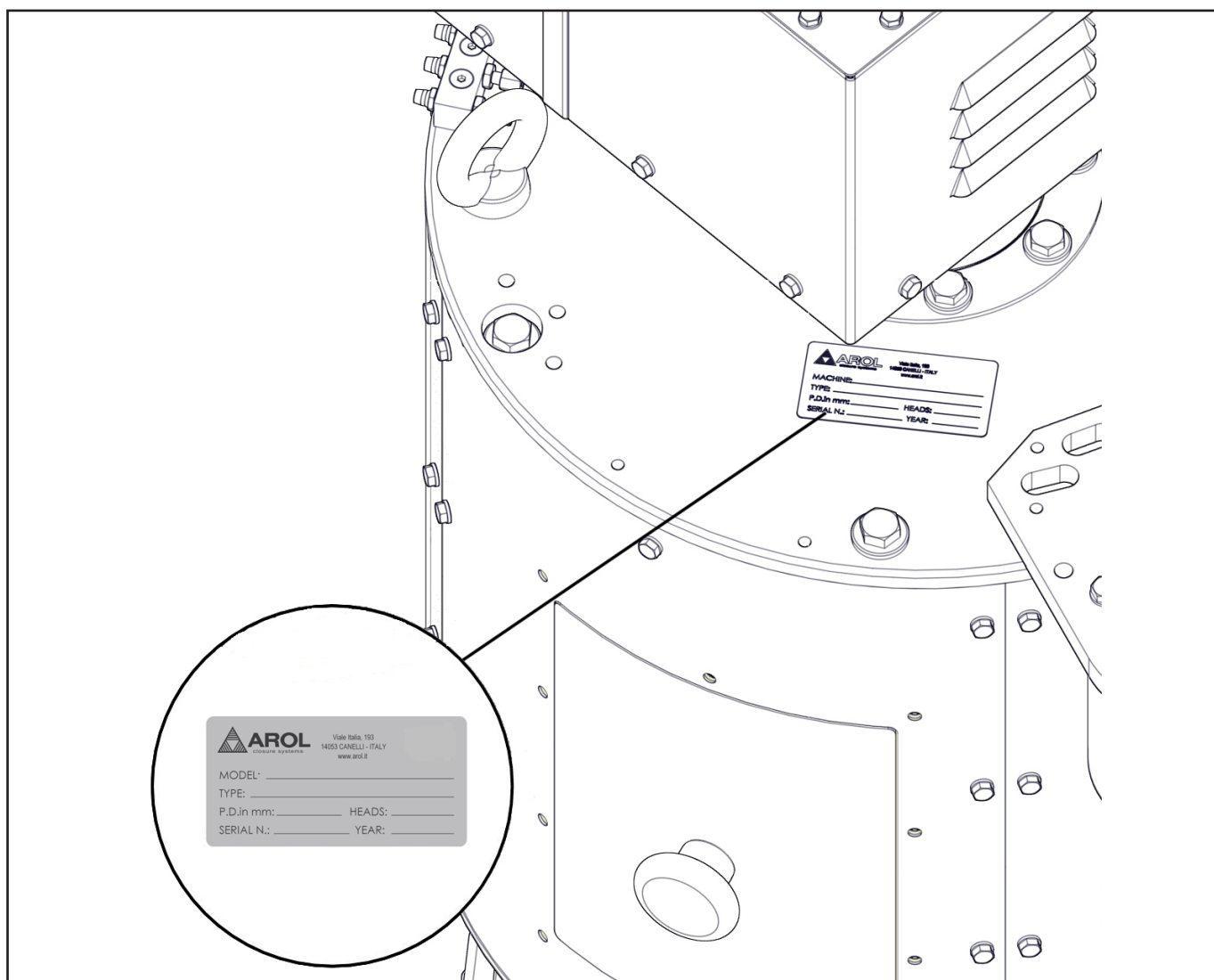


Fig. 1

The data plate is applied using 2 rivets at the opposite corners and should not be removed for whatever reason.

In case of damage, ask for a spare copy to *AROL S.p.A.*.

Clean with neutral detergents paying attention in order not to damage the warning instructions.



3 DECLARATION OF INCORPORATION

The closing machine of this manual is defined as "partly completed machinery" (according to the European Directive 2006/42/EC art. 2 point g), and comes with the *DECLARATION OF INCORPORATION*.

The original copy of the *DECLARATION OF INCORPORATION* is kept at AROL S.p.A..

A compliant copy of the *DECLARATION OF INCORPORATION* is delivered together with the partly completed machinery documents and must be kept till the machine is scrapped.



AROL S.p.A. points out that this statement only and **exclusively** concerns the above-described partly completed machinery, in the conditions and utilisation processes it was placed on the market, thus excluding any added components and/or operations carried out later on without any authorisation by the manufacturer.

In particular, any possible modification or replacement of parts of the closing machine, like as, for example modification to the software of programmable components, made without asking the permission of AROL S.p.A., releases the manufacturer from any responsibility and may cancel the validity of the Declaration of Conformity for the machine where this partly completed machinery is integrated.



4 CLOSING MACHINE DESCRIPTION

The closing machine model EURO VA IES LIGHT has been designed and built for the application of:

- aluminium caps to threading and/or skirting on bottles/containers.

4.1 MAIN UNITS OF THE CLOSING MACHINE

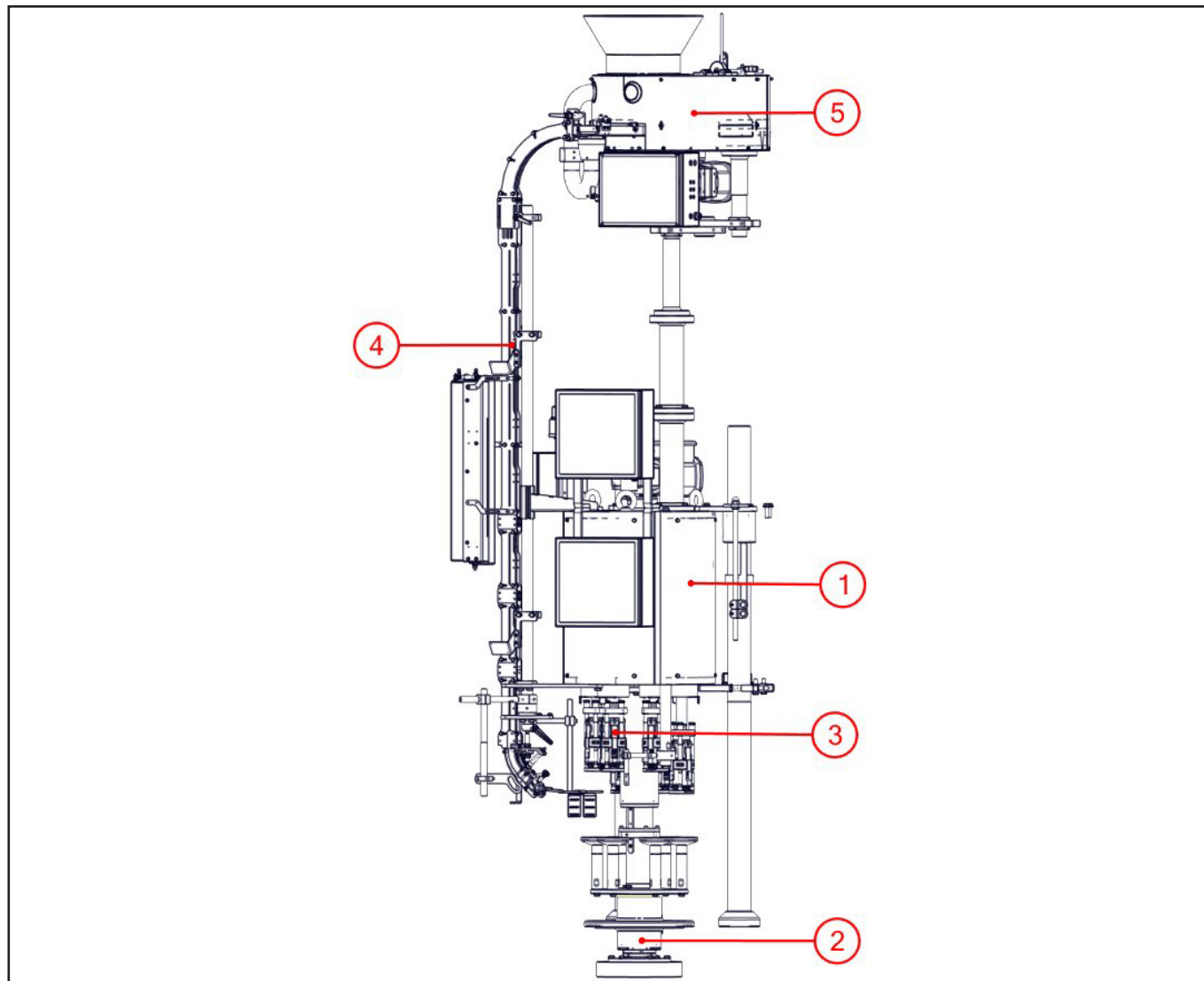


Fig. 2

Legend:

- 1 Upper closure carousel.
- 2 Lower closure carousel.
- 3 Closure heads.
- 4 Chute for closure systems.
- 5 Closure system sorter.



The components forming the closing machine are located inside the main assemblies:

- Closure heads.



Fig. 3

- Closure heads driving spindle.

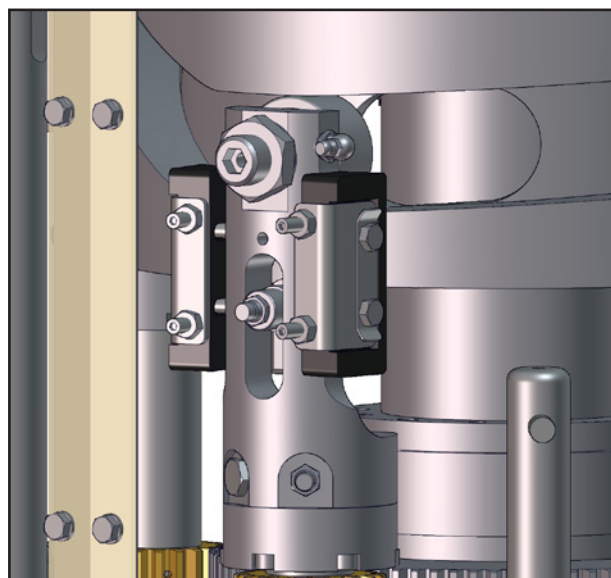


Fig. 4

- Height electrical adjustment of the closure carousel.

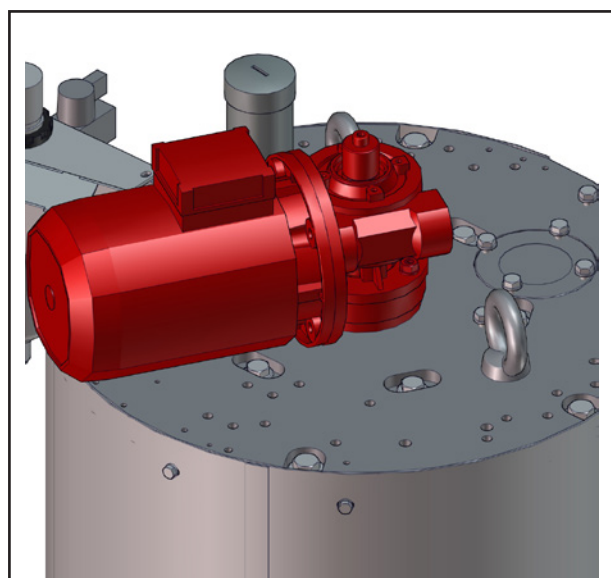


Fig. 5



- Direct radiation sterilisation device to provide, through the use of an UV ray tube, to sterilise caps and air surrounding the device itself.
- These tubes with an emission peak of 235.7 nm (nanometres) have a strong germicidal effect on all micro-organisms (mould, bacteria and viruses).

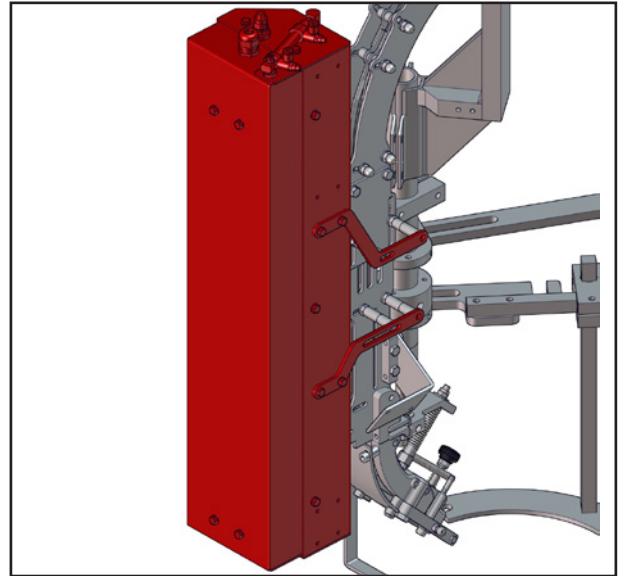


Fig. 6



4.2 DESCRIPTION OF THE PRODUCTION CYCLE

The application and sealing of closure systems on the containers is the following:

- An orientation device selects the caps inside the hopper.

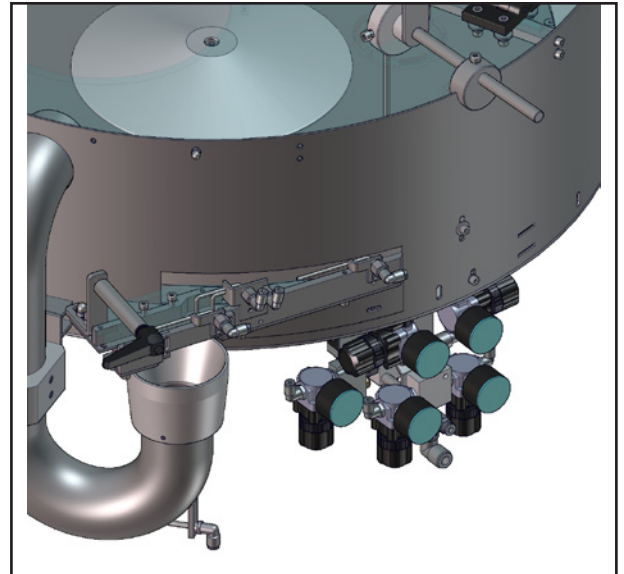


Fig. 7

- The cap placed at about 45°, is taken, by a system calle "pick off", from the container that is coming during the rotation.

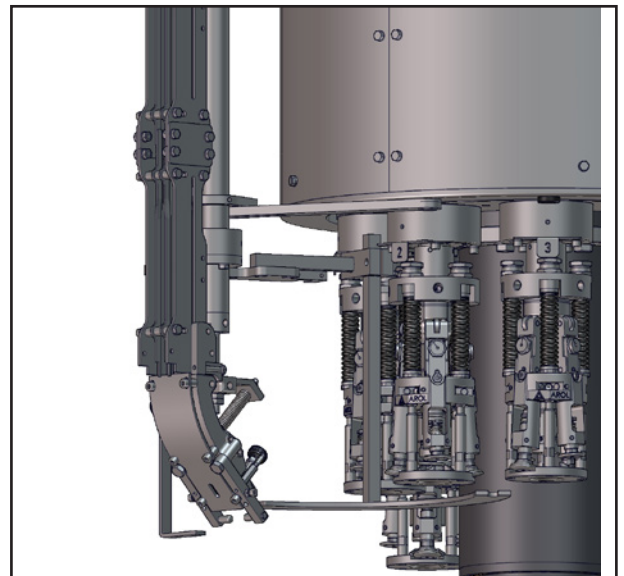


Fig. 8



- The closure head follows the profile of a cam and will thread and seal the cap on the container by the rotation of the threading and skirting rollers.
- After the closure, the exiting container is released, then there is a reset for the following closure.

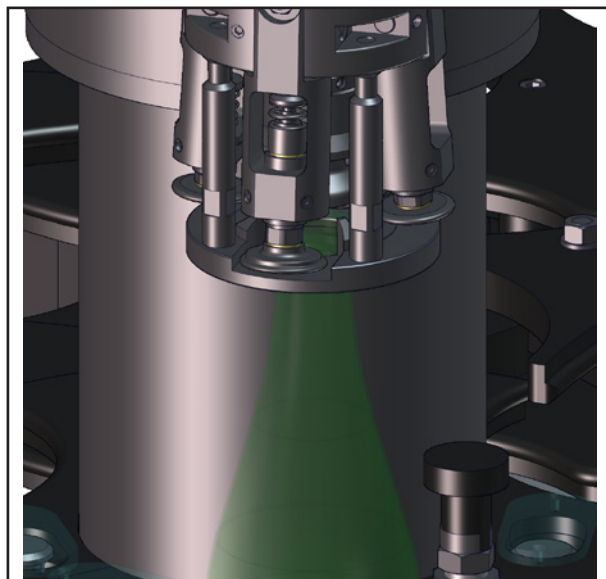


Fig. 9



4.3 CLOSING MACHINE TECHNICAL DATA

TECHNICAL SPECIFICATIONS		
Model	EURO VA IES LIGHT	
Serial number	17203	
Pitch diameter	288 mm	
Nr of heads	6	
Direction of rotation	Clockwise	
Manufacturing year	2018	
NOMINAL PRODUCTION		
Pieces/hour	10000	bph
CLOSING MACHINE WEIGHT		
Weight	870	kg.
ELECTRICAL DATA		
POWER SUPPLY VOLTAGE:		
main	480	V ~
	60	Hz
auxiliary	24	V =
Installed power:		
closing machine lifting motor	0,37	kW
caps sorter motor	0,37	kW
U.V. lamp power	0,04	kW
Total installed power:	0,78	kW
PNEUMATIC DATA		
Sterile air capacity	400 ca	l/min (1 bar 20°C)
Sterile air min. pressure	6	bar
Sterile air max. pressure	8	bar



4.4 USE RESTRICTIONS

4.4.1 PRESCRIBED USE AND FORBIDDEN USES

The closing machine has been designed and made to apply closure systems (caps/corks) on preformed rigid containers with production limits indicated in chapter 4.3 CLOSING MACHINE TECHNICAL DATA page 13.

The closing machine must be used to apply closure systems (caps/corks) on preshaped rigid container already produced according to the order, planning and building stages, as described in the tables attached to this manual.

The closing machine receives the closure systems correctly orientated by its orientation system.

Besides the conduction of the closing machine out of what underlined in the chapter 4.3 CLOSING MACHINE TECHNICAL DATA page 13, just as an example and not limited to, the following uses are forbidden and not expected:

- use of closure systems (caps/corks) and/or preshaped rigid containers whose specifications are different to the ones described in the expected use.
- supply to the closure systems (caps/corks) and/or preshaped rigid hand containers or other means.
- revert the rotation direction of motors.
- tamper guards and/or protection devices.
- overlook the protection means using means to rise on the top or climb on parts of the closing machine.
- perform maintenance operations without insulating and locking all the energy sources and dissipating the residual energies.
- do not comply with the specifications of power sources listed in the table page 13.
- Treat products with characteristics other than those communicated at the order, changing the environmental conditions here the closing machine must be fitted (example: temperature of the treated product; with environmental characteristics other than those communicated at the order).

4.4.2 OPERATING MODES

The closing machine cannot work autonomously.

The installation technician must thus install the closing machine under his own responsibility and according to the European Community directives and take care that all requirements and specifications of the laws in force are complied with.



4.4.3 NUMBER OF OPERATORS

The operations described in this manual, referring the cycle of life of the closing machine, have been analysed by AROL S.p.A.. Therefore, the number of phases and operators indicated is the one most suitable to carry out the functions in a optimal way.

4.4.4 PROFESSIONAL ROLES OF USERS

The closing machine User shall check that the operators have the following requirements:

- had read and understood the indications that are in the manual.
- had received job instruction and training appropriate to their tasks;
- have been properly trained about the correct use of the closing machine, about the residual risks, on the closing machine controls and on the PPM.

Only the operators and the person charged with the maintenance with a suitable training can use the closing machine.

The professional terms used are specified here below to understand the manual fully:

TERM	DESCRIPTION
EMPLOYER	Person legally responsible for the use of the closing machine described herein.
OPERATOR	Person charged with the standard closing machine operation.
MECHANICAL MAINTENANCE OPERATORS	Person charged with the routine and extraordinary mechanical maintenance of the closing machine.
ELECTRICAL MAINTENANCE OPERATORS	Person charged with the routine and extraordinary electrical maintenance of the closing machine.

4.4.4.1 OPERATOR

The Operator must have:

- familiarity with the use of the check system (operator interface), with capacity of checking the correlation between the set data and the effect of these on the production;
- knowledge of the technology and specific experience in operating the closing machine acquired through the basic training of this manual;
- base general knowledge and sufficient technical knowledge to read and understand the manual content and the annexes, included the correct interpretation of the drawings.
- Sufficient technical knowledge to carry out intervention specified in the manual, in safe operation condition.
- Knowledge of the accident-preventing norms.
 - General Health and safety, prevention of the accidents on the work place.
 - specification (for the type of product processed by the closing machine, according the laws in force of the country in which the machine is installed).



Besides the conduction of the closing machine from the productive point of view the operator must:

- Know the operation of all the safety devices (for the operator and for the closing machine) so as to be able to check their efficiency and signal any possible fault to the personnel in charge according to the procedures described herein.

When the operator locates a failure that might compromise safety, he must stop the closing machine using the emergency stop controls.



it is forbidden to execute any intervention on the electric and/or pneumatic equipment.

4.4.4.2 MAINTENANCE MAN

The Maintenance Man (mechanical/electrical), besides the above characteristics, must also have a good technical preparation and a suitable working experience in the relevant field. Further, he must have specific and special (mechanical and electrical) technical knowledge needed for the operations described in this manual.

4.4.5 EMPLOYER'S OBLIGATIONS

the employer shall guarantee appropriate training and information of all the professional roles of the closing machine on the following topics:

- Instruction and specific procedures for use and maintenance of the closing machine.
- The user of the closing machine should comply with the safety and health prescriptions on the workplace, in force in the premises where the closing machine is started; all the specific risk conditions should be taken into account.

He must have acquired a good knowledge of the contents of this manual so that he can correctly process the training procedures of the professional roles of the closing machine.

At the end of the training, the employer must make sure that operators and maintenance technicians have understood the above points.

Assess the risks resulting from installing the closing machine in the workplace.

The employer must supply the PP (Personal Protection Means) required for all operations linked to the useful life of the closing machine.



4.4.6 INFORMATION - JOB INSTRUCTIONS - TRAINING

4.4.6.1 INFORMATION - JOB INSTRUCTIONS

The employer shall inform and train the personnel charged to use the closing machine and make sure that:

- On the work place there are suitable information concerning dangers, risks and prevention and protection solutions including the use of P.P.M., if provided.
- the workers the task of using of work equipment receive a suitable training accordingly.
- The workers the task of using of equipment requiring special knowledge and responsibility receive a suitable and specific training, so that they can use such equipment safely and suitably, also as far as risks caused to other people are concerned.

4.4.6.2 TRAINING

Training serves to explain the correct use of the closing machine, in compliance with the obligations prescribed by national laws transposing 2009/104/EC directive concerning minimum safety and health requirements for the use of working tools by the operators while working.

4.4.7 ARRANGEMENTS BY THE CUSTOMER

Excluding special contract conditions, the user must arrange the following:

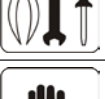
- The closing machine must be electrically supplied by the electric board of the machine to which the partly completed machinery is to be installed. The protection against indirect contact must be envisaged during installation, according to point 6.3 of EN 60204-1.
- Compressed air mains connection.
- Generic tools for the closing machine maintenance and consumables.
- Lifting means suitable for the closing machine handling.



4.4.8 SIGNS FOR WARNING, OBLIGATION, PROHIBITION AND DANGER

SYMBOLS CONTAINED IN THE MANUAL

The graphic symbols inside the manual are described and shown here below.

GRAPHIC SYMBOL	DESCRIPTION
	It indicates a generic warning. Pay utmost care to the text blocks marked by this symbol.
	Indicates the danger of hot surfaces. Pay utmost care to the text blocks marked by this symbol.
	It refers to a warning about key function or useful information. Pay utmost care to the text blocks marked by this symbol.
	Notes, usage tips, or additional information
	Part of the manual dedicated to the OPERATOR.
	Part of the manual dedicated to the MECHANICAL MAINTENANCE OPERATOR.
	Part of the manual dedicated to the ELECTRICAL MAINTENANCE OPERATOR.
	Obligation to insulate all energy sources. The symbol placed close to this description means that all the energy sources must be insulated, dissipating the residual energies by the user, because this involves the risk of an accident.
	Mandatory use of <i>PROTECTIVE GLOVES</i> . The presence of the symbol close to the descriptions informs that the operator must wear safety gloves due to the risk of injury.
	INSPECTION AND OPERATION Visual inspection.
	INSPECTION AND OPERATION Inspection hearing.
	INSPECTION AND OPERATION Intervention with tools
	INSPECTION AND OPERATION Manual intervention.
	INSPECTION AND OPERATION Intervention of lubrication.



SYMBOLS ON THE CLOSING MACHINE

A set of pictograms describing procedures, obligations, cautions and prohibitions are arranged on the closing machine to attract the operator's attention.



The warning and/or prohibition pictograms on the closing machine have an extremely important safety function; the operators must follow the warning indications.



The pictograms of the closing machine cannot be removed.

Should the pictograms wear out occasionally and become illegible, or if they are removed, they must be restored with new ones that must be asked to AROL S.p.A..

The pictograms on the closing machine are described and shown here below.

PICTOGRAM	DESCRIPTION
	Indicates the duty to see instruction manual before to operate on the machine.
	It indicates a generic warning. Pay attention to the specific pictogram that accompanies it.
	Instructs to unlock the handles holding the anti-torsion columns and the central column fastening screw.
	Means that the lubricant cartridge must be activated before starting the machine.
	



The following figures indicate the pictograms on the closing machine and their position.

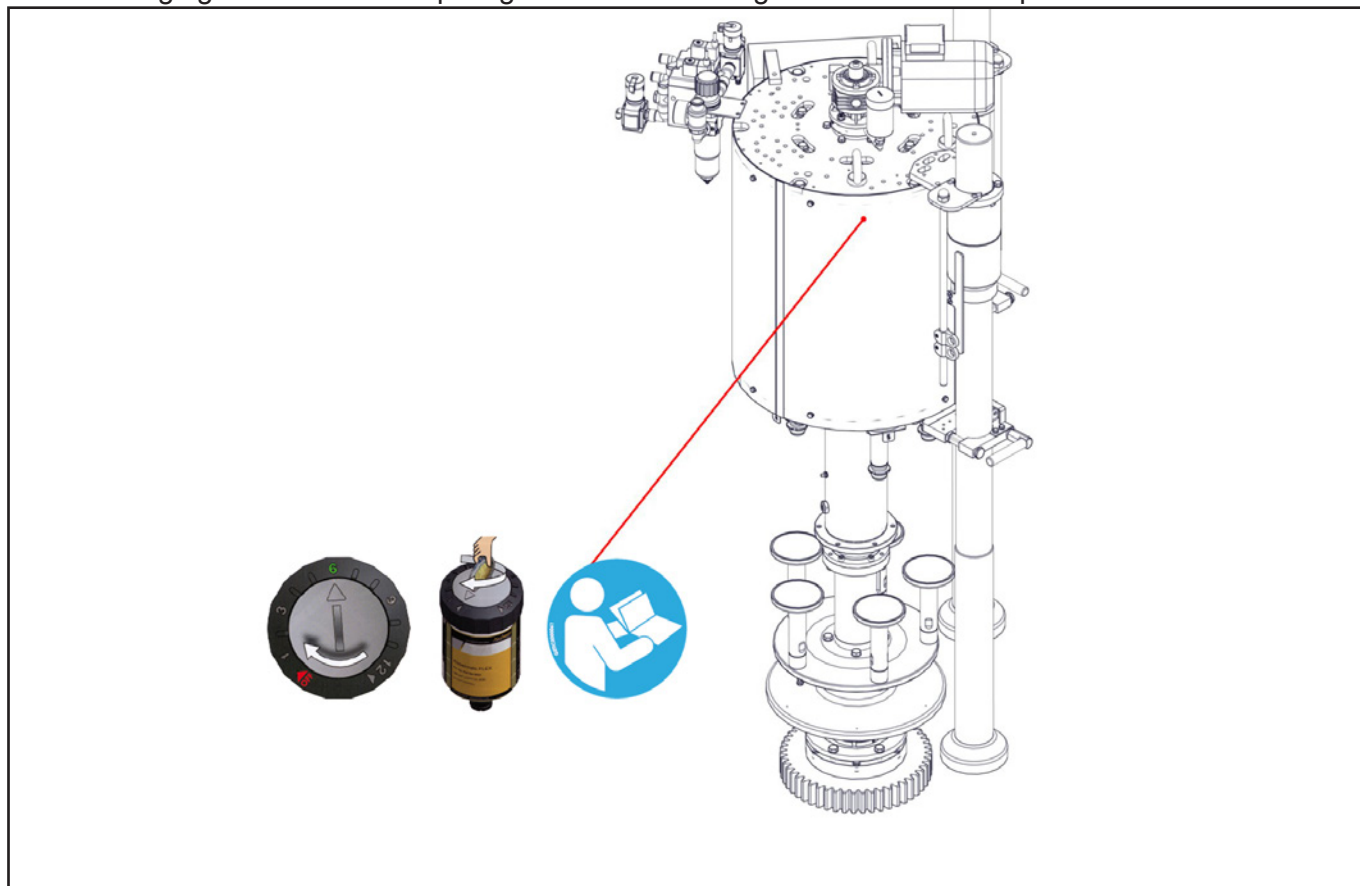


Fig. 10

Re-ordering code: GETIC0000061

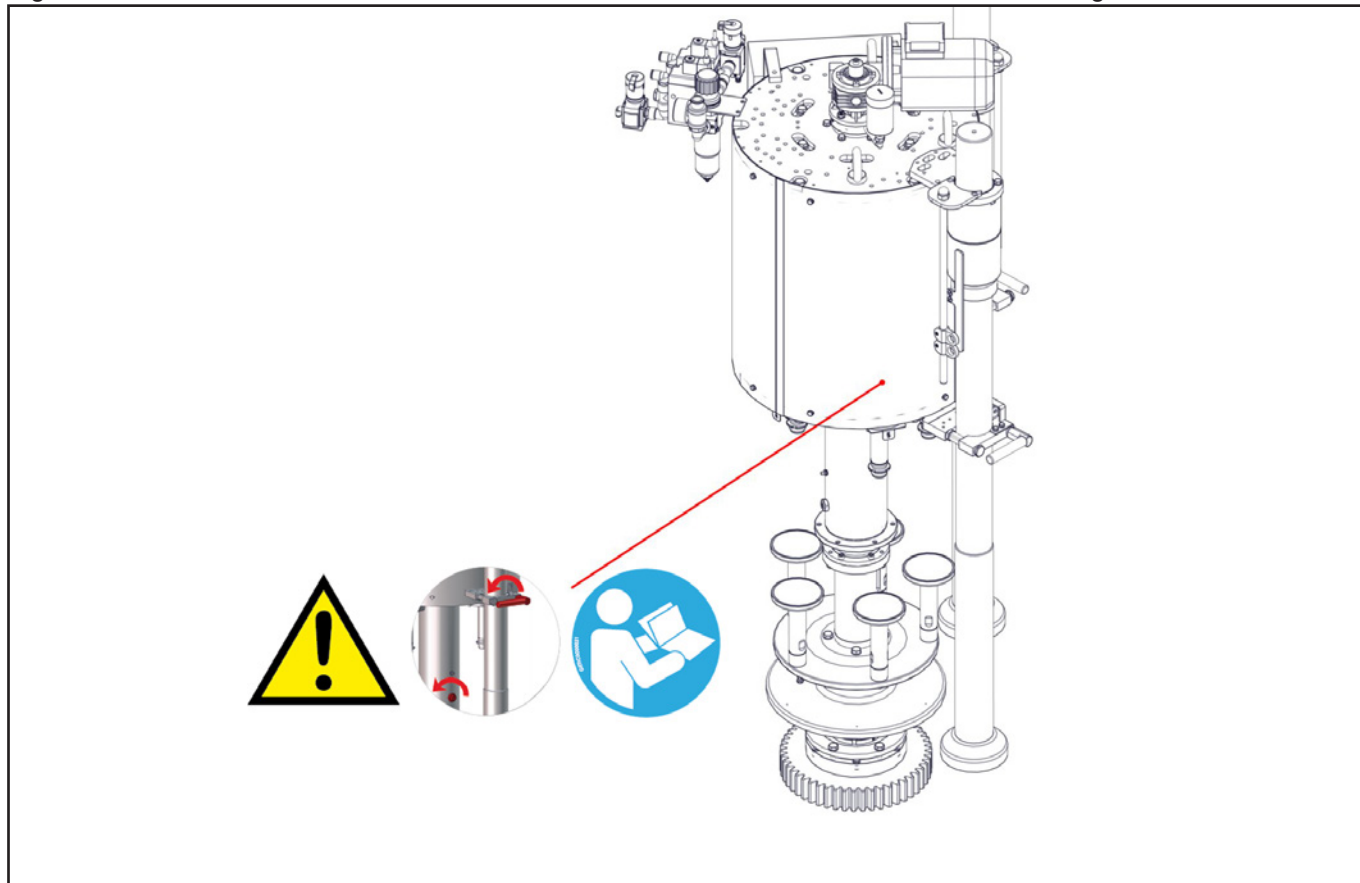


Fig. 11

Re-ordering code: GETIC0000021



4.5 ENVIRONMENTAL LIMITS

4.5.1 ENVIRONMENTAL OPERATING RISKS

USE ENVIRONMENTAL CONDITIONS

The closing machine must work in premises protected by weather agents, with a temperature ranging between 5°C and 40°C, a maximum relative humidity of 80% and height above sea level not over 2000 m.

LIGHTING

The closing machine can work with the standard lighting of the workplace (about 250 lux); local lighting of areas concerned with maintenance is not required very often and must be performed with low voltage lights.

EXPLOSIVE ATMOSPHERE

It is strictly forbidden to operate the closing machine in an area with potentially explosive atmosphere, even if only momentarily.

4.5.2 RESIDUES AND ENVIRONMENTAL CONTAMINATION

No toxic substance is released if the closing machine is used and maintained according to the instructions herein.

The disposal of spent lubricants must respect the norms in force for environment protection in the country of use.



5 SECURITY AND ACCIDENT PREVENTION

5.1 GENERAL INFORMATION

The employer will must provide for the training of the staff about dangers caused by accidents, about equipments arranged for the operator safety, about dangers of noise emission and about the general accident rules provided by international directives and by legislation of the country of designation of the closing machine.

Anyway the behaviour of staff, maintenance operator, cleaning, control, etc. will must scrupulously respect the accident norms of the country of designation of the closing machine.



Before starting the work, the operator must be thoroughly familiar with the location and operation of all controls and characteristics of the closing machine must also have fully read this manual.



The closing machine must be used only from operators that had received a training and understand totally the instructions of this manual.



It is necessary to respect totally instructions, cautions, general accident rules that are in this manual.



Before any maintenance and/or size change operation, the closing machine must be safety conditions:

- **insulate the closing machine from all energy sources dissipating residual energies.**
- **divide and lock all the cutting off devices;**
- **Prevent access to the closing machine to unskilled staff, not charged with maintenance operations, using suitable signs.**



5.2 RISKS - PROTECTIONS - CAUTIONS AND PRECAUTIONS

5.2.1 SAFETY DEVICES OF THE CLOSING MACHINE

To guarantee health and safety of the people exposed, the closing machine is provided with this safeties:

FIXED GUARDS

All the fixed guards installed on the closing machine are connected to the structure by means of fastening devices which do not allow manual removal, since they necessarily require the use of specific tools.

It is forbidden to use the closing machine after having disassembled the fixed guards; operation can resume after the reassembly of guards.

The housing of the fixed guards in vertical position in the structure of the closing machine has been designed so that the guards, once the fastening devices have been removed, do not remain in closed position, falling by gravity.



To access the area protected by the guard, the closing machine must be insulated from all power sources. (See sub-chapter 5.4 SAFETY PROCEDURES FOR THE MAINTENANCE page 25).

5.2.1.1 POSITION OF THE SAFETIES

FIXED GUARDS

- Head protection guard.

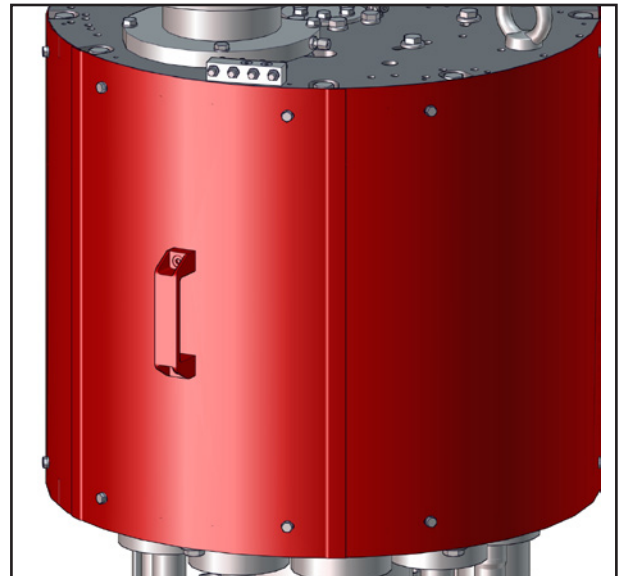


Fig. 12



5.3 SAFETY PROCEDURES FOR THE OPERATION



Wrenches and tools used for the maintenance or the access to closing machine hazardous sections must not be left at the disposal of operators not suitably trained for the size change or the maintenance.

All check, adjustment, maintenance and lubrication steps must be carried out by staff previously trained and instructed.

All check, adjustment and maintenance steps of the electric system must be carried out by staff previously trained and instructed.

People on drugs, alcohol or drowsiness-causing medicines are not allowed to use or maintain the machine.

All check, adjustment, maintenance and lubrication steps must be carried out with the closing machine stopped and separated and split electric power supply.



The cap/cork feeding system has been designed and manufactured to be loaded by an automatic elevator with ground loading.

If the cork/cap loading is manual, it should be performed leaving the closing machine at a standstill, with separated and split electric power supply.



Before accessing the top of the machine, separate and split the energy sources and dissipate the residual energies.

Access at any height must occur using certified means according to the norm UNI EN ISO 14122.



5.4 SAFETY PROCEDURES FOR THE MAINTENANCE



BEFORE CARRYING OUT ANY MAINTENANCE OPERATIONS, the closing machine must be in safety conditions to avoid untimely screwing and/or electrocution.



It is therefore compulsory to insulate all energy sources dissipating residual energies.

The insulation for power sources must be charged to qualified technicians and be performed as follows:



Wrenches and tools used for the maintenance or the access to closing machine hazardous sections must not be left at the disposal of operators not suitably trained for the size change or the maintenance.

After the maintenance works and before starting the closing machine again it is always necessary to:

- check that the possibly replaced parts and/or used tools have been removed from the closing machine.
- check that all guards, protections and safety devices possibly removed during the intervention have been re-installed correctly and are operating.



Replace the electrical safety components within their life cycle, approximately 20 years.



The standard operating conditions will be restored only after the intervention and the checks.



5.4.1 GENERAL SAFETY WARNINGS

The safety for the closing machine and the operators also depends on a regular maintenance according to the manufacturer's indications.

The routine and extraordinary maintenance interventions must be exclusively charged to skilled staff, that is to technicians qualified according to the specifications in this manual.

If outsourced operators intervene, they must ensure the same performance of the inhouse qualified technicians and must work in safe operating conditions.

The access to high levels is allowed only after separating and disconnecting any residual energy and using means certified according to UNI EN ISO 14122 standard.

The qualified technicians must:

- respects the limits of their tasks (mechanical, electrical, etc.).
- Follow the procedures and warnings in the manual, within their tasks.
- Follow the times and intervals indicated in the manual for scheduled maintenance interventions.



The following general warnings must be noted:

- Electrocution hazard due to direct contact with the closing machine power supply terminal boards and in the electrical system connector blocks (the hazard is signalled by plates with the suitable triangle with yellow bottom).
- Be extremely careful during the interventions on the electrical system, if the system must be left live. Such interventions must be carried out by staff skilled in this type of maintenance only, trained and correctly equipped for the steps to be made. The intervention with live system must always be authorized in advance by the employer who must oversee its operations by assessing the specific risk situations.
- Carry out maintenance operations on pneumatic systems and components only after cutting off and separating the electric power and pneumatic sources and checking that the residual pressures have been relieved.
- It is forbidden to carry out maintenance and lubrication interventions on moving components.
- While working on heated parts, the operator must always wait until they cool to a temperature below 55°C to prevent serious burns.

5.5 RESIDUAL RISKS

The residual risks are shown in Annex VII B of the European Directive 2006/42/EC about "Technical documentation for partly completed machinery".



6 HOW TO ASK FOR INTERVENTIONS

AROL S.p.A. puts its own Technical Service at the disposal of its customers to solve any problem regarding the machine use and maintenance.

The interventions must be required after carefully evaluating the faults and their reasons, and stating:

- The details on the occurred faults.
- The performed checks.
- The performed adjustments and resulting effects.
- Any further information deemed useful.

7 HOW TO ORDER SPARE PARTS

All orders for spare parts must be sent to AROL S.p.A..

The installation of not original spare parts will invalidate the warranty.

To order spare parts, fill in the suitable form in the “*SPARE PART CATALOGUE*” for the machine.



The replacement of safety devices must be performed with the same components. The 2006/42/EC directive provides that the components supplied by the machine genuine manufacturer do not need the Declaration of Conformity. Vice versa for all components purchased on the market, the Declaration of Conformity is required, along with the control that the replacement component has the same features and performance of the replaced one. Unsuitable components might cause the decay of the safety performance expected by AROL S.p.A.. It is therefore recommended to contact AROL S.p.A..



8 INSTALLATION AND CONNECTIONS

8.1 INSTALLATION OF THE CAPS CHUTE WITH “PICK-OFF” CAP DISTRIBUTION



To install the caps chute on the machine it is necessary:

- Loosen the threaded fasteners located on the support chute.

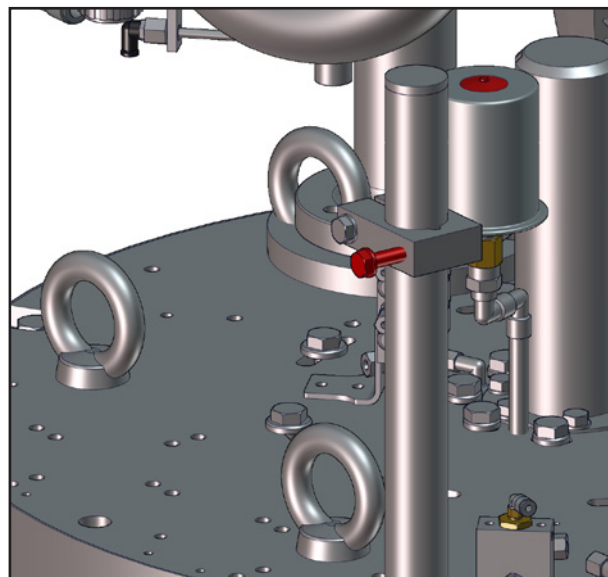


Fig. 13

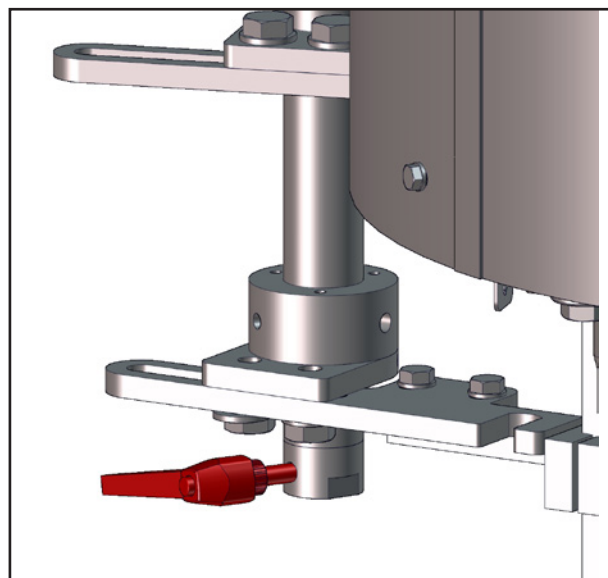


Fig. 14



- Insert the head height adjustment pin into the hole of the chute support and lift the complete caps chute.

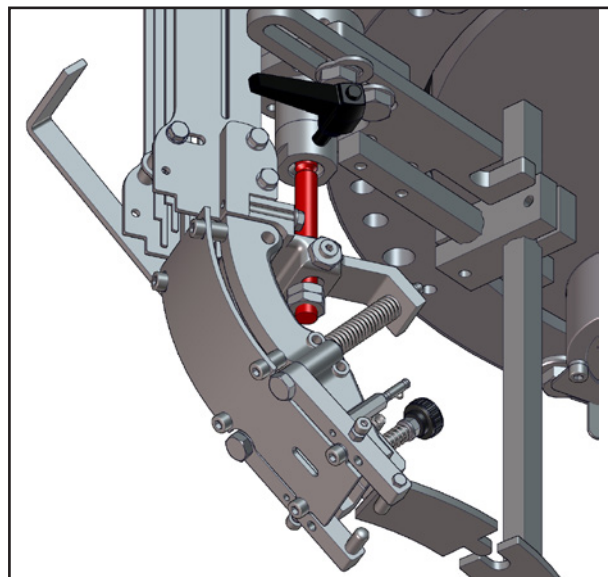


Fig. 15

- Place the caps chute so that the fixing brackets are positioned on threaded fasteners.
- Lock threaded fasteners located on the support chute.

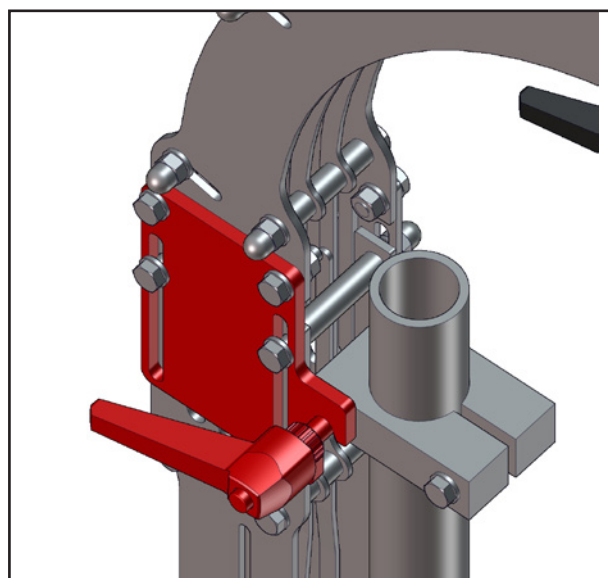


Fig. 16

- Lock the threaded fixing system of the caps distribution head.

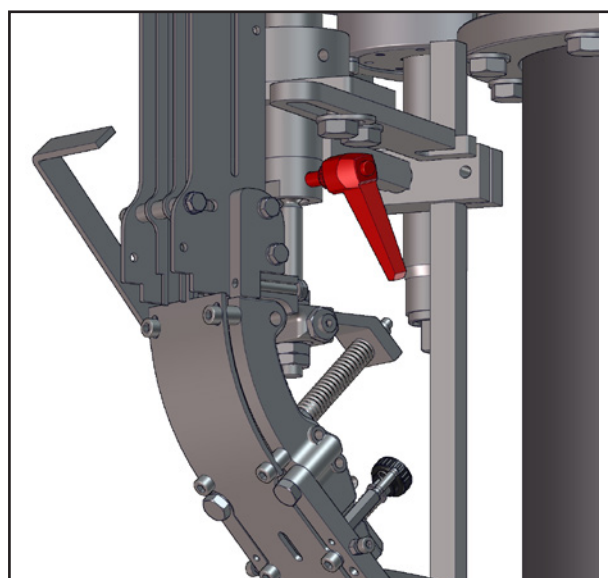


Fig. 17



- Connect the electrical connection of the caps presence photocell (one or more).



Fig. 18



8.2 INSTALLATION OF THE "FLAT" CENTRIFUGAL CAPS SORTER



For the installation of the centrifugal caps sorter see the listed below operations:

- Lift and extract the hopper lid.

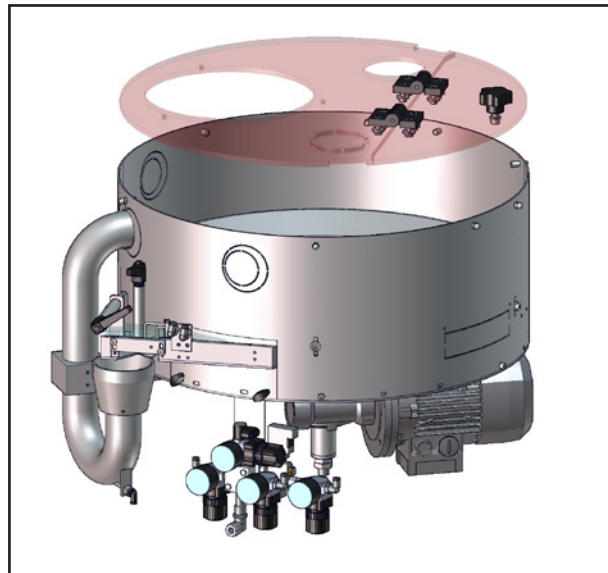


Fig. 19

- Unscrew and remove the central screw locking cone.

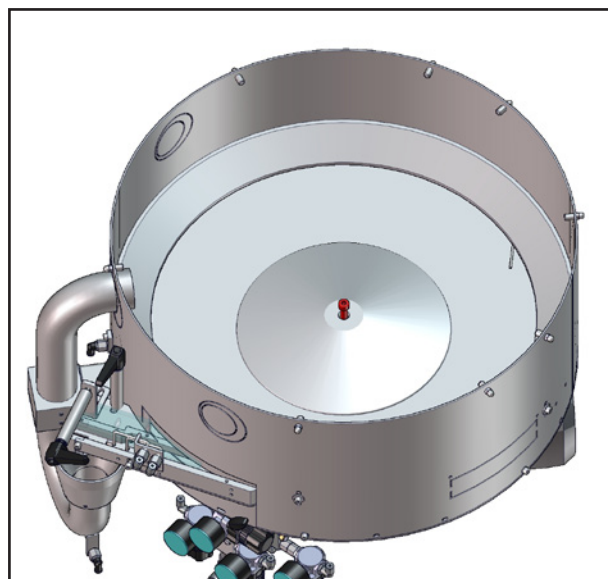


Fig. 20



- Slide out the central cone.

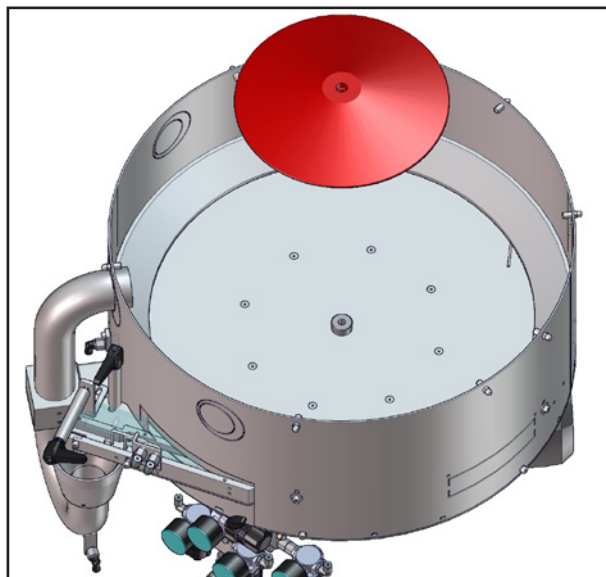


Fig. 21

- Screw at the center of the rotating disc an M8 eyebolt

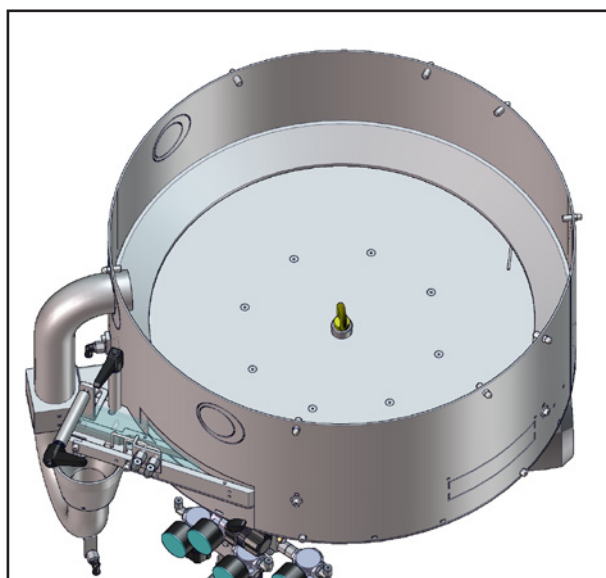


Fig. 22

- Hook with a belt or a chain with hooks to the M8 eyebolt.

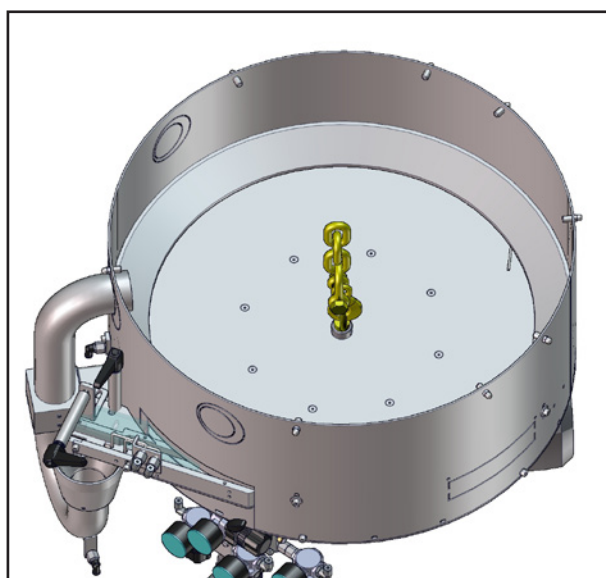


Fig. 23



- Unscrew the bolts that fasten the caps sorter to the transport platform.
- Lift the caps sorter with adequate lifting means.



The caps sorter mass is not symmetrically distributed so it overturns when lifted up. Do not stand or walk under the caps sorter when it is hoisted!

- Position the caps sorter complete with the support in such way that the references on the support and the references of the flange are aligned and that the exit mouth is aligned with the caps chute.
- Fix the caps sorter by bolts.

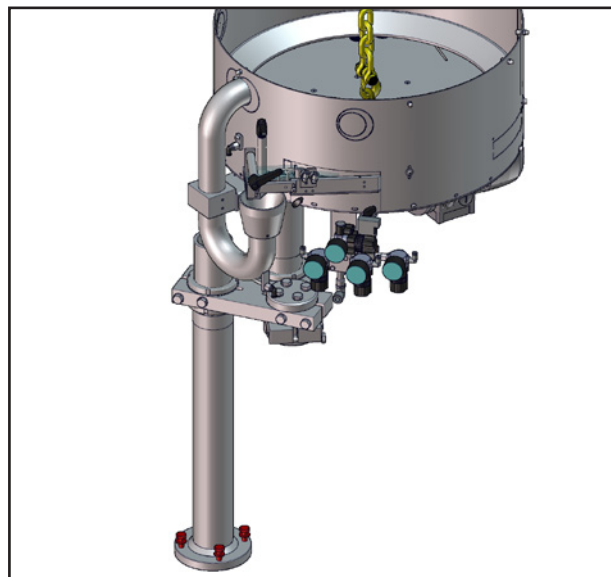


Fig. 24

- Unhook the belt or the chain with hooks from the eyebolt.

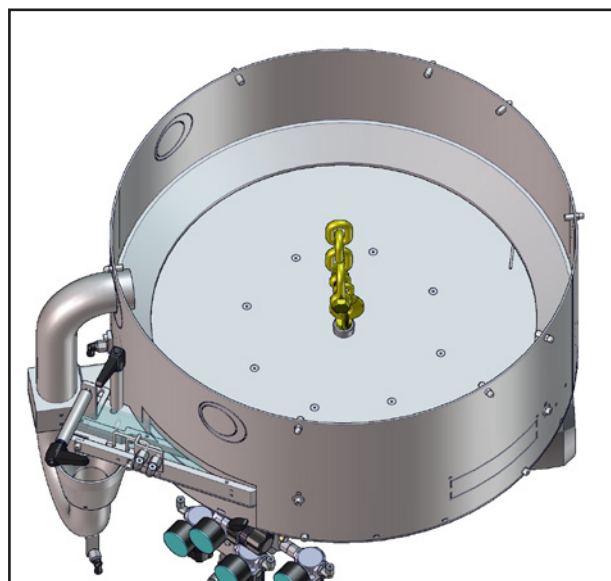


Fig. 25



- Remove the eyebolt.

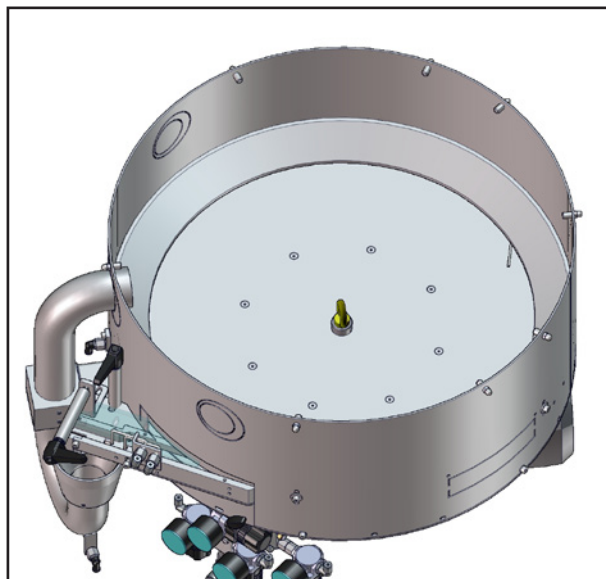


Fig. 26

- Remount the central cone and fix it with the screw.

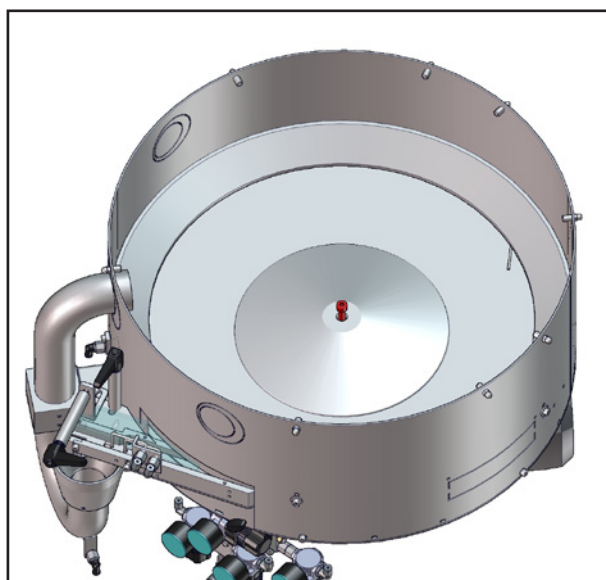


Fig. 27

- Place and fasten the caps cover on the caps sorter proceeding as described below:
 - Lean the complete caps sorter cover on the support pins set on the hopper in such way the hole for the caps level device is in the opposite of the caps exit mouth.

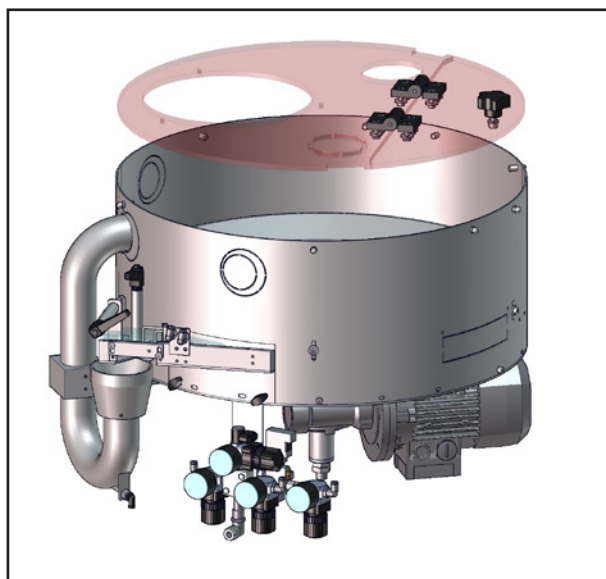


Fig. 28



- Lock threaded fasteners located on the support chute.

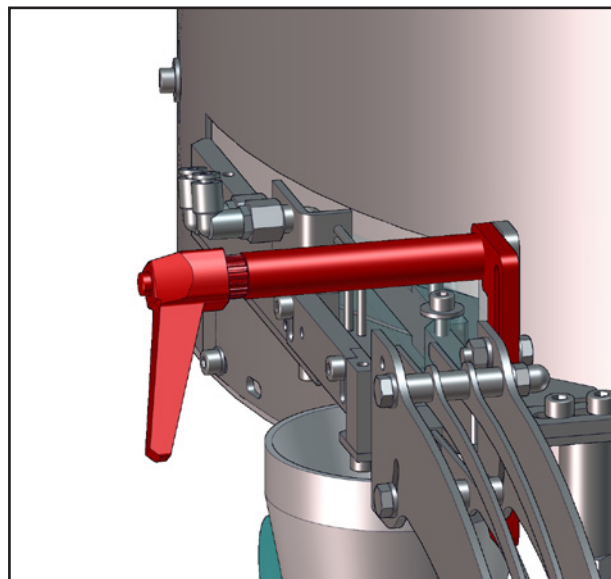


Fig. 29

- Place the caps level device on the caps sorter proceeding as described below:
 - Place the level blade in the hole of the cover and the rod fixing bush on the hopper.

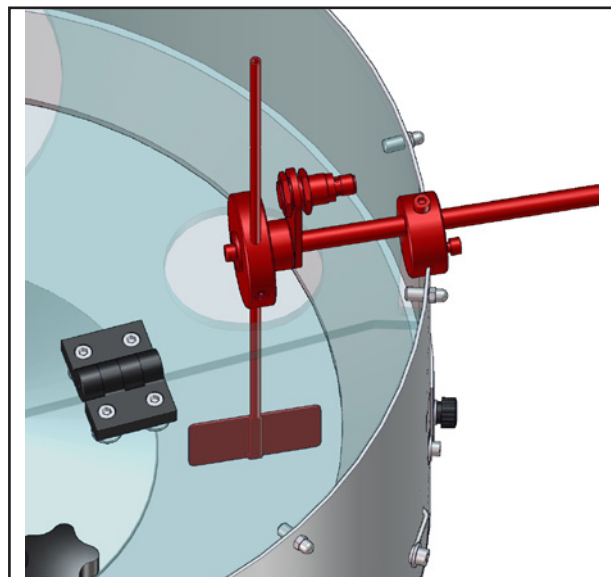


Fig. 30

- Fasten all things locking the screw against the hopper.

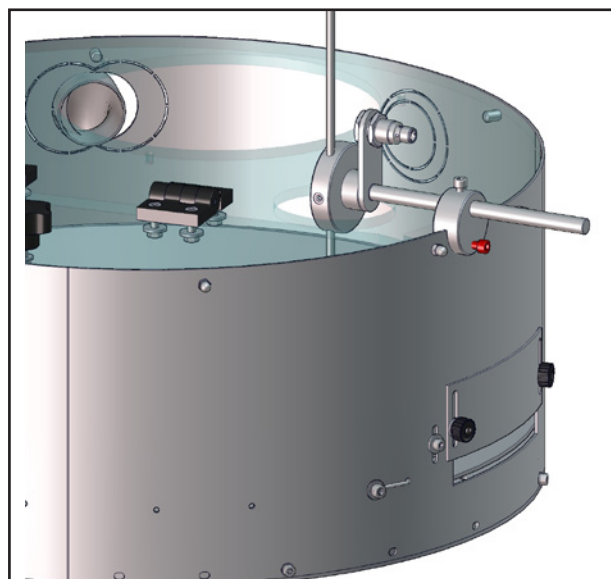


Fig. 31



8.3 PNEUMATIC CONNECTION



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.



The air used in the pneumatic installation must be filtered, so without impurities, and dried.

For pneumatic data see the table page 13.

Connect with the mains as follows:

- connect the coupling of the pneumatic assembly to the main distribution network.

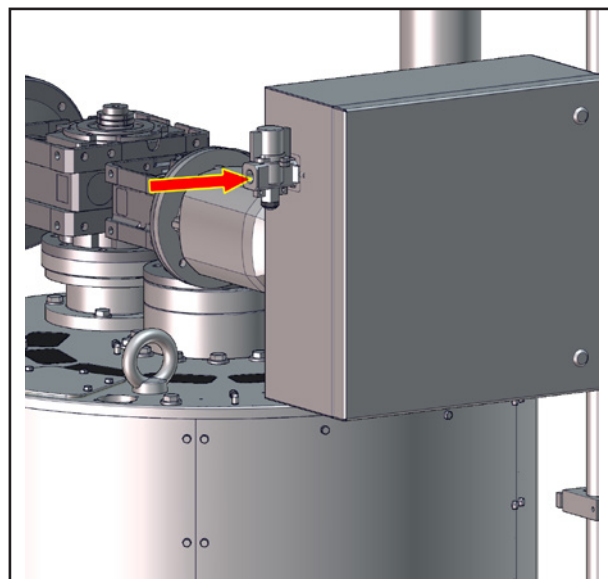


Fig. 32

- Connect the flow rate pipe coming from the main pneumatic unit to the caps orientation unit coupling

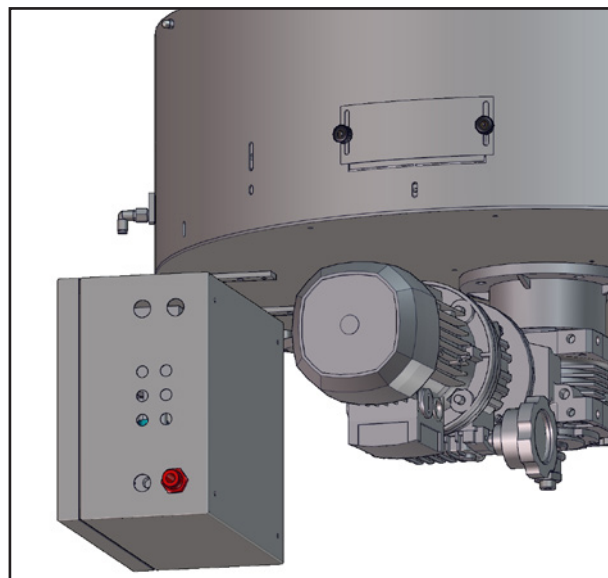


Fig. 33



- Enable the sectioning device to allow air to flow in.

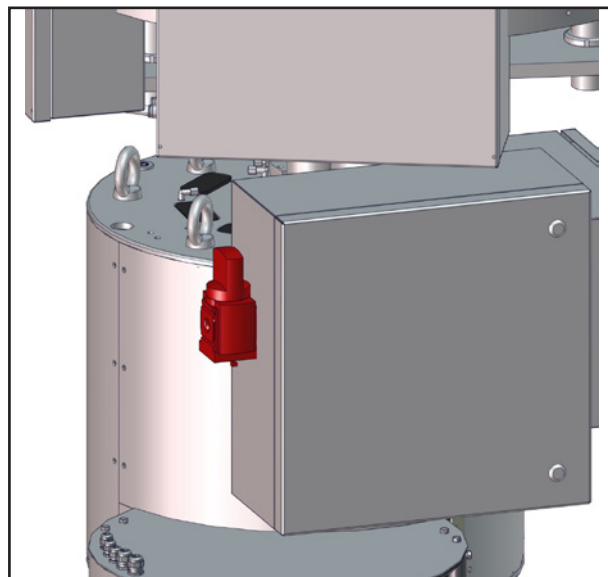


Fig. 34



9 CLOSING MACHINE USE

9.1 SETTING UP, CHECKS AND TEST FOR THE FIRST START UP



The setting up operations of the closing machine for the first start up must be carried out by qualified technicians.

The closing machine has been tested by AROL S.p.A. premises before the shipping; all the calibration and required adjustments have been carried out during the test.



Before carry out operations on the closing machine, it is necessary to install and activate the lubricant cartridge, proceeding as described in sub-chapter 11.4.6.2 CARTRIDGE REPLACEMENT page 130.

9.1.1 CHECKS BEFORE THE START UP



In order to prevent errors and/or accidents, before the start up of the closing machine it is necessary to carry out some checks:

- Check the sequence of the motor supply phases.
- Check that there are no leaks from the pneumatic circuit.



9.2 PRELIMINARY CHECKS BEFORE STARTING



Before starting the closing machine up, check the following:

- Check that all other safety devices installed on the closing machine are operating properly. (See also sub-chapter 5.2.1 SAFETY DEVICES OF THE CLOSING MACHINE page 23).
- Check all closing machine components are lubricated (See Chapter 11.4 LUBRICATION page 122).
- Make sure the specific installed equipment are suitable for the size to be processed.
- Check the correspondence between container and cork to be treated.



Each set of equipment is marked as indicated in the relevant tables enclosed in this manual. Each mark corresponds to a size to be processed.

- Check the height of the closure carousel according to the container height.
- Check the caps sorter is sufficiently full.
- Check that the working pressure is correct (the pressure is indicated on the pressure gauges).



9.3 PREPARATION FOR STARTING

9.3.1 CLOSING MACHINE HEIGHT ADJUSTMENT FOR "PICK OFF" DISTRIBUTION CAPS



The height must be adjusted exclusively by lowering the closure carousel. Before operating, take the closure carousel at a height higher than the one of the bottle to be processed.

For the adjustment and check, operate as described here below:

- Release the handles fastening the anti-torsion columns.

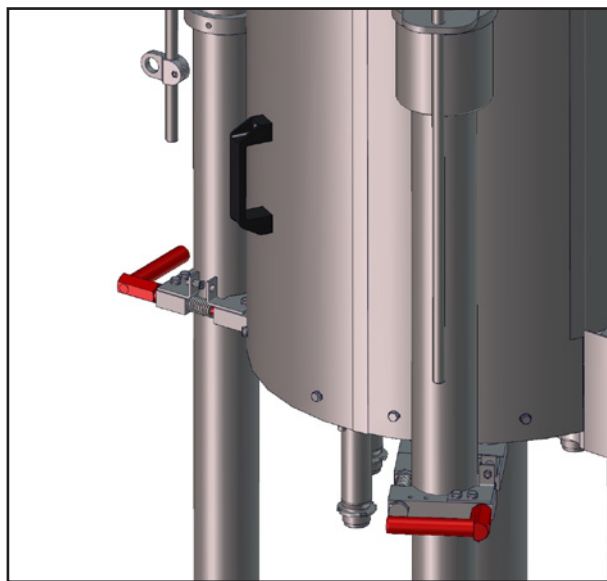


Fig. 35



Fig. 36



- Release the screw locking the central column.

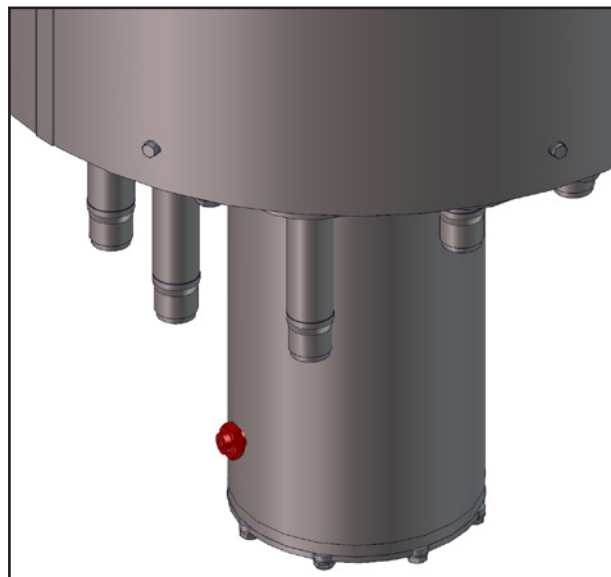


Fig. 37

- Loosen the fixing knob and lower the reference bracket.



Fig. 38

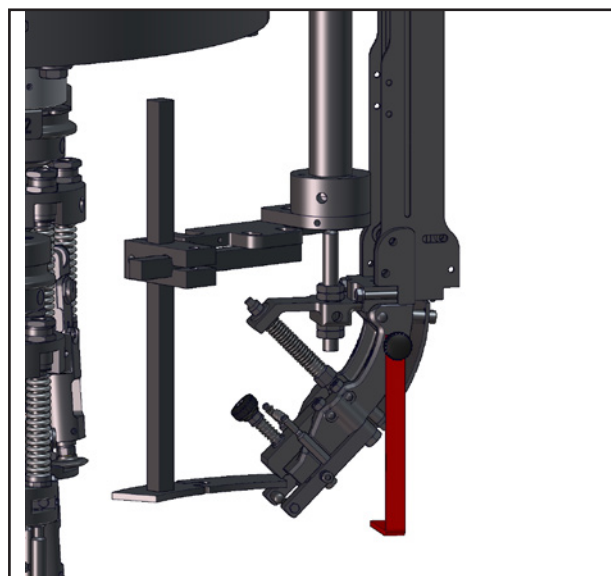


Fig. 39



- Lift the closure carousel using the control till passing the height of the bottle to be processed.
- Insert a bottle to be closed under the caps distribution head.

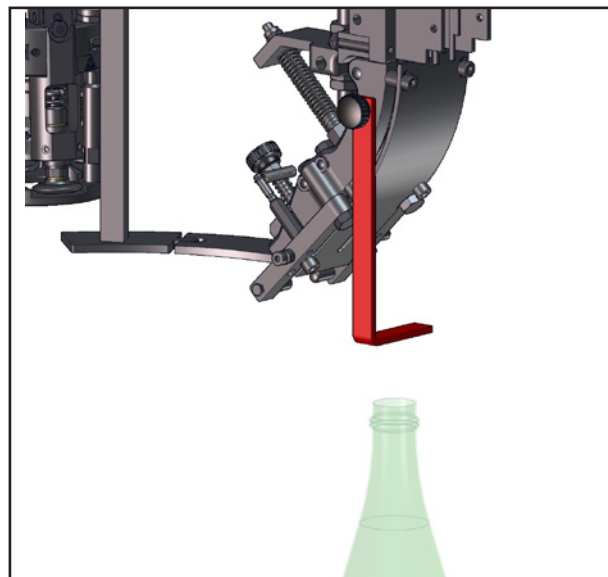


Fig. 40

- Lower the closure carousel until the reference bracket lays on the bottle neck.

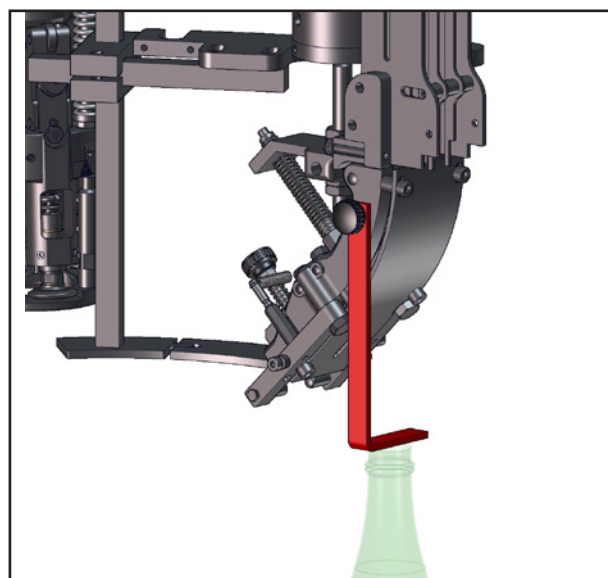


Fig. 41



- Loosen the fixing knob and turn upwards the reference bracket in such way it is not causing obstruction.



Fig. 42

- Lock the handles of the anti-torsion columns.

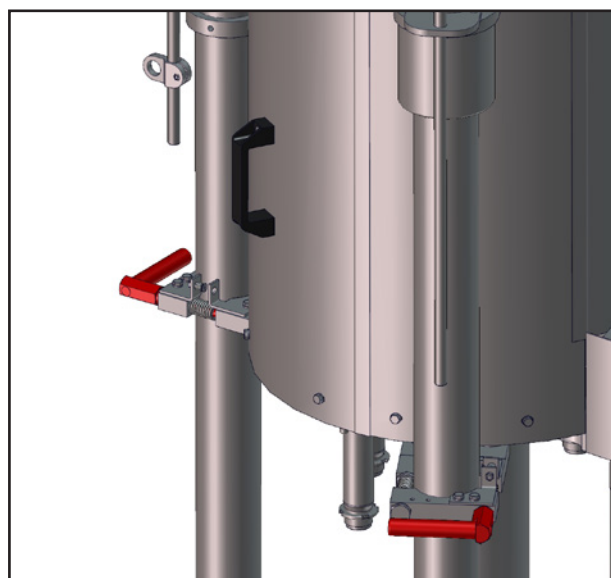


Fig. 43

- Lock the fixing screw of the central column.

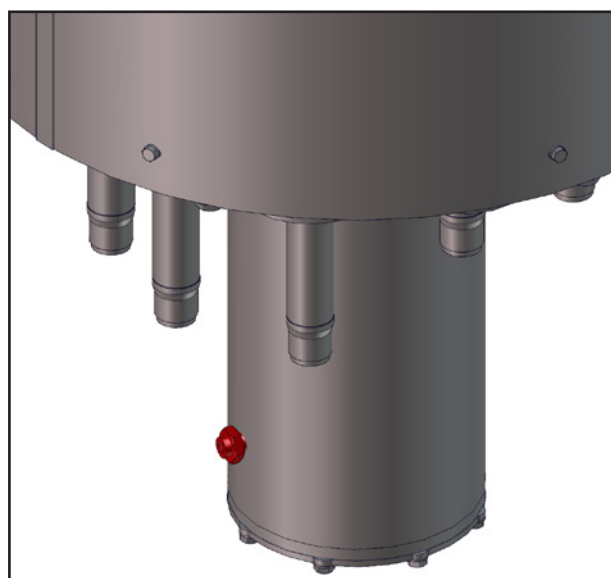


Fig. 44



9.3.2 UV RAY STERILIZER USE



9.3.2.1 INSTRUCTIONS FOR STERILIZER USE



Avoid exposure to the light emitted by the ultraviolet ray tube, even if only for a few seconds. The emitted light causes strong cases of conjunctivitis and erythema.



It is recommended to wear appropriate P.P.M. (Personal Protection Means) to protect the eyes, and do not leave gaping holes in the most sensitive areas of the body (neck, décolleté, arms, etc..), for hands, we recommend the use of cotton gloves with cuff up.



Optimal lamp performance is established after 100 hours of continuous operation. It is therefore recommended, as soon as the device is installed, to make it work without interruption for at least 100 hours.



Each switch on shortens the life of the lamp. It is therefore recommended, keeping it on always.

More ignitions and more power surges will undergo the lamp and the lower its life.

Therefore recommend that:

- never turning off the device unless as an exception for stops that last several hours.
- Do not exceed the number of three switch daily interspersed with several hours.



Do not use the lamp for more than 6,000 service hours, as beyond this limit, the germicidal lamp can no longer guarantee its sterilising effect. Make note of the date and time it was started up.



9.3.2.2 FUNCTIONING

The equipment works with direct radiation.

This device can remain on 24 hours on 24. Therefore, the microbic load can be maintained constantly low.

The lamp sterilizer is located within the safety guard.

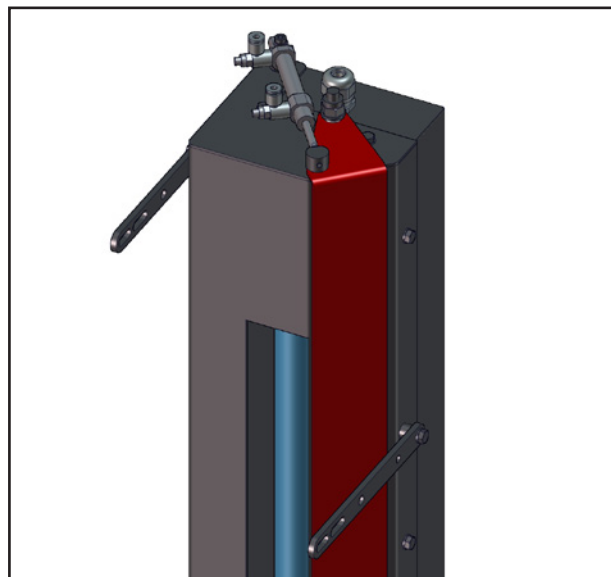


Fig. 45

In order to avoid exposure to UV lamp during breaks in production is necessary to operate the special protective cover provided by *AROL S.p.A.*

Such door, driven pneumatically by an appropriate control solenoid valve, will have to be closed (so as to obscure the lamp) during the stops of the machine and reopened before the restart.

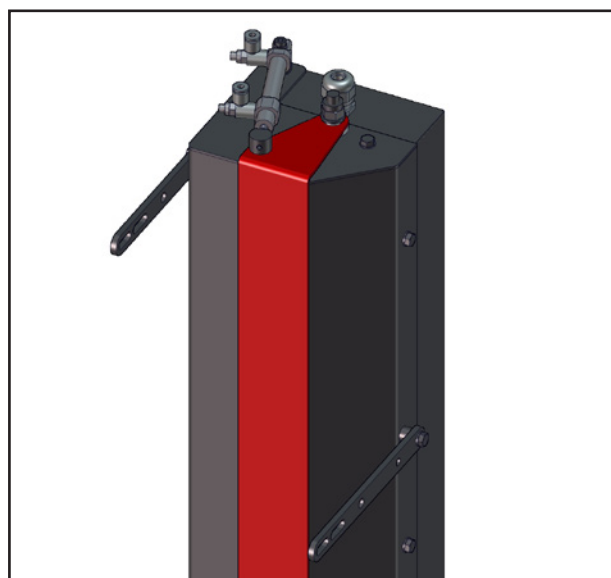


Fig. 46

9.3.2.3 STARTING UP

Supply power to the device, the UV tube will light up.

Verify the correct functioning of the lamp protecting door.



9.3.3 CAPS SORTER LOADING



On the caps sorter is installed a caps level sensor having the task to signal to the automatic elevator when it is necessary to put a new quantity of caps.

For correct operation, the device levelling rod must be adjusted so that it is about 52 mm above the height of an only layer of caps.

For this adjustment, loosen the grub screw on the levelling rod bush.

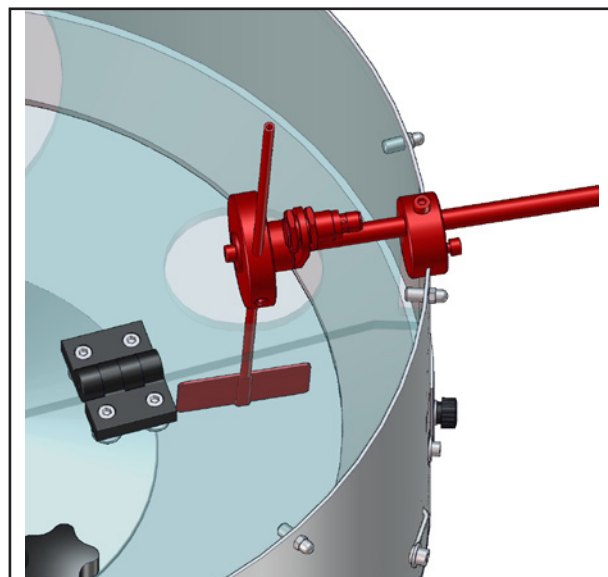


Fig. 47

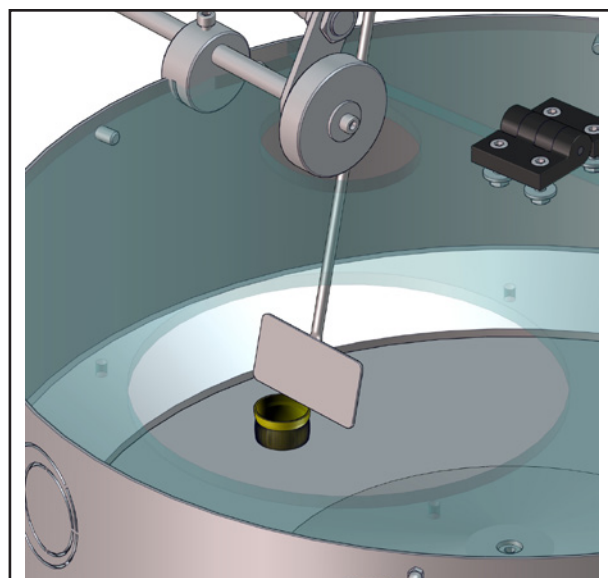


Fig. 48

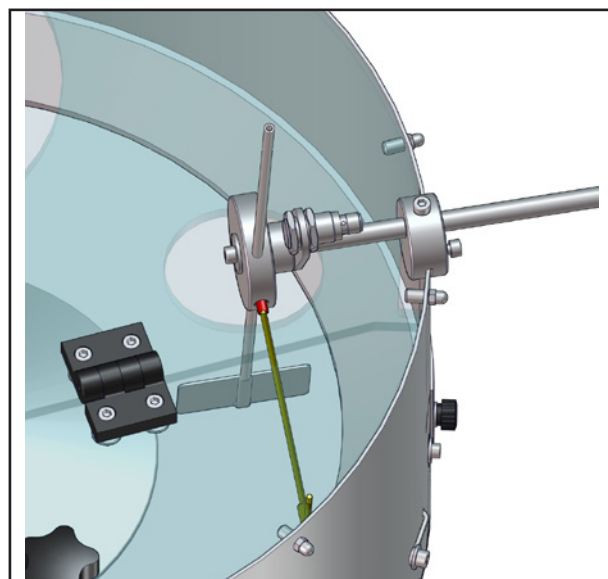


Fig. 49



The levelling rod must be positioned so that there is a distance of one cap between it and the hopper ring.

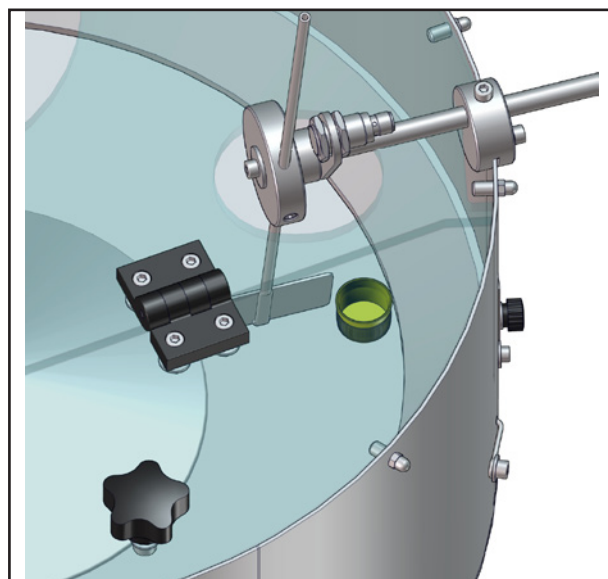


Fig. 50

For this adjustment, loosen the fixing screw on the rod fixing bush.

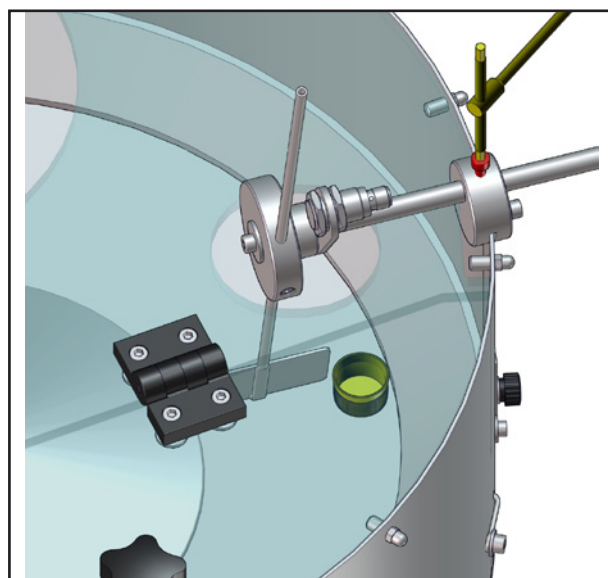


Fig. 51



For correct operation of the device, with empty orientator the sensor must be oriented so as to be in the center of the levelling rod.

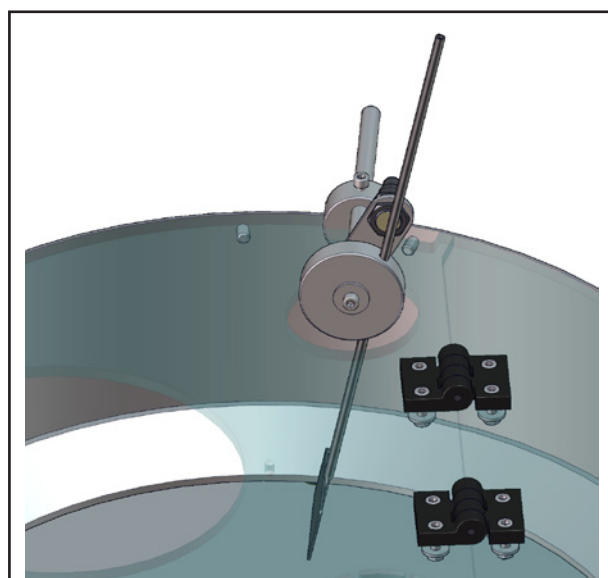


Fig. 52



9.4 EMPTYING OF THE CAPS DISTRIBUTION SYSTEM

9.4.1 CAPS SORTER EMPTYING



To discharge the caps sorter it is necessary to rotate it till totally empty.

9.4.2 CAPS CHUTE DISCHARGE



To discharge the caps chute, operate as follows:

- Insert a collection container under the caps distribution head.
- Remove the cap taking head spring.

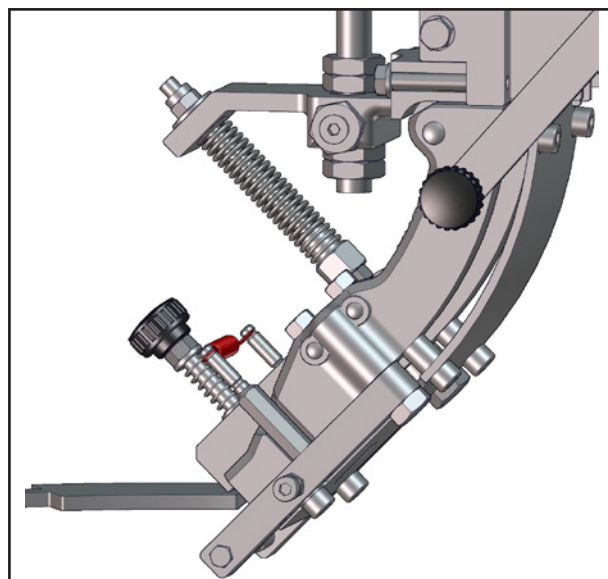


Fig. 53

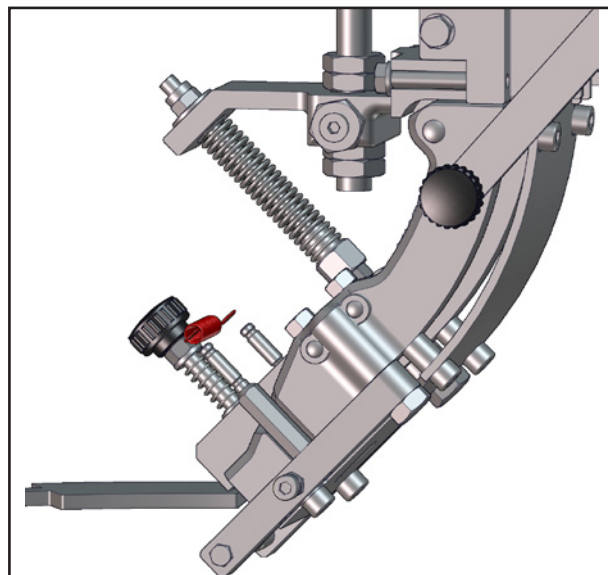


Fig. 54

- Rotate externally the caps stopping hammers so as to allow the discharge of the caps.
- If necessary, empty the cap sorter.
- Empty the cap chute.
- Restore the position of the caps stopping hammers.
- Mount the head taking cap springs on the spring hooking pins.

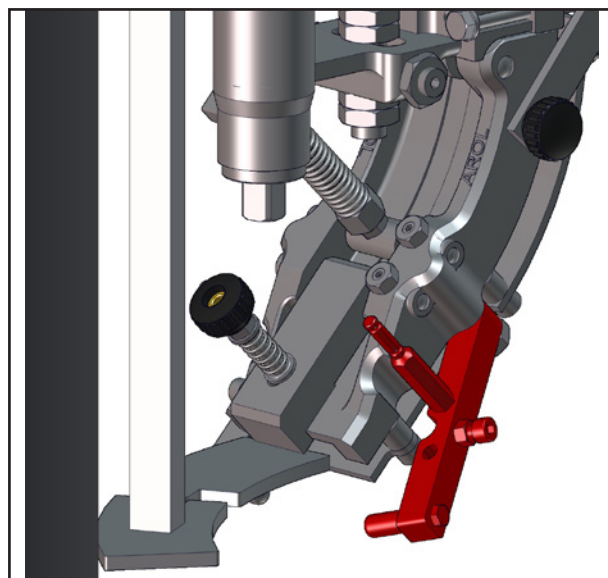


Fig. 55



9.5 CLOSING MACHINE CLEANING



The determination of the frequency of cleaning operations during the closing machine continuous use is to be checked by the user, depending on the environment is being operated.

The following subchapter describes the cleaning intervals of the closing machine and its components, when considering their use in a clean industrial production area.

9.5.1 CLOSING MACHINE CLEANING INTERVALS

The following table shows the cleaning intervals of the closing machine, expressed in hours.

Cleaning	Work shift (hours)		
	1 shift	2 shifts	3 shifts
Closing machine external washing. See sub-chapter 9.5.2 EXTERNAL WASHING OF THE CLOSING MACHINE page 51.	8	16	40
Closure head washing. See sub-chapter 9.5.2 EXTERNAL WASHING OF THE CLOSING MACHINE page 51.  After having washed the closure heads, proceed with lubrication as shown in the sub-chapter 11.4.7 LUBRICATION OF THE CLOSURE HEAD on page 131.	8	16	40
Cork/cap sorter cleaning. See sub-chapter 9.5.2 EXTERNAL WASHING OF THE CLOSING MACHINE page 51.	120	120	120
Cleaning the steriliser tube. Clean the steriliser's U.V. tube using a cloth and alcohol, without touching the glass parts with your hands (pure quartz).	1000	1000	1000



9.5.2 EXTERNAL WASHING OF THE CLOSING MACHINE



The description below concerns: cleaning areas, cleaning systems and specifications, products allowed for cleaning the closing machine and its components.

Legend:

- 1 Closure system direction area
- 2 Upper area of the closure carousel



The direction of the jet coming from cleaning system must be directed downwards. Do not direct air jet towards area "1".

- 3 Closure area
- 4 Bottle-guide and conveying area

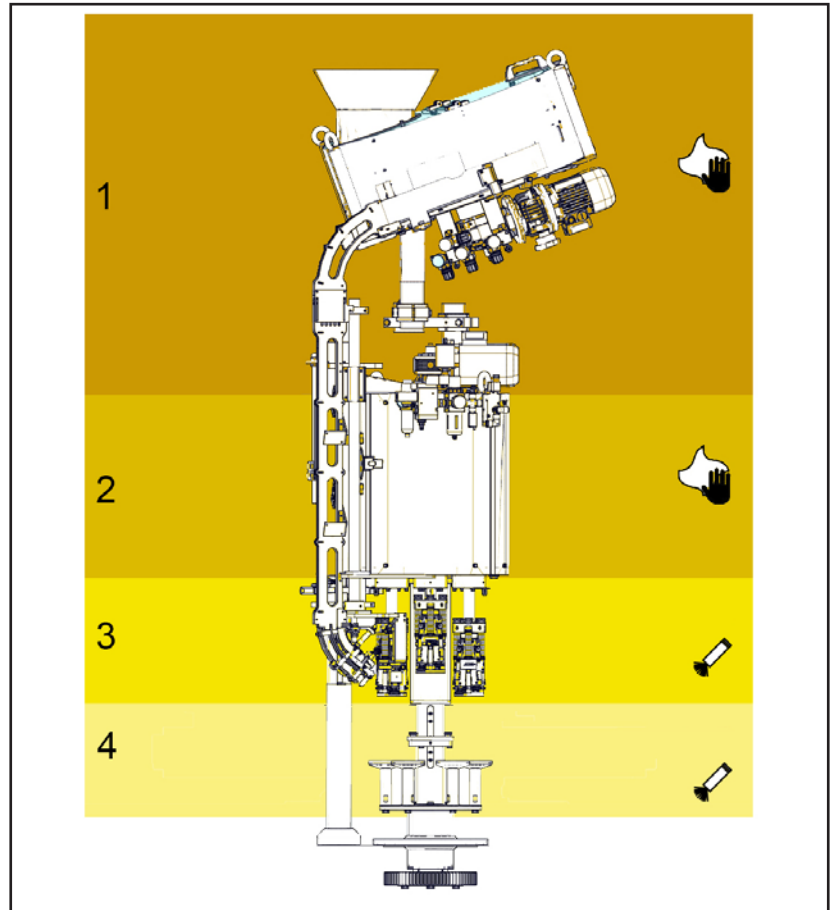
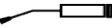


Fig. 56

CLEANING SYMBOL		MAX PRESSURE OF THE CLEANING [bar]	DISTANCE FROM THE SURFACE [mm]	WASHING SYSTEM TILTING [°]
	fixed squirting nozzles or spray balls	≤ 5	≥ 250	
	cloth.	-	-	-
	washing lance	≤ 20	≥ 500	



PRODUCT ZONE 3 - 4	CONCENTRATION	MAX. TEMP. (°C)	pH
Alkaline solutions of hydroxide sodium (NaOH) and/or potassium (KOH)	1 ÷ 2 [%]	35	12 ÷ 13
Acid detergents containing nitric acid (HNO ₃)	1 ÷ 2 [%]	35	1 ÷ 2
Acid detergents containing phosphoric acid (H ₃ PO ₄)	1 ÷ 2 [%]	35	1 ÷ 2
Disinfectant non chlorinated (Aldehyde, quaternary ammonium)	1 ÷ 2 [%]	35	2 ÷ 6

PRODUCT ZONE 1 - 2	MAX. TEMP. (°C)	pH
Water	35	
Neutral detergents	35	5 ÷ 9



Before starting production, the closing machine should be suitably rinsed, to remove all the detergents used, then dried up.

Make sure the the upper flange of the closure carousel is correctly dried.

It is absolutely forbidden to clean the closing machine with:

- steam.



Before carrying out cleaning, you should refer to the product data sheets relating to detergents.



IMPORTANT : AROL S.p.A. supplies instructions for washing in order to clean the parts are remove coarse grease stains, powder and scale. On the other hand, it does not guarantee the hygiene of the surfaces, which must be checked by the user depending on the type of treated products, pollutants and disinfection, sanitisation or sterilisation procedures.



Detergents residuals or residuals of liquids used for the cleaning of the closing machine must be disposed according the laws in force of the country in which it is used.



9.5.2.1 AUTOCLAVE CLOSURE HEADS STERILIZATION

The closure heads thanks to their specific materials, following the instructions listed below, can be subjected to a sterilization process in autoclave.

The frequency of the sterilization in autoclave depends on the specifications and needs of the bottling process and must be determined by the user according to his target and experiences. *AROL S.p.A.* suggests, anyway, that closure heads in bottling lines of sensitive products must be subjected to a sterilization process at least every time one proceeds to their maintenance, manipulation or after a long inactivity period.

AROL S.p.A. suggest that these procedures must be stricter by the increasing the frequency of the sterilization cycles, considering some conditions linked to the context and specific needs or targets that only the costumer can define.

9.5.2.2 CLOSURE HEADS STERILIZATION PROCEDURE

To carry out a sterilization cycle, it is necessary:

- Carry out the ordinary maintenance and closure heads components lubrication.
- Insert the closure heads previously cleaned, into the fitting sacks.
- Close and hermetically seal the sack containing the cleaned closure head.
- Position the closure head into the autoclave.
- Close the autoclave door.
- Carry out a 20 minutes duration sterilization cycle having a pressure of 1 bar and a temperature of 121 °C.
- After the sterilization cycle, remove the closure head from the autoclave.



The closure head must be removed from the sack **ONLY** in a sterile environment.

Remove the envelope bag. Install the closure head on the closing machine.



10 SIZE CHANGE



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.



The closing machine has been designed to handle containers and caps / corks as indicated in the relevant tables enclosed in this manual.



Every container and/or cap must be processed exclusively with the equipment there indicated.

Each series of equipment is marked according to the size to be processed:

- the container guide equipment are marked with coloured inserts.
- The cap guide equipment, in case of multiple sizes, are marked with an identification initials.

In order to run properly and without obstructions the format change operations, it is necessary to raise the closing turntable at its maximum height.



10.1 CONTAINERS BODY GUIDES EQUIPMENTS SIZE CHANGE



Having to change the size of the container should be treated:

- lift the closure carousel so as to allow the removal of bottles body guide equipment.
- Replace the central star-wheel. (See sub-chapter 10.1.1 CENTRAL STAR-WHEEL page 56).



To identify the correspondence between container guide unit and the same container, see the TOOLING COLOUR TABLE attached to this manual.



Fig. 57



10.1.1 CENTRAL STAR-WHEEL



10.1.1.1 REMOVAL

To remove proceed in the following way:

- Make the closing machine rotate in manual jog until a semi star-wheel is totally in the the closure carousel rear part.

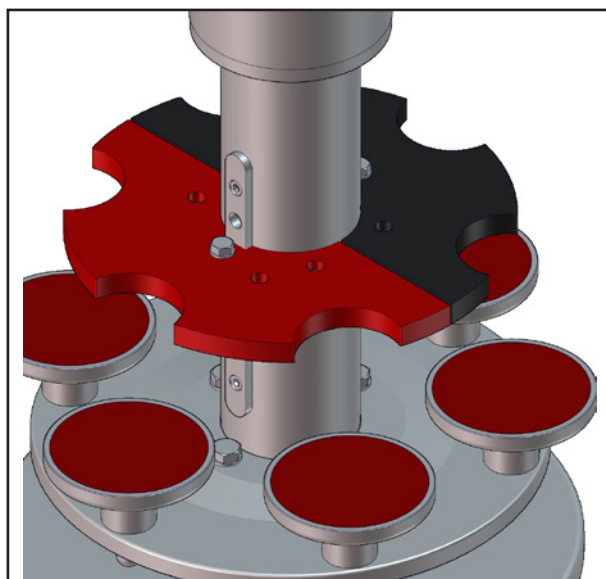


Fig. 58

- Unscrew and remove the fixing bolts.

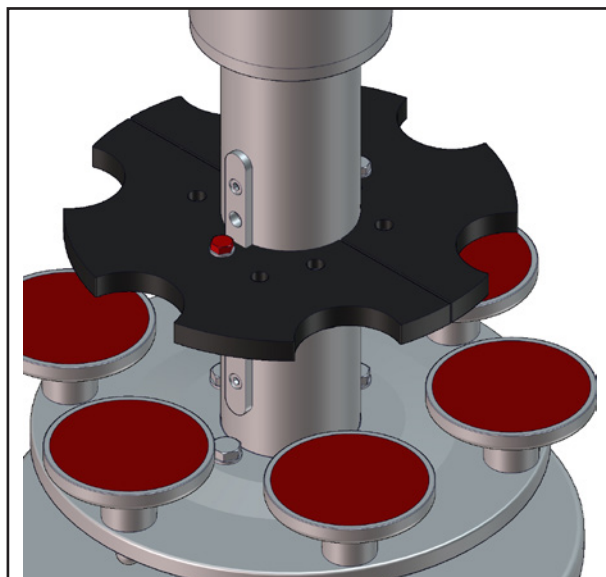


Fig. 59

- Slide the star-wheel out removing it from the centering pins

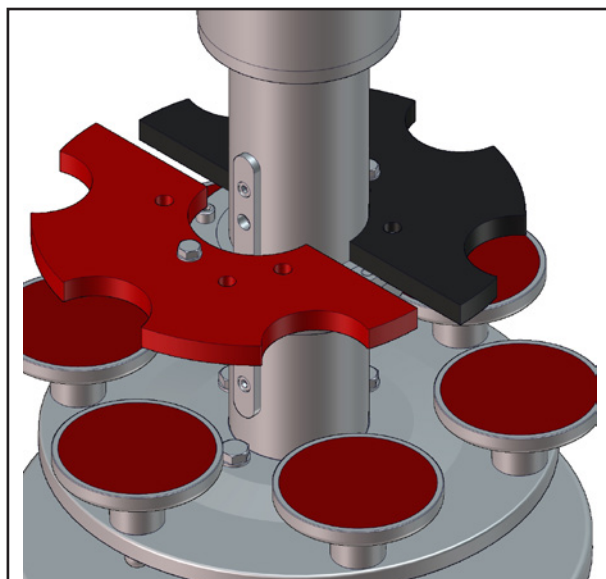


Fig. 60



- Make the closing machine rotate in manual jog until the second semi star-wheel is totally in the the closure carousel rear part
- Unscrew and remove the fixing bolts.
- Slide the star-wheel out removing it from the centering pins

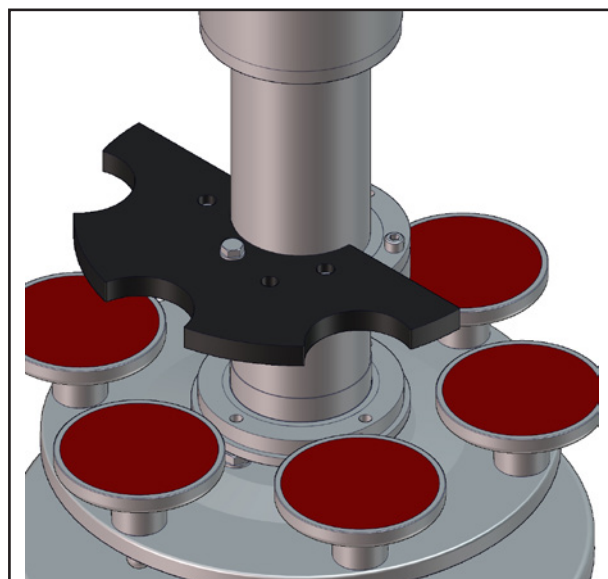


Fig. 61

10.1.1.2 INSTALLATION

For installation proceed as follows:

- Position the half star-wheel on the centering pins.

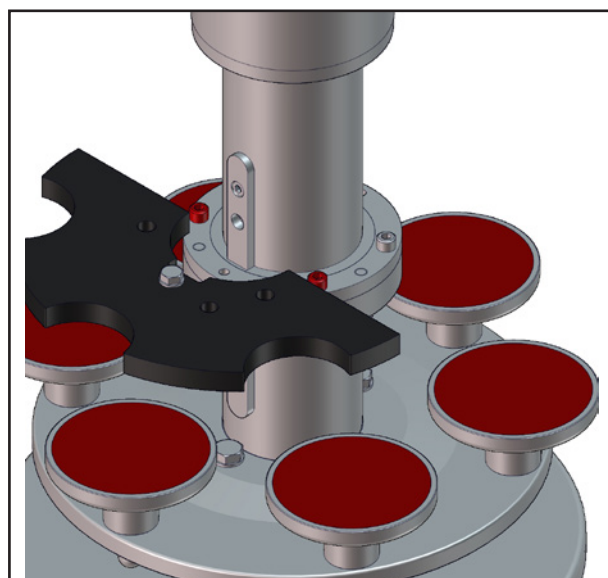


Fig. 62

- Fasten the half star-wheel by fixing bolts.

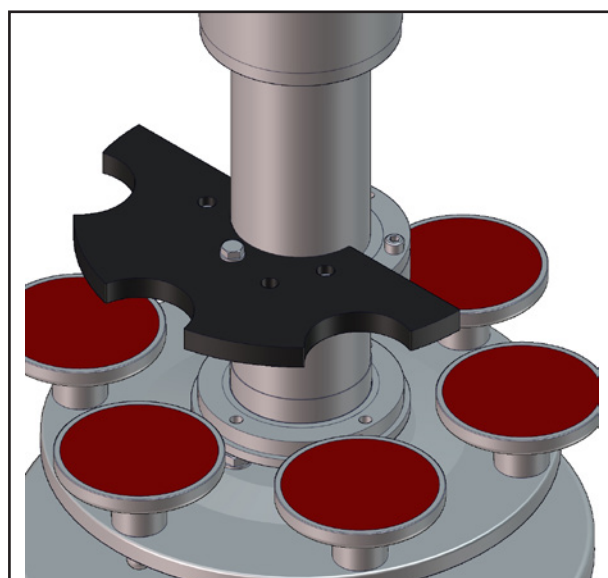


Fig. 63



If it is necessary to adjust the height of the star-wheel, loosen the locking grub screw and adjust the height of the star-wheel holder ring. When the adjustment is completed, lock the locking grub screw

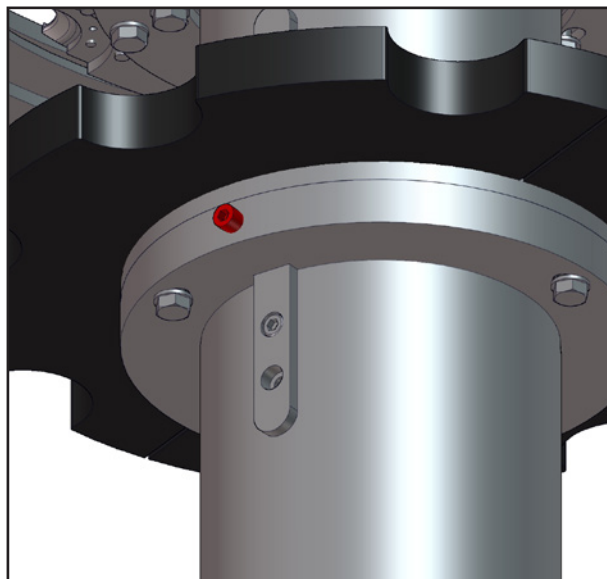


Fig. 64

10.2 CAP SIZE CHANGE



Having to change the format of the caps to be treated, you must:

- replace the closure heads. (See sub-chapter 10.2.1 REMOVAL AND INSTALLATION OF THE CLOSURE HEADS page 60).
- Replace the caps distribution group, if provided. (See sub-chapter 10.2.2 "PICK OFF" CAPS CHUTE page 64).
- Replace the caps selection units, if provided. (See sub-chapter 10.2.3 CAPS SELECTION EQUIPMENT FOR CENTRIFUGAL CAPS SORTER page 69).



Fig. 65

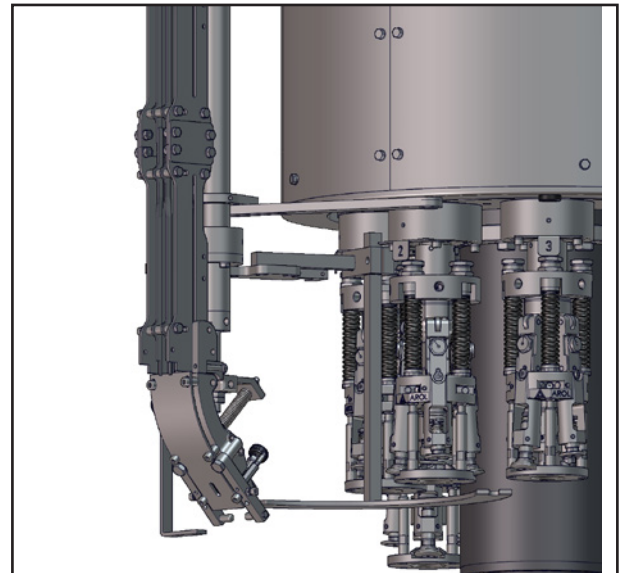


Fig. 66

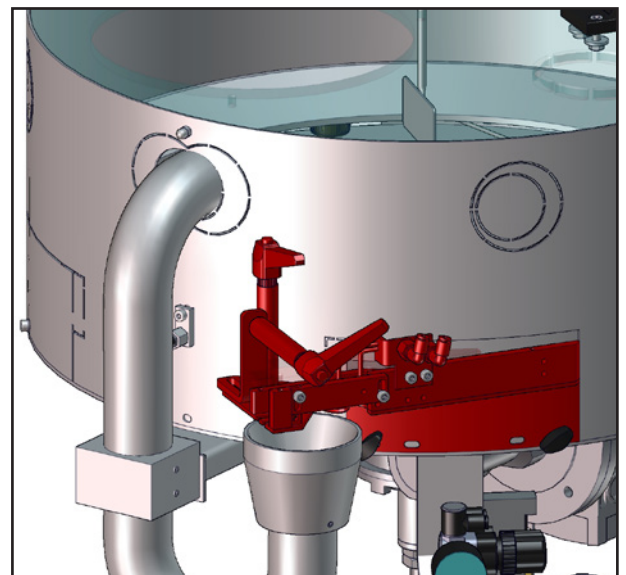


Fig. 67



10.2.1 REMOVAL AND INSTALLATION OF THE CLOSURE HEADS

The table below shows the dedicated tools for disassembly and assembly of the closure head.

TOOLS	
	Head mounting wrench
	Removing threaded ring wrench
	Installation threaded ring wrench

10.2.1.1 REMOVAL OF SSD HEAD

To remove the closure head from the spindle, proceed as follows:

- Insert the removing threaded ring wrench in the head fixing threaded ring seats.
- Unscrew the head fastening threaded ring with the removing threaded ring wrench till releasing the closure head.



Fig. 68



- Insert the head mounting wrench in the seats of the closure head.
- Untighten the closure head with the head mounting wrench and remove it.



Fig. 69

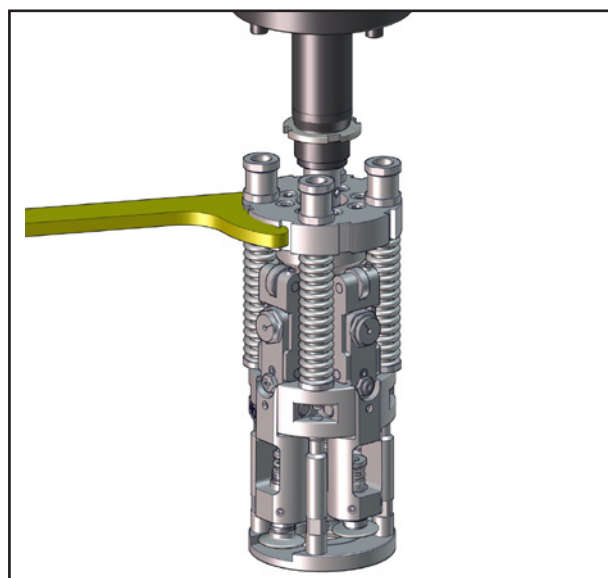


Fig. 70



10.2.1.2 INSTALLATION OF SSD HEAD

To install the closure head on the spindle, proceed as follows:

- Insert the closure head on the spindle.
- Screw the head on the spindle, manually as far as possible.



Fig. 71



Fig. 72



- Insert the head mounting wrench in the seats of the closure head.
- With the head mounting wrench complete the screw to the bottom of the thread.

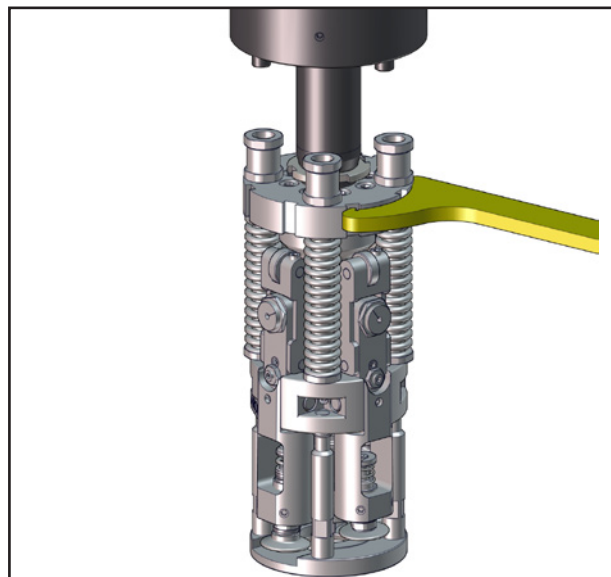


Fig. 73

- Insert the installation threaded ring wrench in the head fixing threaded ring seats.
- Tighten the head fixing threaded ring with the threaded ring installation wrench.



Fig. 74



10.2.2 "PICK OFF" CAPS CHUTE

10.2.2.1 REMOVAL OF THE CAPS CHUTE WITH "PICK-OFF" CAP DISTRIBUTION

Remove the caps chute in the following way:

- Disconnect the electrical connection from the caps presence photocell (one or more).
- Loosen the threaded fasteners located on the support chute.



Fig. 75

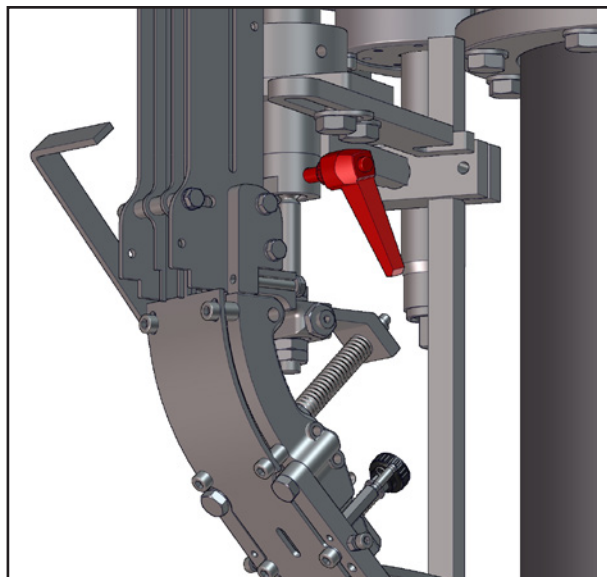


Fig. 76

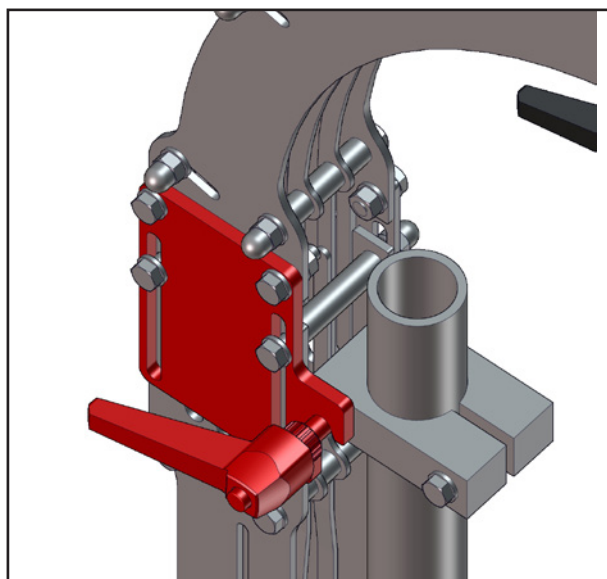


Fig. 77



- Loosen the threaded fastening placed on the caps sorter

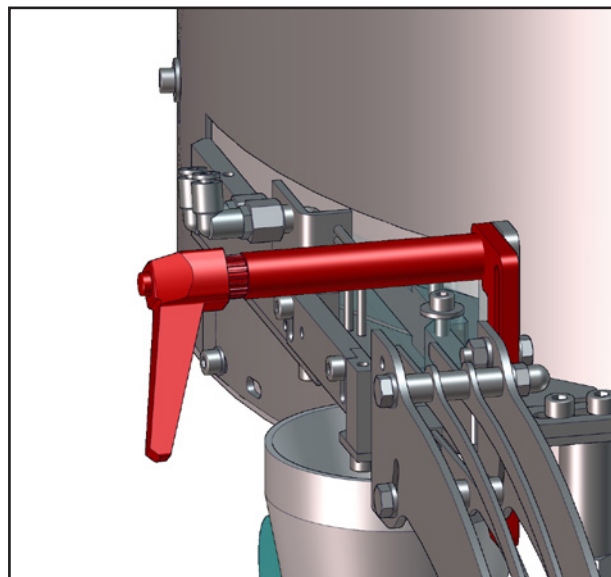


Fig. 78

- Lift the complete caps chute, so as to extract it from fixing systems.
- Successively lower the caps chute so that the head height adjustment pin come out from the hole of the chute support, then remove the chute.

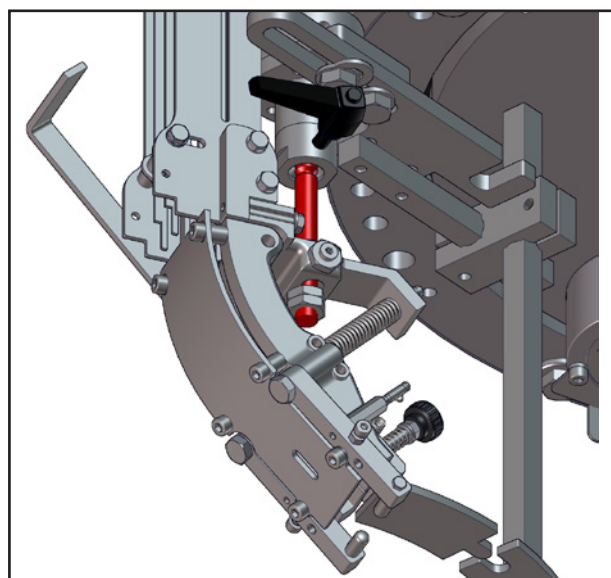


Fig. 79



10.2.2.2 INSTALLATION OF THE CAPS CHUTE WITH “PICK-OFF” CAP DISTRIBUTION

To install the caps chute on the machine it is necessary:

- Loosen the threaded fasteners located on the support chute.

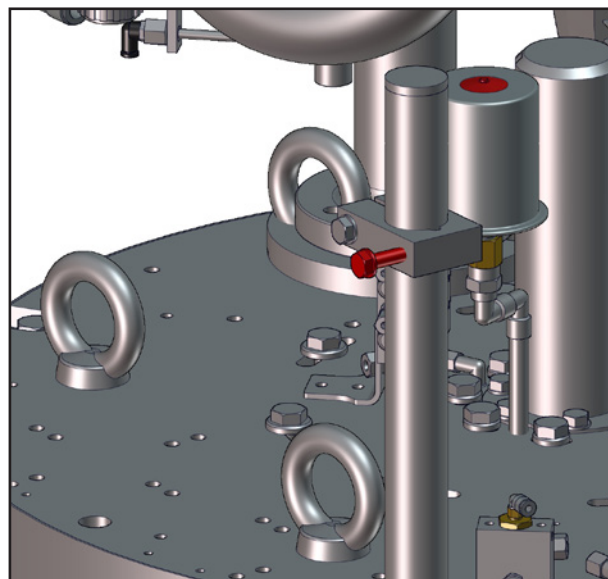


Fig. 80

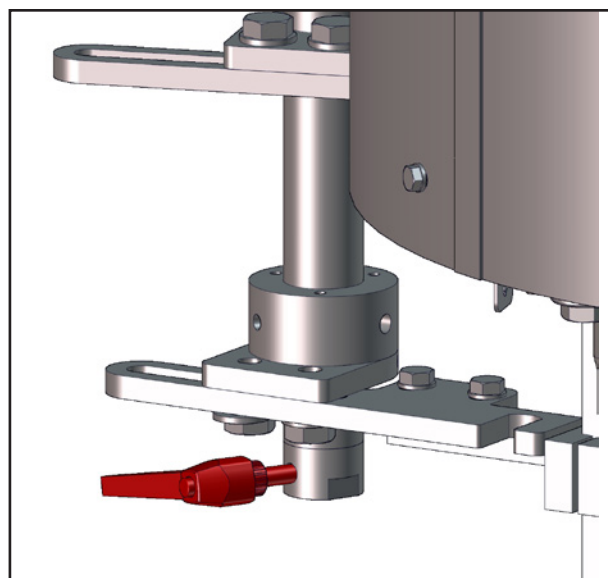


Fig. 81

- Insert the head height adjustment pin into the hole of the chute support and lift the complete caps chute.

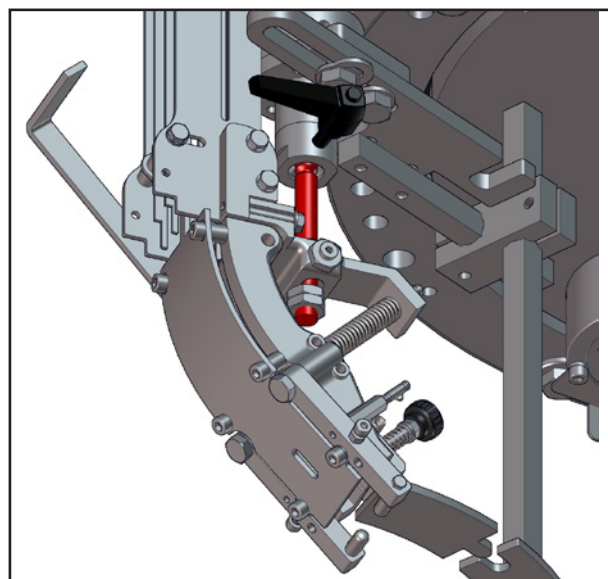


Fig. 82



- Place the caps chute so that the fixing brackets are positioned on threaded fasteners.
- Lock threaded fasteners located on the support chute.

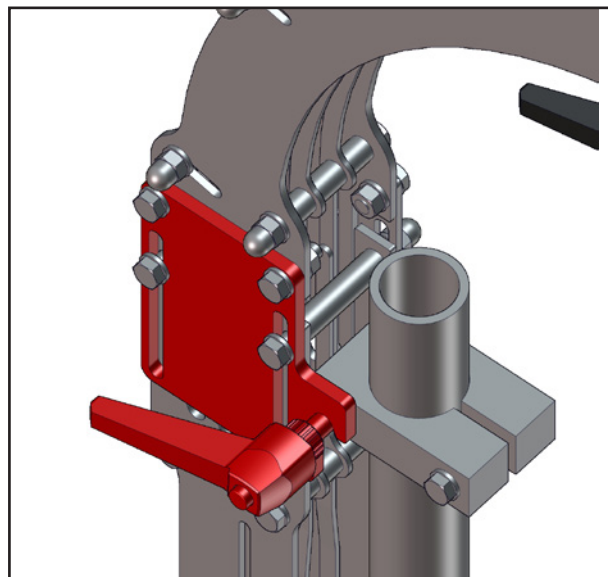


Fig. 83

- Lock the threaded fixing system of the caps distribution head.

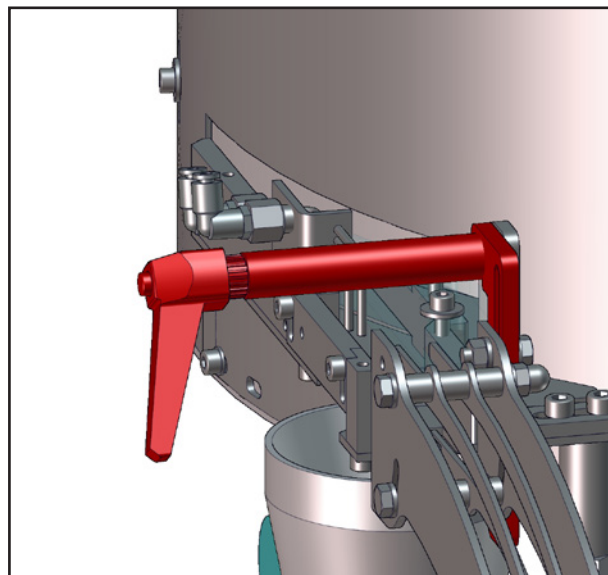


Fig. 84

- Lock the threaded fixing system of the caps distribution head.

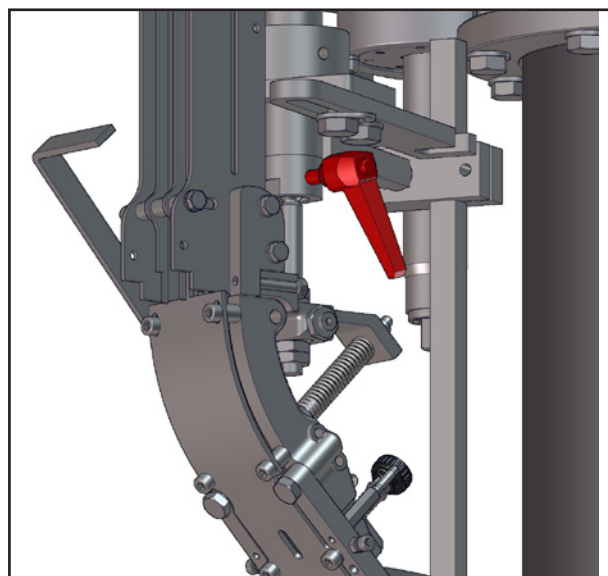


Fig. 85



- Connect the electrical connection of the caps presence photocell (one or more).



Fig. 86

- If necessary, adjust the height of the caps upper guide:
 - loosen the screw of the clamp.
 - Align the caps upper guide with the guide of the caps distribution head.
 - Lock the screw of the clamp.



When necessary, the height relative to the size of cap to be treated, is engraved on the rod of the guide itself. The engrave should be aligned with the upper side of the clamp.

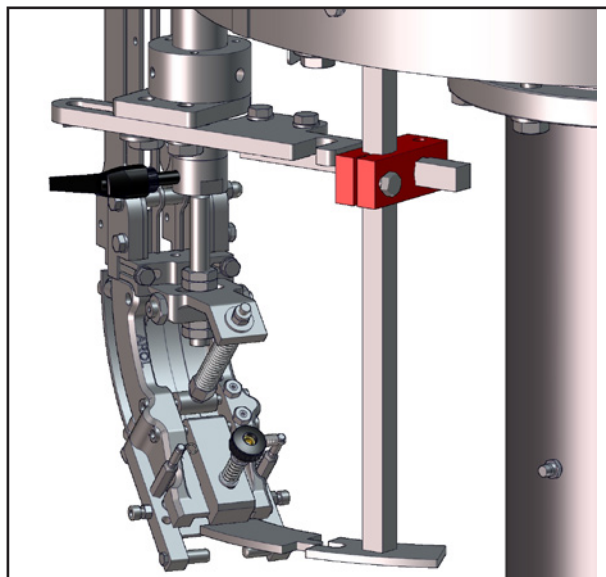


Fig. 87



10.2.3 CAPS SELECTION EQUIPMENT FOR CENTRIFUGAL CAPS SORTER



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

10.2.3.1 CAPS EXIT MOUTH



10.2.3.1.1 REMOVAL

To remove proceed as follows:

- disconnect the pneumatic connections from the caps exit-mouth.

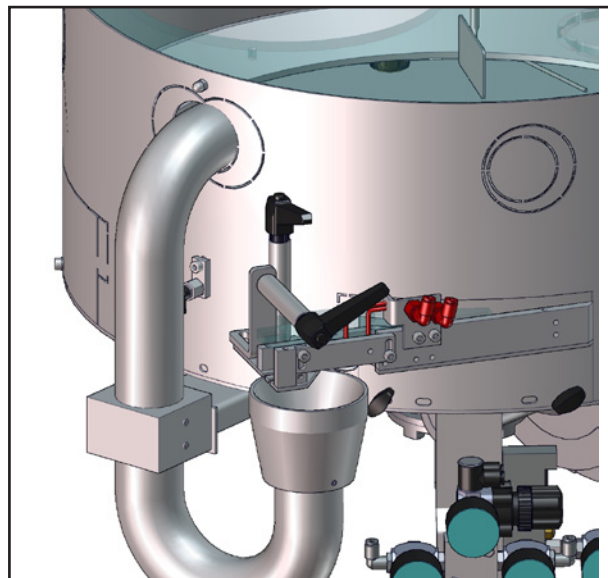


Fig. 88

- Unscrew and remove the fixing systems of the caps exit mouth.

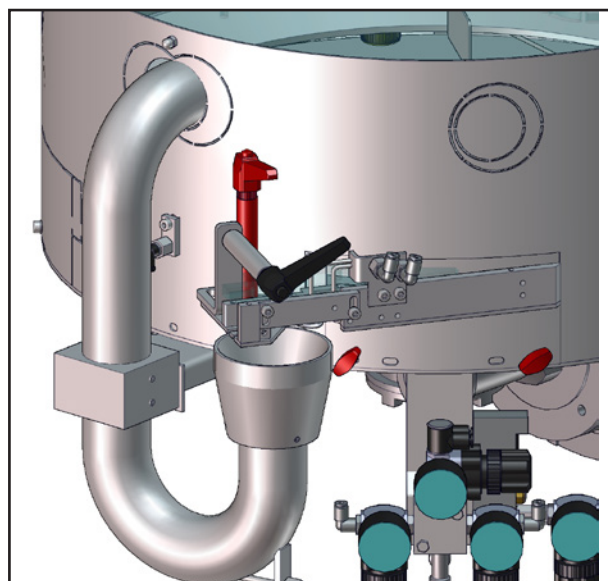


Fig. 89



- Remove the whole caps exit-mouth.

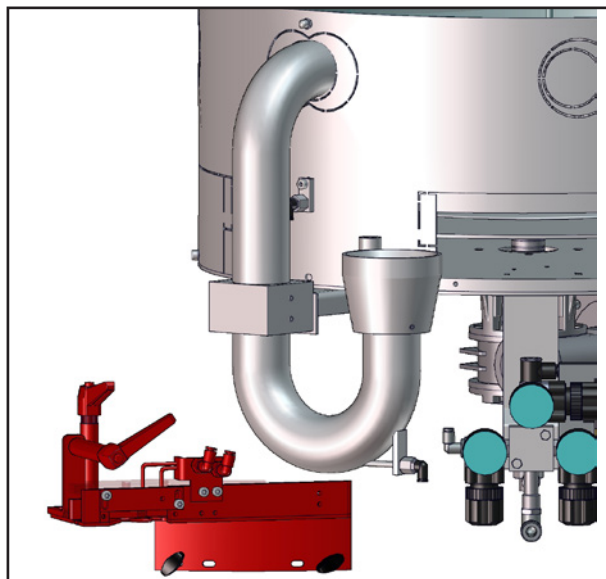


Fig. 90

10.2.3.1.2 INSTALLATION

To install proceed as follows:

- Position the exit mouth and fasten it into the holes by the suitable fastening systems

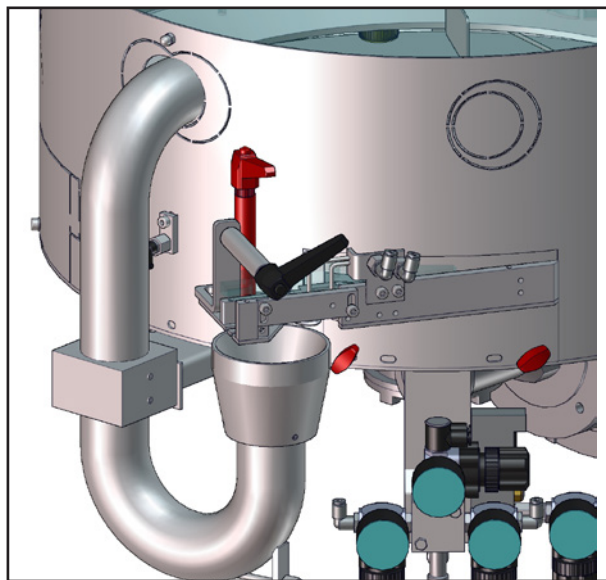


Fig. 91

- Connect the pneumatic connections to the caps exit-mouth.

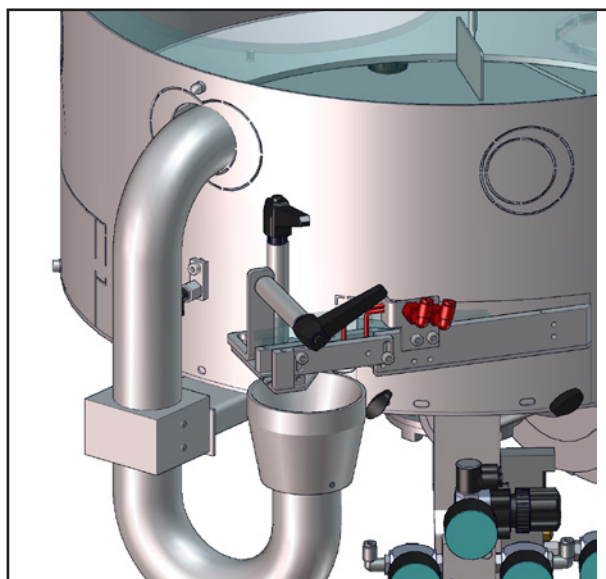


Fig. 92

10.2.3.2 ADJUSTMENT OF THE CAPS SORTER FOR SIZE CHANGE



- If varies the height of the cap to be treated must be adjust the height of the inner hopper ring in the following way:
 - loosen the screws located on the outer perimeter of the hopper.
- Adjust the height of the hopper ring in such a way between the new cap to be treated and the ring there is a clearance of about 3 mm.
- Tighten the screws previously loosened.



Fig. 93

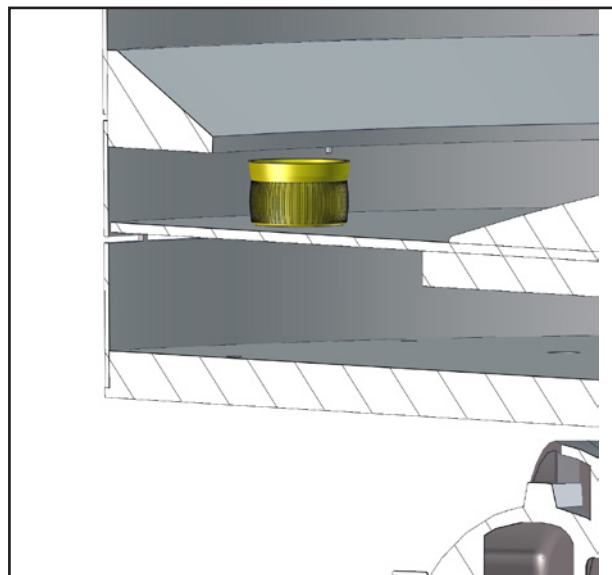


Fig. 94

- On the caps sorter there is a door that allows, thanks to the centrifugal force, the ejection of caps without seaming and seaming on them.
- If varies the height of the cap to be treated must adjust the height of the adjustable door in the following way:
 - loosen the fixing knobs.

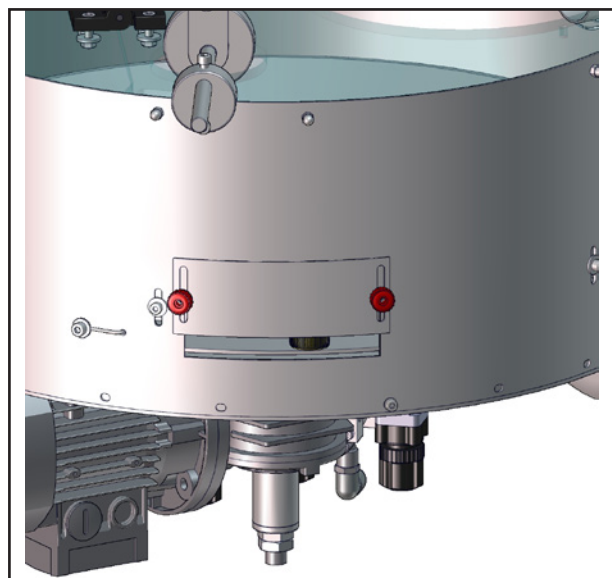


Fig. 95



- Adjust the height of the adjustable door so that they can be expelled only the caps without seal.
- After adjustment, tighten the fixing knobs.

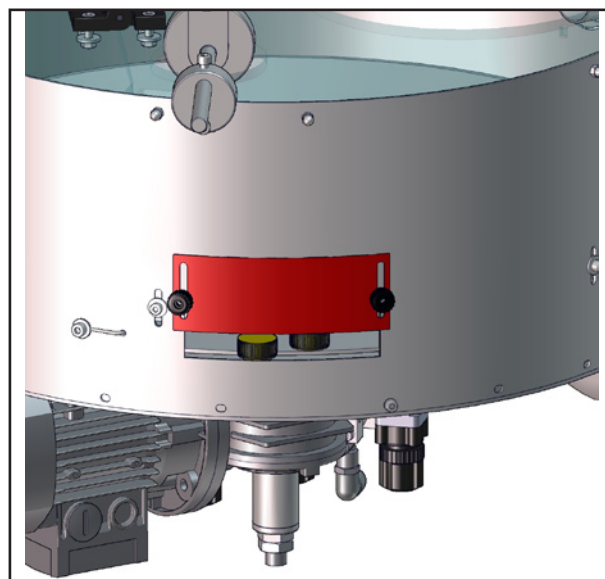


Fig. 96

If the cap is not correctly shifted by the nozzles it is sufficient to increase or decrease the air pressure by the fitting pressure reducers.



Fig. 97



10.3 ADJUSTMENTS OF THE CLOSURE HEAD

10.3.1 "SSD" CLOSURE HEAD

The table below shows the necessary tools for adjustment of the closure head.

TOOLS	
	Springs adjustment wrench
	Rollers height adjustment wrench
	Adjustment pad

10.3.1.1 CLOSURE HEAD PREPARATION

Proceed as follows:

- Insert the the closure head supporting equipment into a jaw.

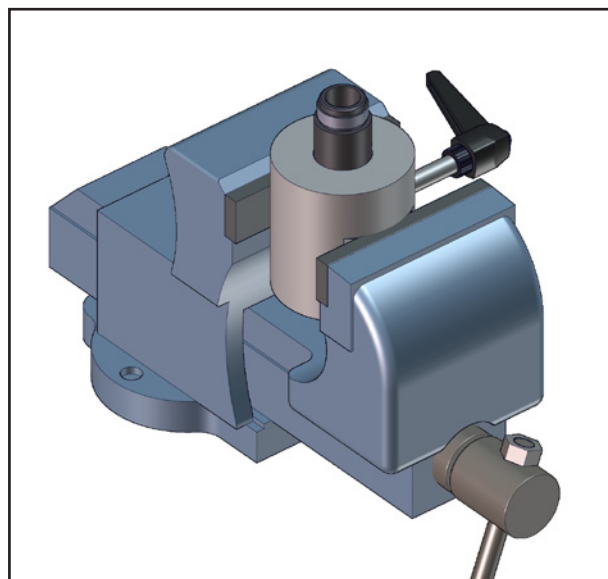


Fig. 98



Before working, measure and make a note of the distance between the adjustment bush and the head holder flange.

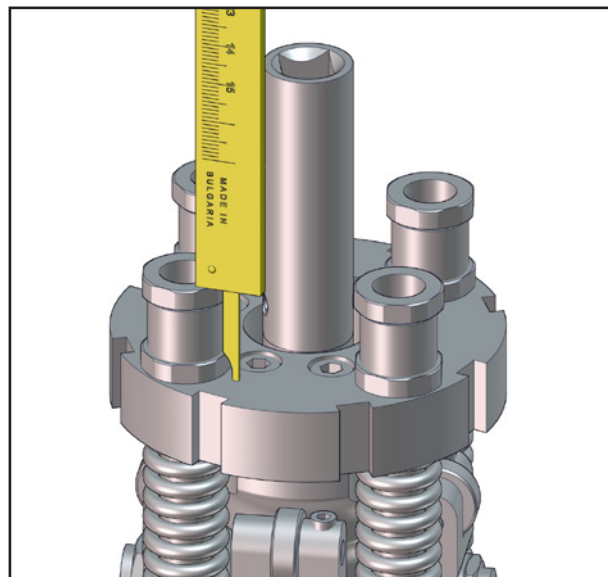


Fig. 99

- Loosen the bush fixing nuts with the spring adjustment hexagonal wrench.

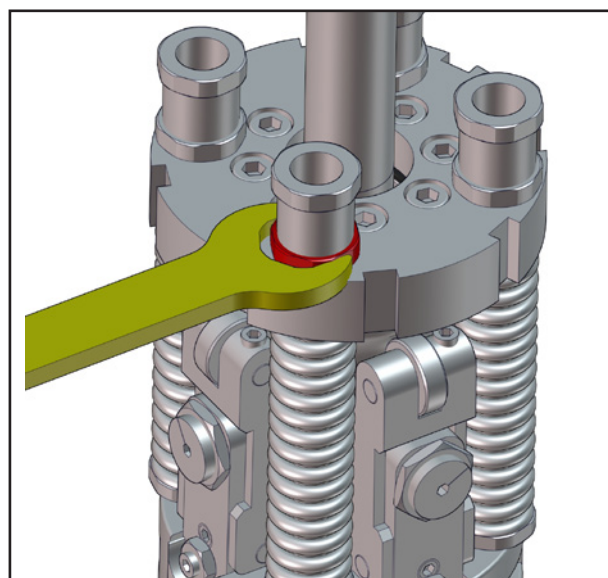


Fig. 100



- Completely unscrew the adjustment bushes with the spring adjustment hexagonal wrench.

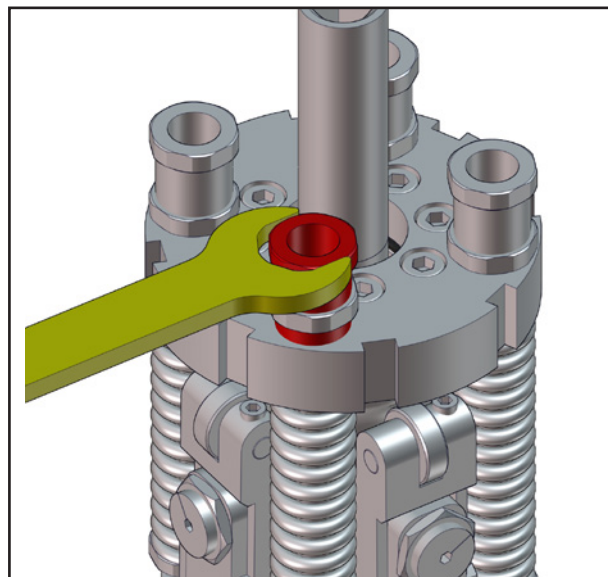


Fig. 101

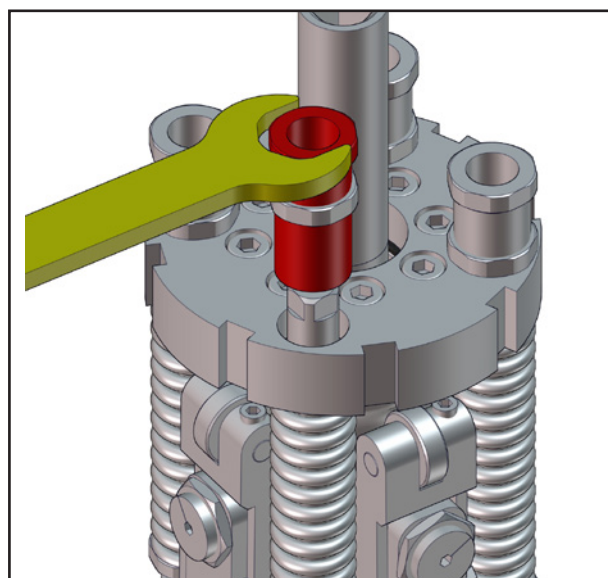


Fig. 102



- Hold the spring guide tie-rods with a 10 mm hexagonal wrench and loosen the tie-rod fastening nuts with a springs adjustment wrench.

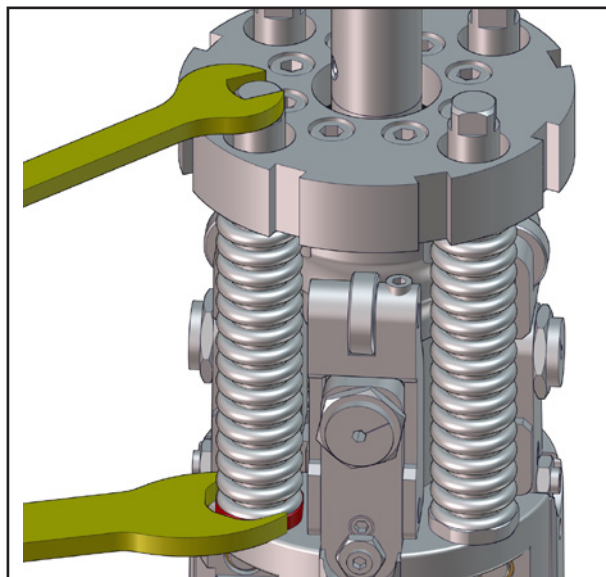


Fig. 103

- Completely unscrew the spring guide tie-rods with a 10 mm hexagonal wrench.

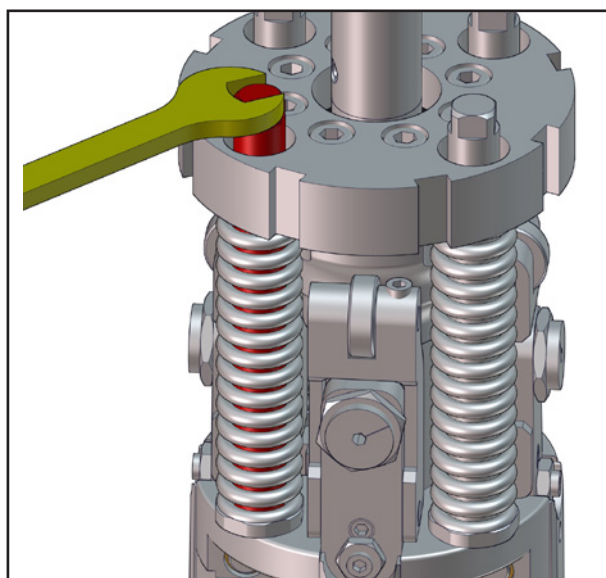


Fig. 104

- Slide out the spring guide tie-rods and tie-rod locking nuts.

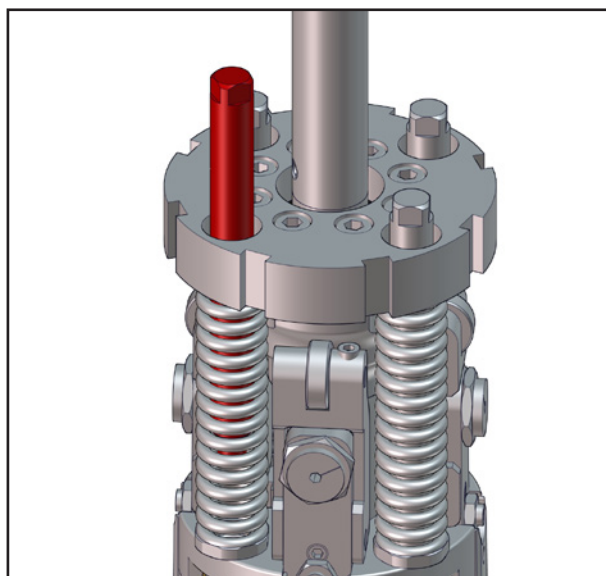


Fig. 105



- Remove the compensating springs, tie rod locking nuts and springs supporting washers.

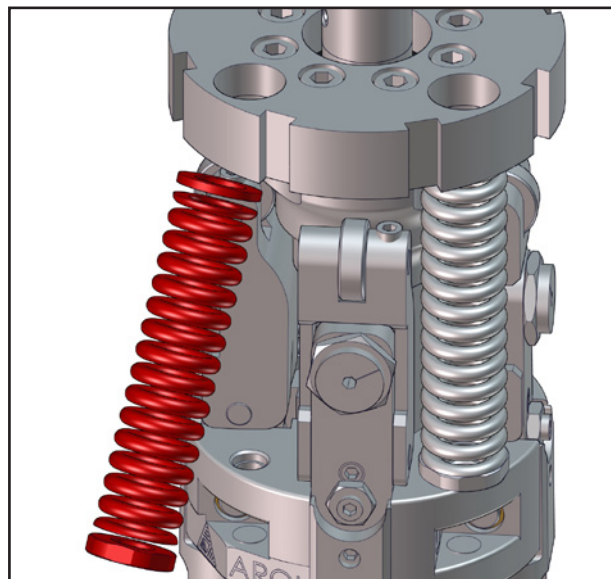


Fig. 106

- Screw the closure head on the sleeve of the supporting equipment.

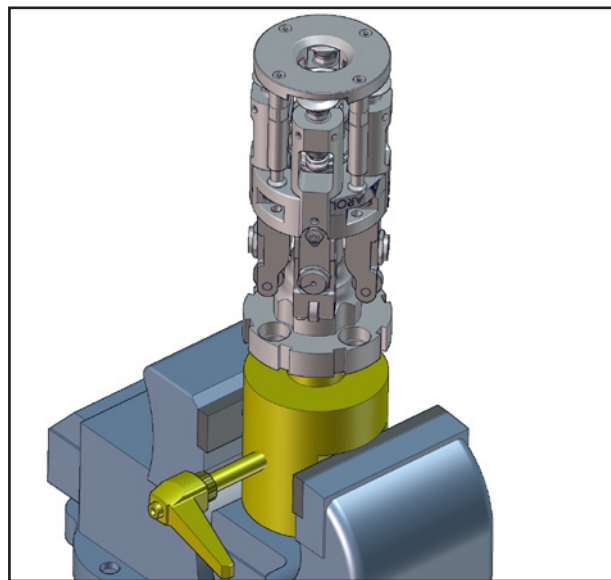


Fig. 107



- Unscrew the centering device fixing screws with a 3 mm Allen wrench.

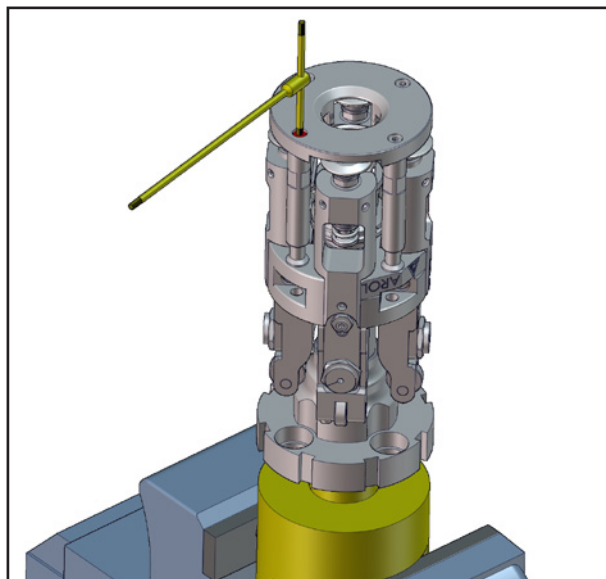


Fig. 108

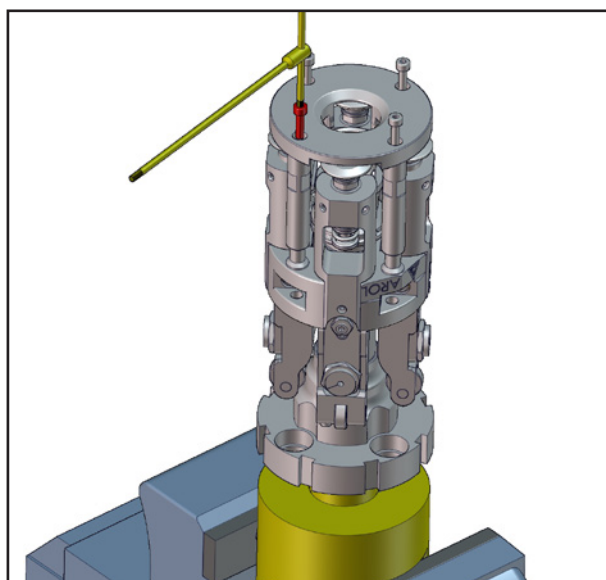


Fig. 109

- Remove the centering device.

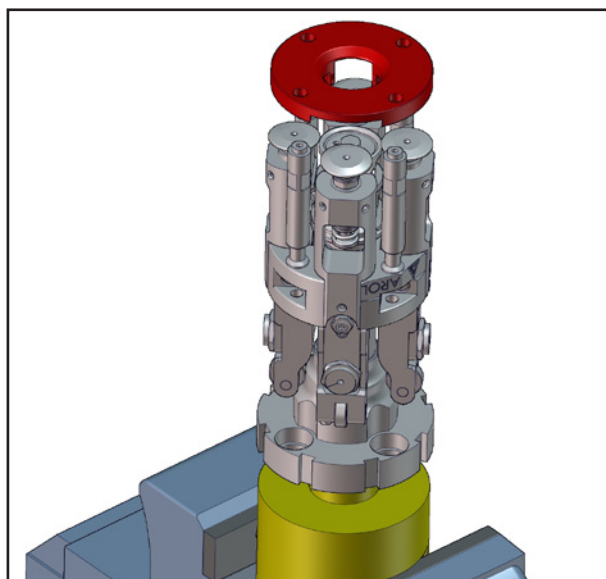


Fig. 110



- Unscrew the centering device's support columns with a 10 mm exagonal wrench, and remove the centring device's support columns.

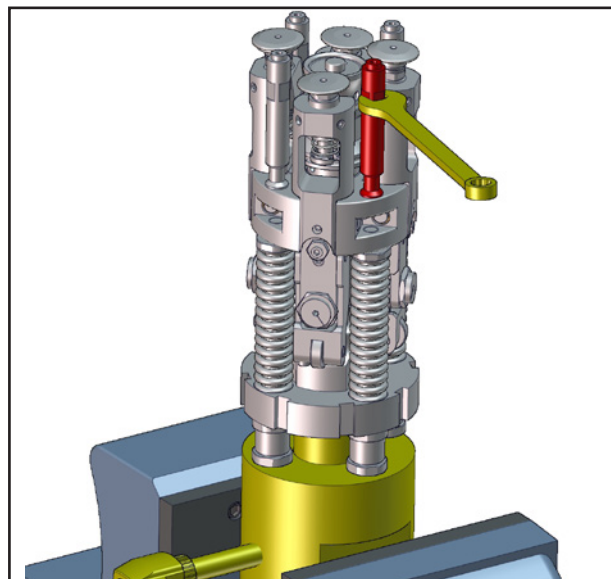


Fig. 111



10.3.1.2 ADJUSTING THE THREADING ROLLER

For the adjustment of the threading roller, proceed in the following way:

- Insert the specific adjuster pad, indicated in the "CLOSURE HEAD CALIBRATION TABLE" enclosed in this manual, on the bottle mouth-presser.

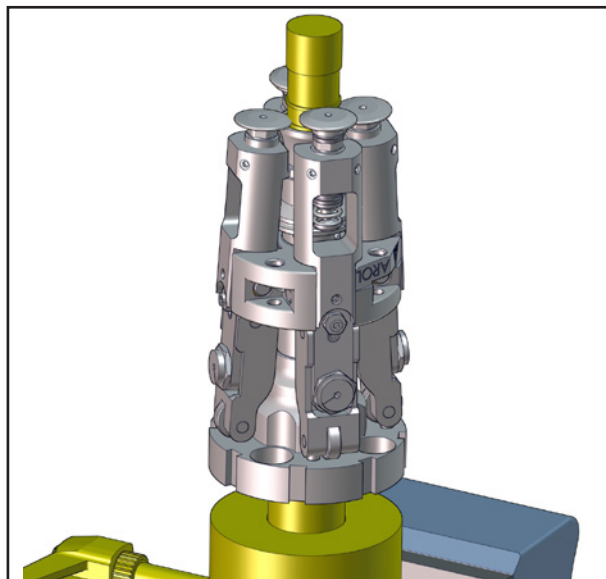


Fig. 112

- Loosen the fixing grub screws with a 3 mm Allen wrench, in order to free the threading roller bush.

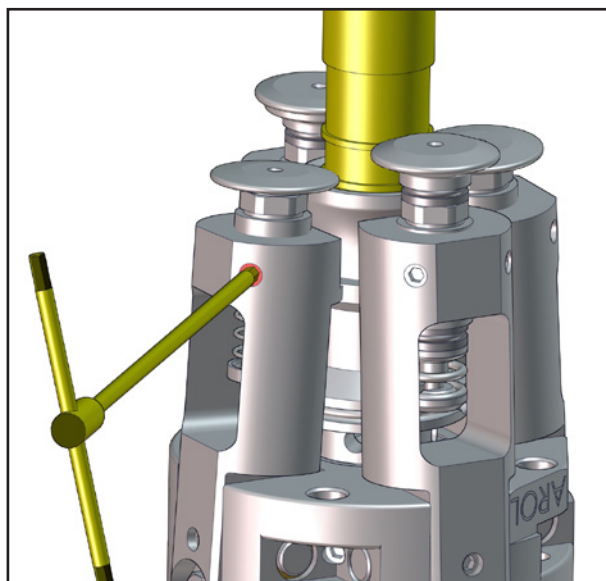


Fig. 113



- Screw or unscrew the threading roller bush with the rollers height adjustment wrench, in order to bring the threading roller to the height of the adjuster pad surface.
- For a further delicate adjustment, when provided, it is necessary to screw or unscrew the threading roller bush of $n^{\circ}/6$ of angle as described in "CLOSURE HEAD CALIBRATION TABLE" enclosed in this manual.

EXAMPLE:

Towards pressure block	Towards centering device
1/6	3/6

- Towards pressure block: screw the threading roller bush in such way to move the roller towards the mouth-presser.
- Towards centering device: unscrew the threading roller bush in such way to move the roller towards the centering device.
- Each 1/6 equals to 60° of rotation. To make easy the operation use as reference of rotation the 6 faces of the nut into the threading roller bush.



f in the table the movement towards the mouth-presser and centering device are not indicated, it means that the adjustment must be carried out with the adjustmend pad.

- After adjustment, secure the fixing grub screws with a 3 mm Allen wrench.
- Insert a 3 mm Allen screw wrench in the seat of the fixing grub screw.

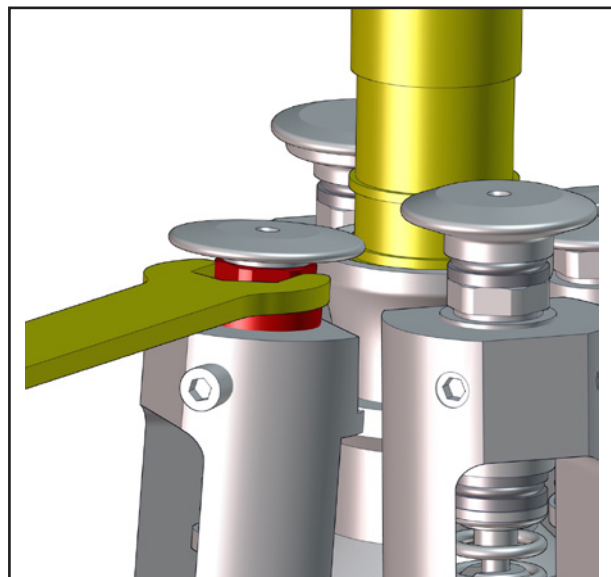


Fig. 114

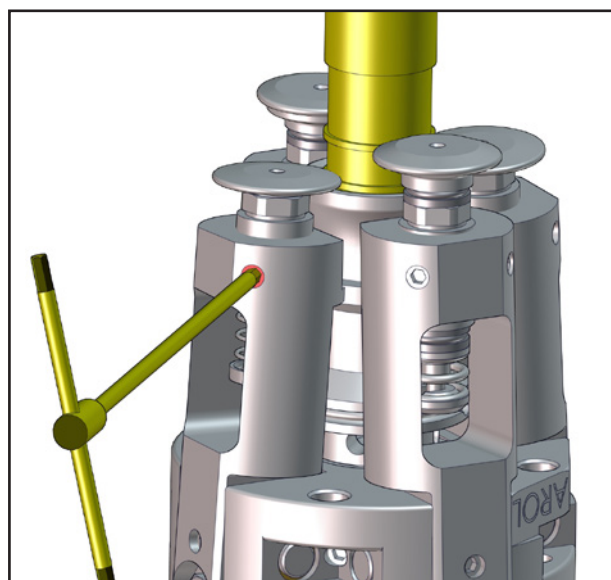


Fig. 115

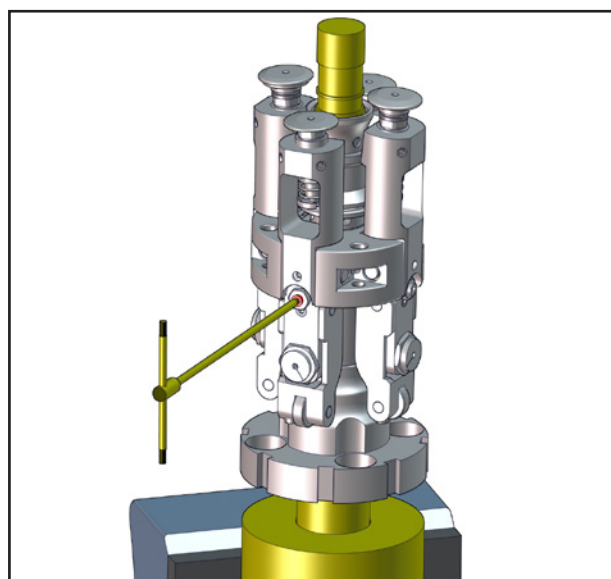


Fig. 116



- Insert the 10 mm hexagonal wrench on the locking nut.
- Holding a 3 mm Allen wrench, loosen the locking nut with a 10 mm hexagonal wrench.

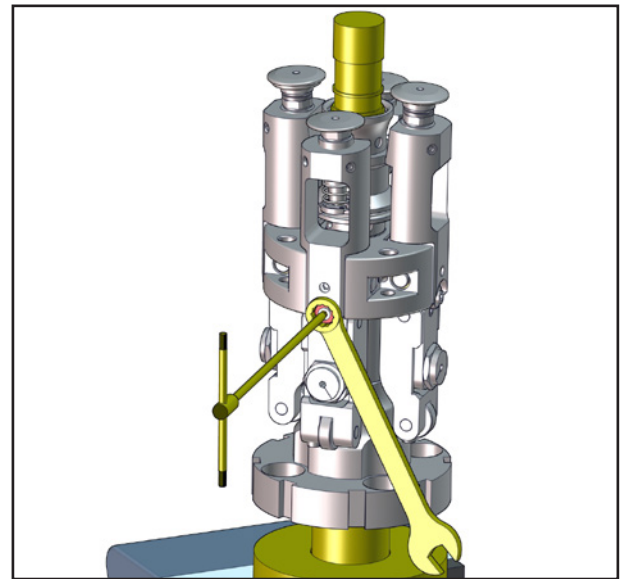


Fig. 117

- Move the roller support arms in a closed position by pushing the head downward.

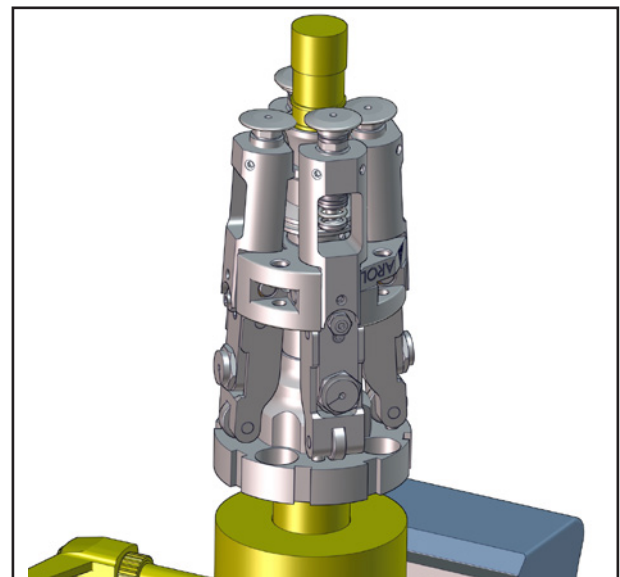


Fig. 118

- Screw or unscrew the locking grub screw with a 3 mm Allen wrench, until the threading roller touches the pad diameter.
- Repeat the above steps also for all the threading rollers.

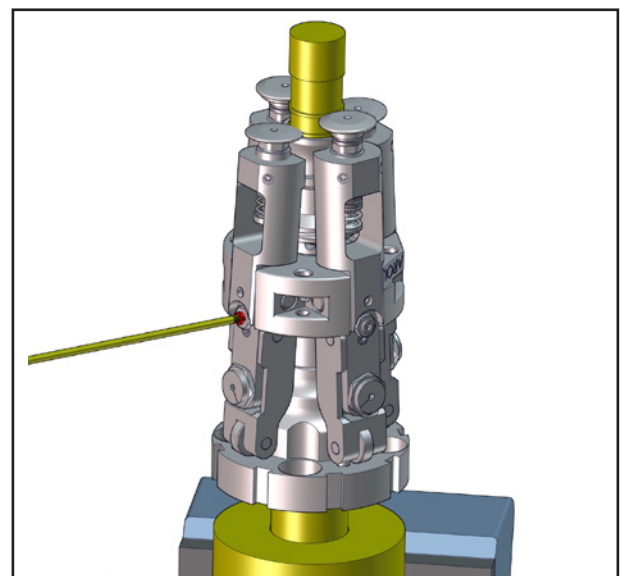


Fig. 119

- Make the pad rotate on the mouth-presser, during the pad rotation the threading rollers must turn freely too.

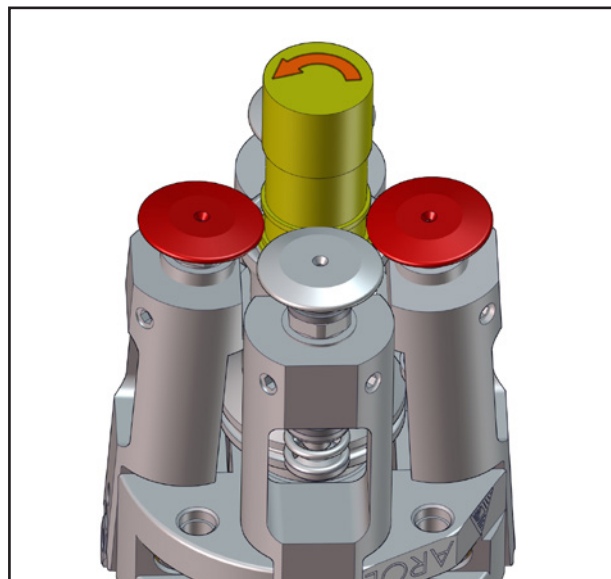


Fig. 120

- Fasten the blocking nut with a 10 mm exagonal wrench, making sure that the grub screw does not move.

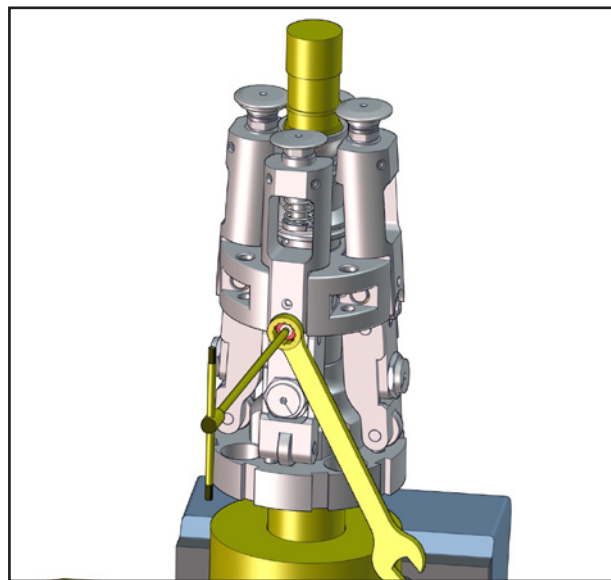


Fig. 121

- Repeat the same operation for the remaining threading arms.
- Remove the pad used for the adjustment.
- Free the arms driving cam putting it in rest position (the rollers holder arms widen).

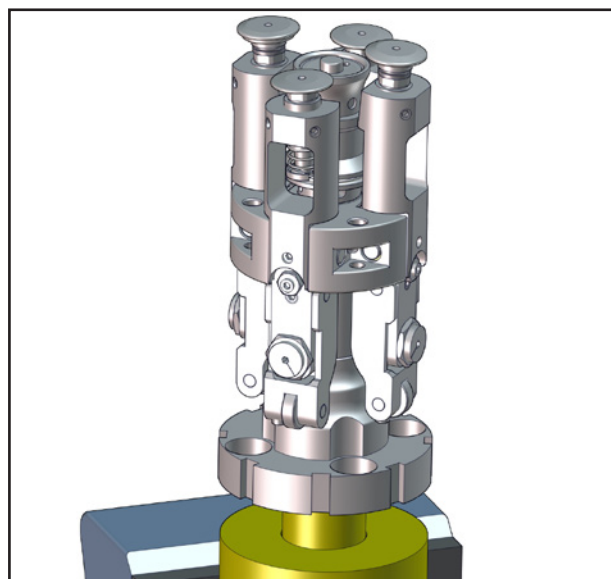


Fig. 122



- Insert a 3 mm Allen screw wrench into the seat of the spring holder bush

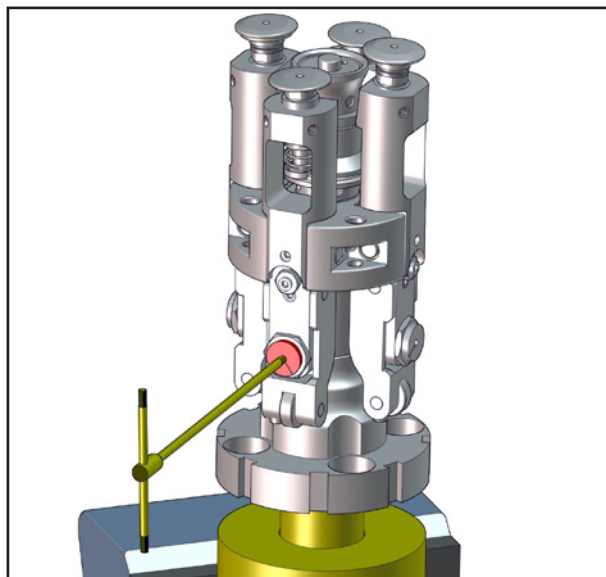


Fig. 123

- Insert the 19 mm hexagonal wrench on the locking nut.
- Holding a 3 mm Allen screw, loosen the locking nut with a 19 mm hexagonal screw.

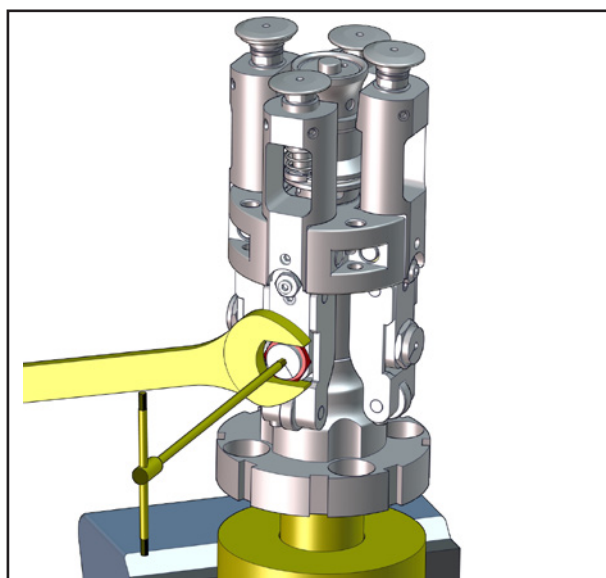


Fig. 124

- Using a dynamometer measure the side pressure strenght of the threading rollers. The measurement must takes place when threading arm moves.

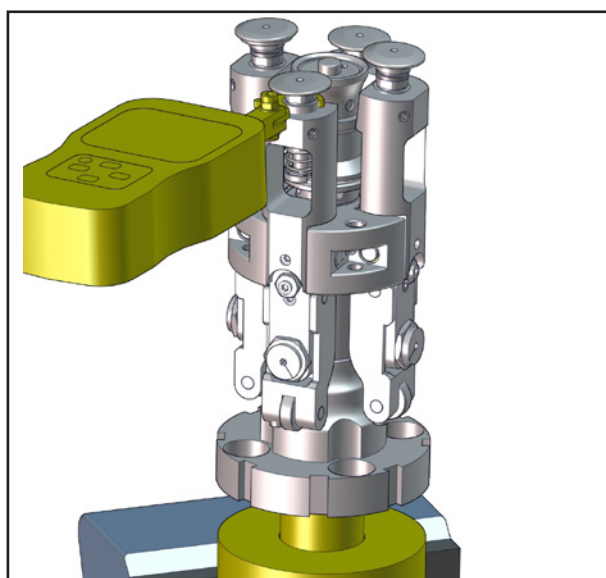


Fig. 125



If the carried out measurement is different from the one indicated in "CLOSURE HEAD CALIBRATION TABLE" enclosed in this manual, it is necessary to carry out as follows:

- Screw the spring holder bush with a 3 mm Allen wrench to increase the side pressure.
- Unscrew the spring holder bush with a 3 mm Allen wrench to decrease the side pressure.

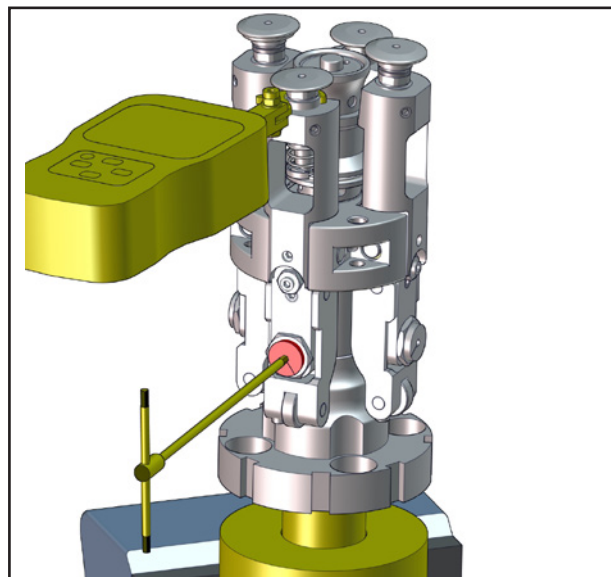


Fig. 126

- Keeping a 3 mm Allen wrench still, screw the locking nut with the 19 mm hexagonal wrench.
- Repeat the same operation for the remaining threading arms.

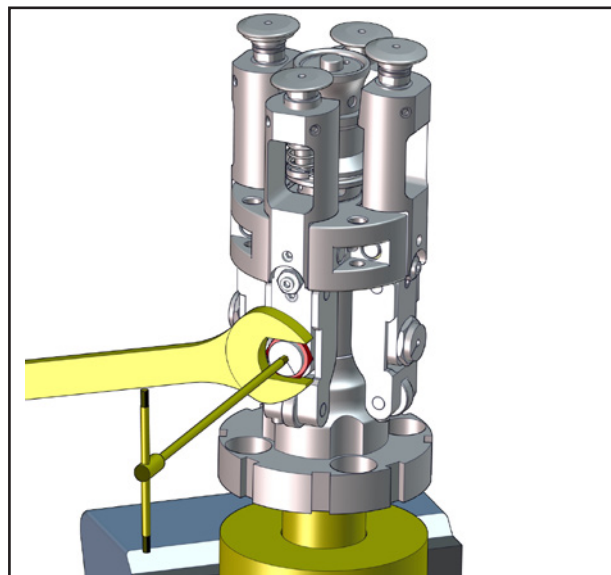


Fig. 127



10.3.1.3 ADJUSTING THE SKIRTING ROLLER

For the adjustment of the skirting roller, proceed in the following way:

- Insert the specific adjuster pad, indicated in the "CLOSURE HEAD CALIBRATION TABLE" enclosed in this manual, on the bottle mouth-presser.

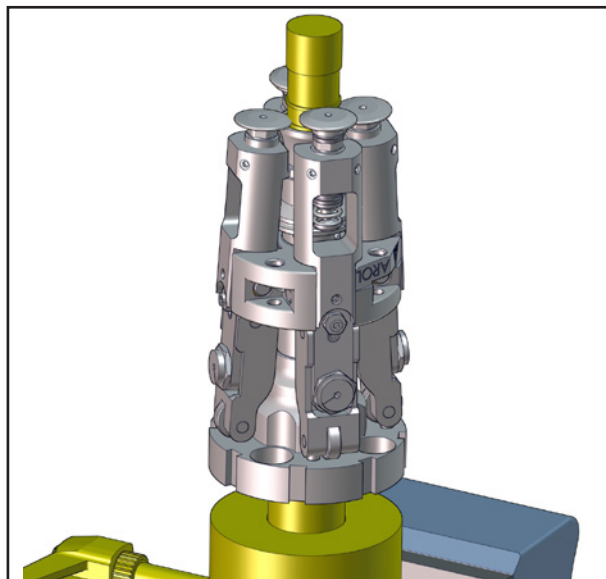


Fig. 128

- Loosen the fixing grub screws with a 3 mm Allen wrench, in order to free the skirting roller bush.

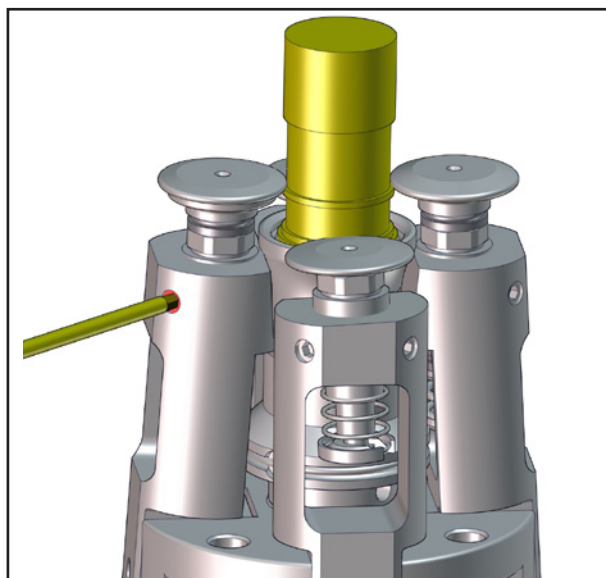


Fig. 129



- Screw or unscrew the skirting roller bush with the rollers height adjustment wrench, in order to bring the skirting roller to the height of the adjuster pad surface.
- For a further delicate adjustment, when provided, it is necessary to screw or unscrew the skirting roller bush of $n^{\circ}/6$ of angle as described in "CLOSURE HEAD CALIBRATION TABLE" enclosed in this manual.

EXAMPLE:

Towards pressure block	Towards centering device
1/6	3/6

- Towards pressure block: screw the skirting roller bush in such way to move the roller towards the mouth-presser.
- Towards centering device: unscrew the skirting roller bush in such way to move the roller towards the centering device.
- Each 1/6 equals to 60° of rotation. To make easy the operation use as reference of rotation the 6 faces of the nut into the skirting roller bush.



f in the table the movement towards the mouth-presser and centering device are not indicated, it means that the adjustment must be carried out with the adjustmend pad.

- After adjustment, secure the fixing grub screws with a 3 mm Allen wrench.
- Insert a 3 mm Allen screw wrench in the seat of the fixing grub screw.

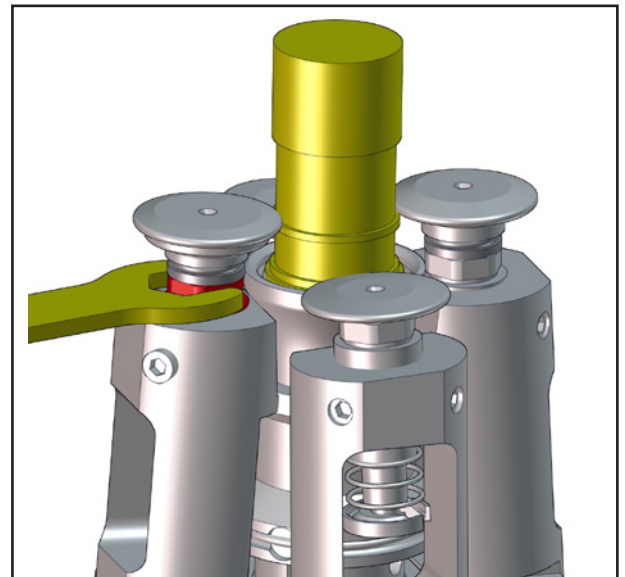


Fig. 130

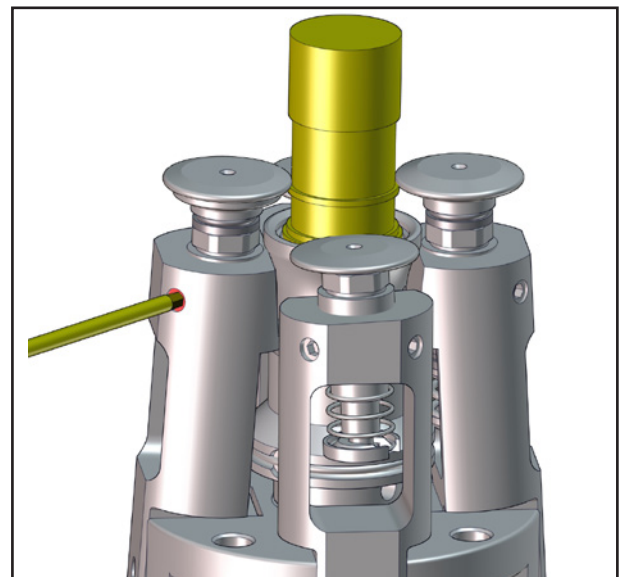


Fig. 131

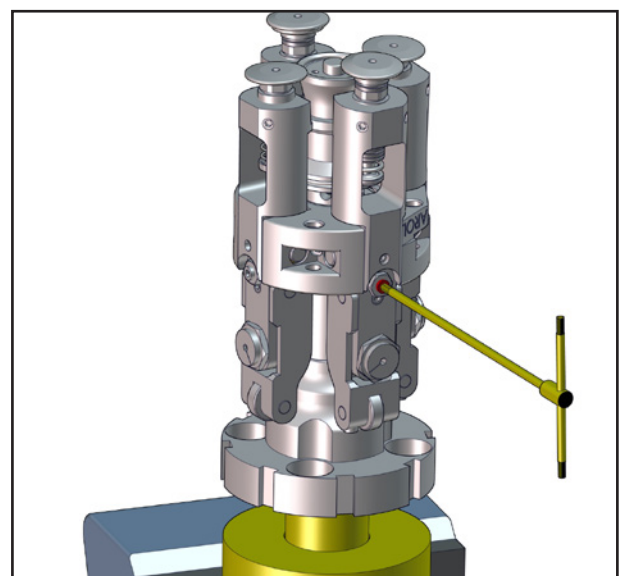


Fig. 132



- Insert the 10 mm hexagonal wrench on the locking nut.
- Holding a 3 mm Allen wrench, loosen the locking nut with a 10 mm hexagonal wrench.

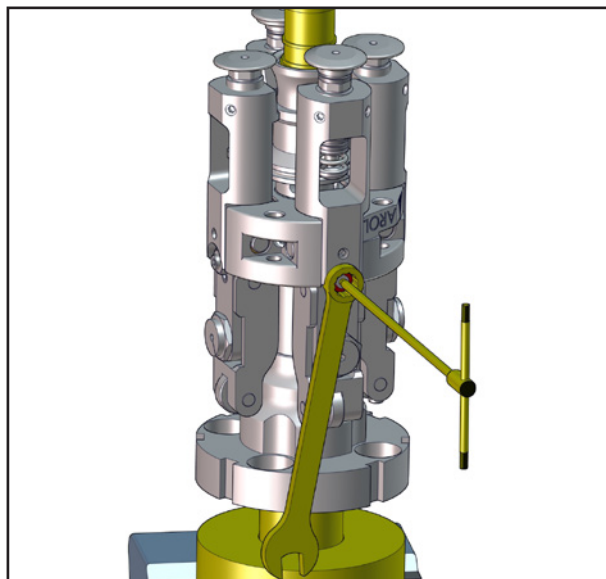


Fig. 133

- Move the roller support arms in a closed position by pushing the head downward.

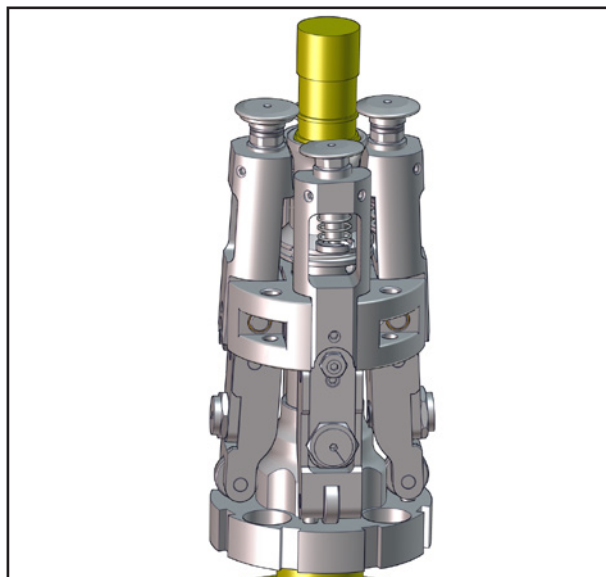


Fig. 134

- Screw or unscrew the locking grub screw with a 3 mm Allen wrench, until the skirting roller touches the pad diameter.
- Repeat the above steps also for all the skirting rollers.

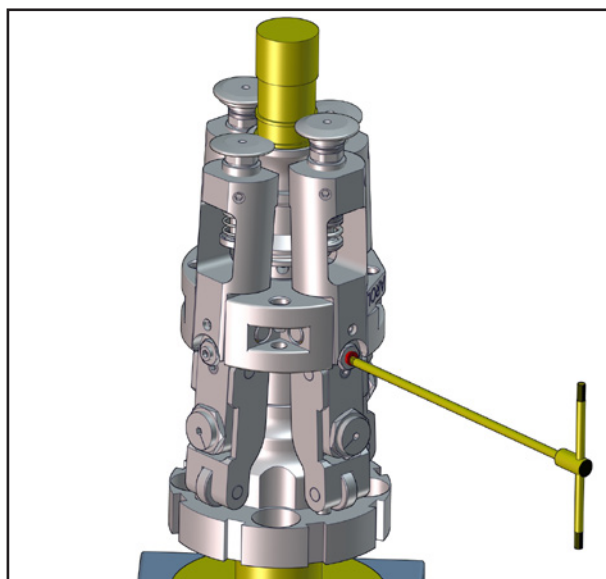


Fig. 135



- Make the pad rotate on the mouth holder, during the pad rotation the skirting rollers must turn freely too.

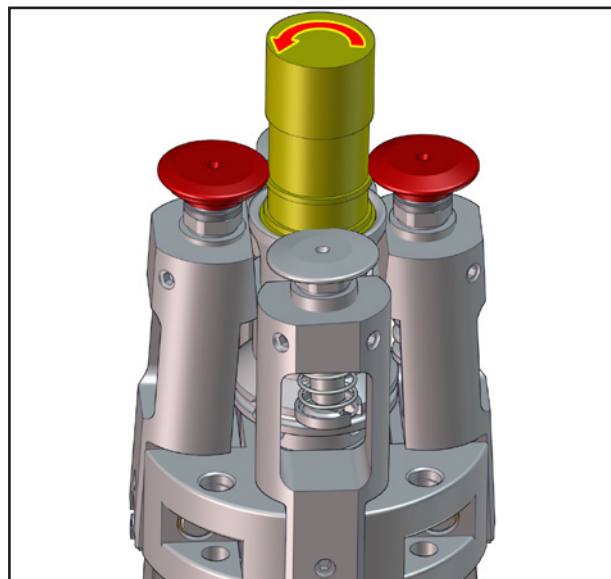


Fig. 136

- Fasten the blocking nut with a 10 mm exagonal wrench, making sure that the grub screw does not move.

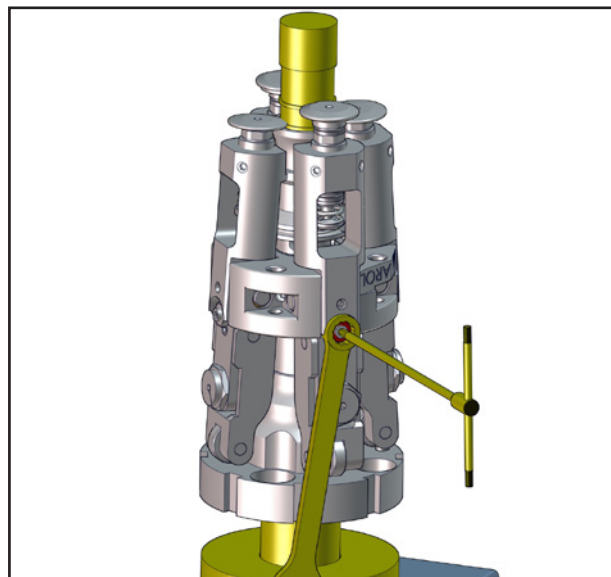


Fig. 137

- Repeat the same operation for the remaining skirting arms.
- Free the arms driving cam putting it in rest position (the rollers holder arms widen).

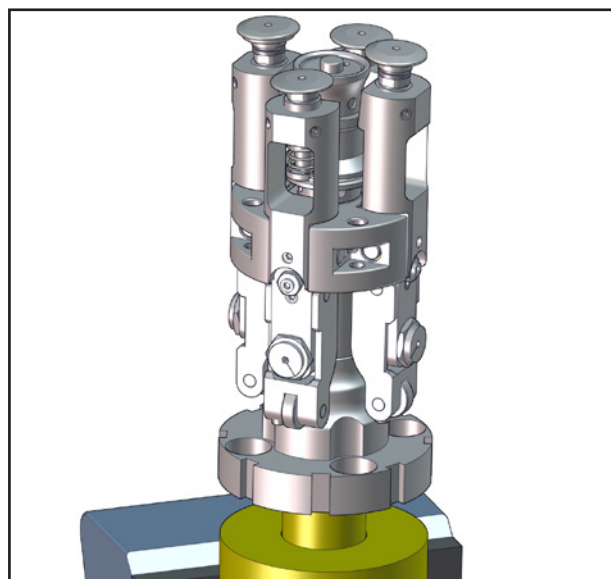


Fig. 138



- Insert a 3 mm Allen screw wrench into the seat of the spring holder bush

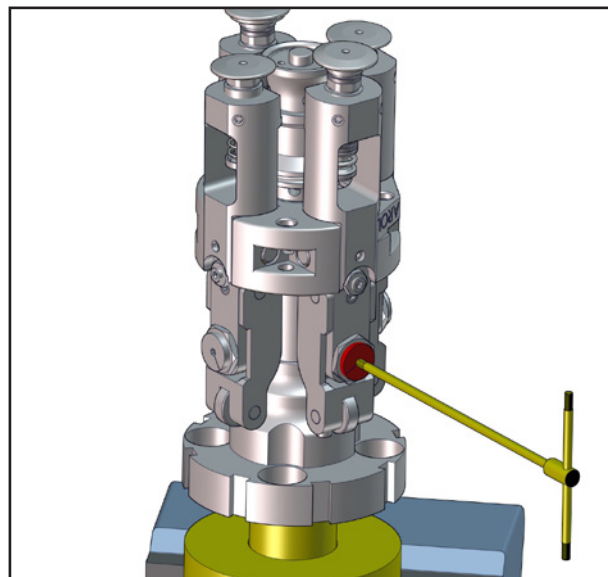


Fig. 139

- Insert the 19 mm hexagonal wrench on the locking nut.
- Holding a 3 mm Allen screw, loosen the locking nut with a 19 mm hexagonal screw.

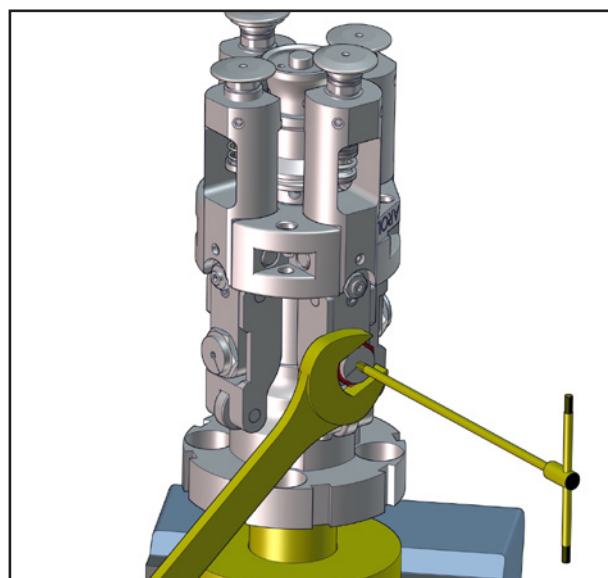


Fig. 140

- Using a dynamometer measure the side pressure strenght of the skirting rollers. The measurement must takes place when threading arm moves.

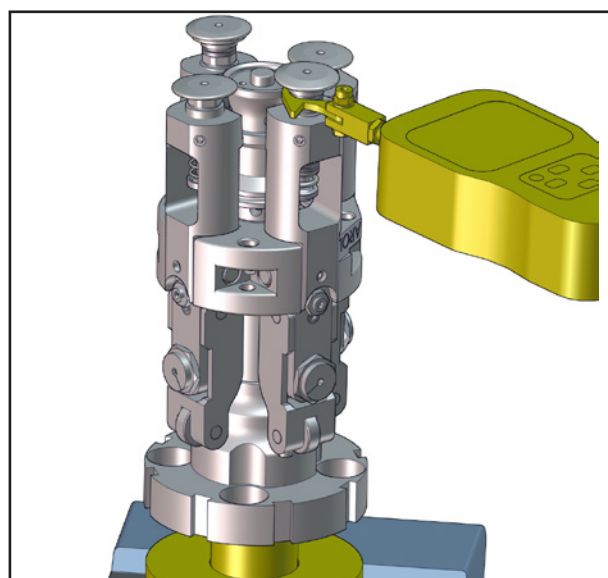


Fig. 141



If the carried out measurement is different from the one indicated in "CLOSURE HEAD CALIBRATION TABLE" enclosed in this manual, it is necessary to carry out as follows:

- Screw the spring holder bush with a 3 mm Allen wrench to increase the side pressure.
- Unscrew the spring holder bush with a 3 mm Allen wrench to decrease the side pressure.

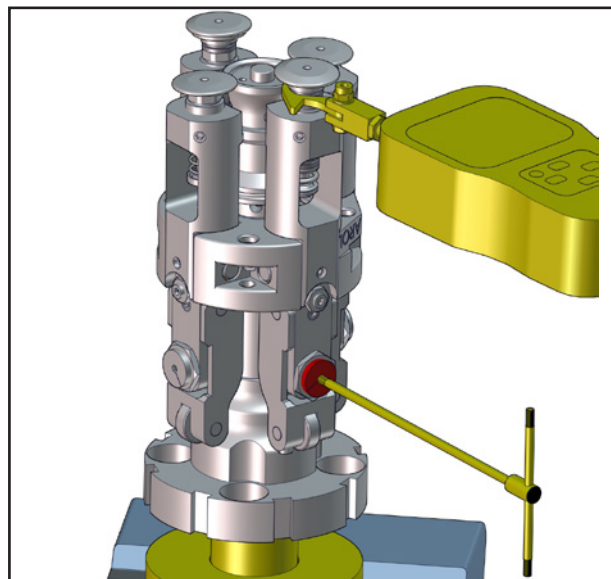


Fig. 142

- Keeping a 3 mm Allen wrench still, screw the locking nut with the 19 mm hexagonal wrench.
- Repeat the same operation for the remaining skirting arms.

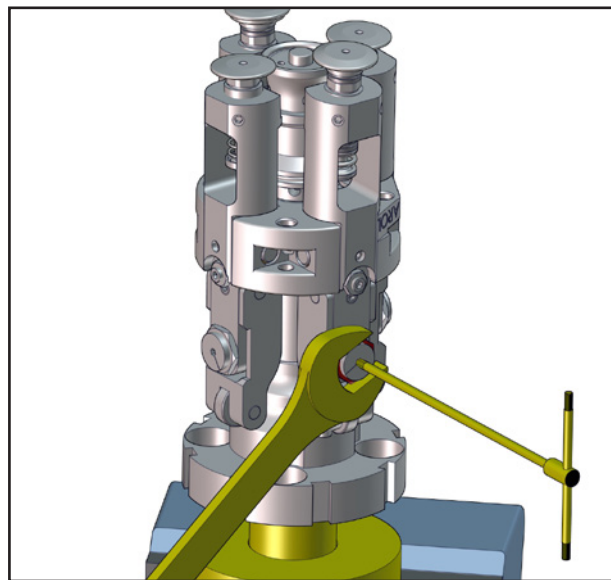


Fig. 143



11 MAINTENANCE OF THE CLOSING MACHINE

11.1 MAINTENANCE GENERAL INFORMATION



11.1.1 MAINTENANCE SCHEDULE

From the constructive point of view, interventions concern mechanical, electrical and pneumatic parts.

The envisaged operations are grouped according to work time intervals. Each intervention or group of interventions can concern about mechanical, electrical and pneumatic aspects.

The ordinary maintenance is divided in two categories:

- scheduled ordinary maintenance (or preventive).
- Maintenance according to the conditions (caused by wear and tear).

The scheduled ordinary maintenance (so called periodical or preventive) includes inspections, checks and interventions that, to prevent stops and faults, keep under control:

- the lubrication state of the closing machine.
- The state of parts that can be affected by wear and tear.

The maintenance according to special conditions recommends a regular inspection of components, to check their conditions and perform the maintenance only if it is required.

11.1.2 MAINTENANCE INTERVALS



The following tables show the maintenance deadlines and the operations needed for a correct operation and long life of the closing machine.

The interventions described below have been arranged at intervals of time expressed in hours.

Interval (hours)	Working in 1 shift	Working in 2 shifts	Working in 3 shifts
8	-	-	-
40	1 week	-	-
120	3 weeks	-	1 week
250	6 weeks	3 weeks	2 weeks
500	3 months	6 weeks	1 month
1000	6 months	3 months	2 months
3000	12 months	9 months	6 months
6000*	*	*	12 months

(*) every 6000 hours or anyway once a year.



11.1.3 GENERAL INDICATIONS ON THE SCHEDULED MAINTENANCE PLAN

A logbook of the scheduled maintenance and lubrication interventions must be arranged according to the maintenance indications in the sub-chapters below and to the closing machine working conditions.

Example:

MAINTENANCE SCHEDULED AT 1000 HOURS					
INTERVENTION	DATE	MAINTENANCE MAN	SIGNATURE	INTERVENTION RESULT	NOTES
Check of the height sliding of the closure carousel.	XX/XX/XX	Mario Rossi		positive	
Control gear lubrication	XX/XX/XX	Mario Rossi		lubricated	

11.2 MAINTENANCE AND LUBRICATION PLANNING



Interval	Intervention type
40 hours	Ordinary maintenance
120 hours	lubrification
	Ordinary maintenance
250 hours	lubrification
500 hours	lubrification
	Ordinary maintenance
1000 hours	lubrification
	Ordinary maintenance
3000 hours	Ordinary maintenance
5000 hours	Ordinary maintenance
6000 hours	Ordinary maintenance
12000 hours	Ordinary maintenance
18000 hours	Ordinary maintenance



11.3 SCHEDULED MAINTENANCE



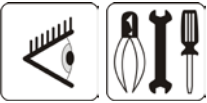
11.3.1 ROUTINE MAINTENANCE

11.3.1.1 EVERY 40 WORKING HOURS

OPERATION	AIM	NOTES
Pneumatic system check. 	Check the air supply pressure to the various utilization points. Check by the gauges, that the pressure reducers are adjusted at the work pressure (the pressure is indicated on the pressure gauges) Drain the possible condensate collected under the discharge filter. Checking the filter clogging. If necessary, replace it.	



11.3.1.2 EVERY 1000 WORKING HOURS

OPERATION	AIM	NOTES
Closure heads check. 	Check the state of wear of springs and if it is necessary, replace them. Check the state of wear of rollers and mouth-presser and if it is necessary replace them.	See sub-chapter 11.3.3 MAINTENANCE OF THE CLOSURE HEAD page 100.
Check of the springs of the pick off device. 	Check the state of wear of springs of pick off. If worn or damaged, replace them.	
Check of the height sliding of the closure carousel. 	Check the correct vertical sliding, making the closure carousel slide from the min. to the max. value, so as to ensure a correct component sliding in time.	
Equipment timing check. 	Check the timing of the whole equipment and, if necessary, re-time it.	See sub-chapter 11.5.3 TRANSPORT EQUIPMENT TIMING page 138.
Electric connection check. 	Carefully check all the connectors and connection cables and if necessary replace all the damaged parts.	
Sight and sound check of possible noises, clearances and unbalancing. 	Check, starting and stopping the closing machine several times without bottles, for possible noises, faults, clearances and vibrations in the motor-drive components (motors, reducers, gears, bearings). Replace possible worn components. Lubricate the new installed components. Clean and lubricate the disassembled components. Report any anomalies you need to contact <i>Technical Support</i> .	



OPERATION	AIM	NOTES
Pneumatic system check.  	Check the correct operation and possible leaks (from fittings, lines, etc.): ▪ verify the air pressure at the closing machine entry. ▪ Supplying the pneumatic system with compressed air, check there is no leak, broken or disconnected pipe. ▪ Check that the closing machine pressure gauges indicate the applied pressure on the suitable data plate.	

11.3.1.3 EVERY 6000 WORKING HOURS

OPERATION	AIM	NOTES
Check of the head control spindles.  	Check the clearance of the spindle pads.	See sub-chapter 11.3.2.1 SPINDLE PADS CLEARANCE ADJUSTMENT page 98.
Replacement of the UV-rays sterilizer. 	Replace the UV-rays sterilizer.	See sub-chapter 11.3.5.1 U.V. TUBE REPLACEMENT page 120.



11.3.1.4 EVERY 12000 WORKING HOURS

OPERATION	AIM	NOTES
Overhaul of the cork inserting pistons. 	Complete overhaul of pistons, including replacement of: ▪ Bearings. ▪ Rotating and static sealing. ▪ Scraper. ▪ Snap rings. ▪ Upper pressure roller. ▪ Lower pressure roller. ▪ Bearing fixing nut. ▪ Piston-guide pads. ▪ Pad fixing plates. ▪ Hardened bush for lower roller. ▪ Main sealing. ▪ Spindle key. ▪ Head fixing threaded ring. ▪ Roller rotation pin. ▪ Spindle closing threaded ring. ▪ Spindle gear. ▪ Gear connection sleeve.	For the overhaul of the piston, contact the <i>Technical Service</i>
Check the wear of the cap/cork driving punch unit cam race. 	Visually check the wear of the roller sliding race.	See sub-chapter 11.5.1 SPINDLES CONTROL CAM WEAR CHECK page 135.
Closing machine general check. 	Check the correct operation of the closing machine; if faults are found, contact the <i>Technical Service</i>.	



11.3.2 HEADS CONTROL SPINDLES MAINTENANCE



11.3.2.1 SPINDLE PADS CLEARANCE ADJUSTMENT

To adjust the pad clearance, operate as follows:

- make the closing machine rotate in jog until the spindle to be adjusted is in the flat (rear) section of the cam.
- Lift the ends of the pads fixing safety plates so as to release the bolts.

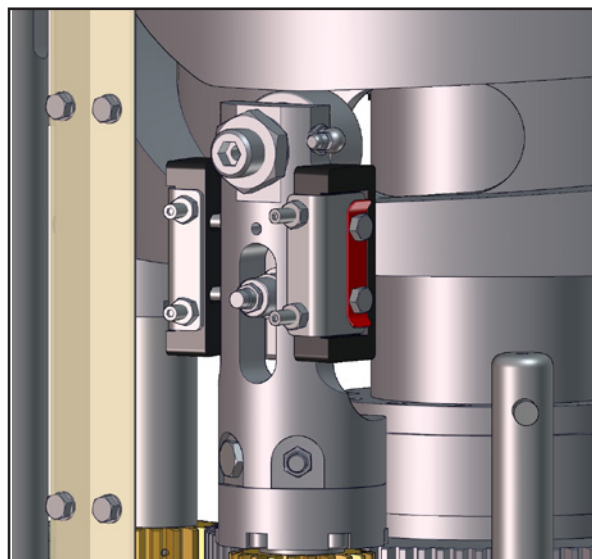


Fig. 144

- Loosen the fixing bolts of both pads.

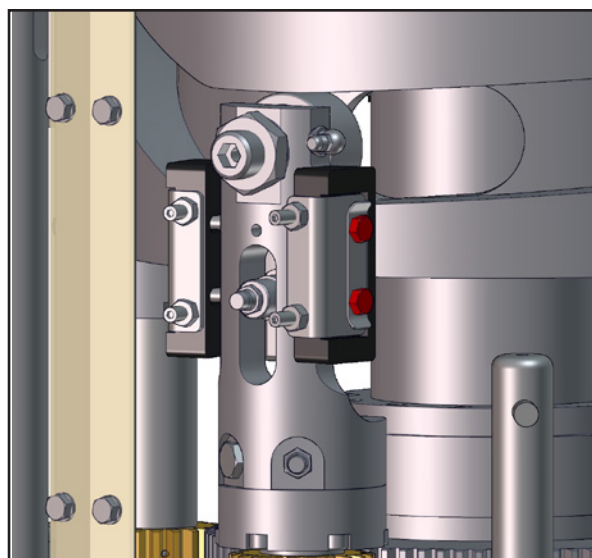


Fig. 145

- Loosen nuts and adjustment grub screws for both pads.

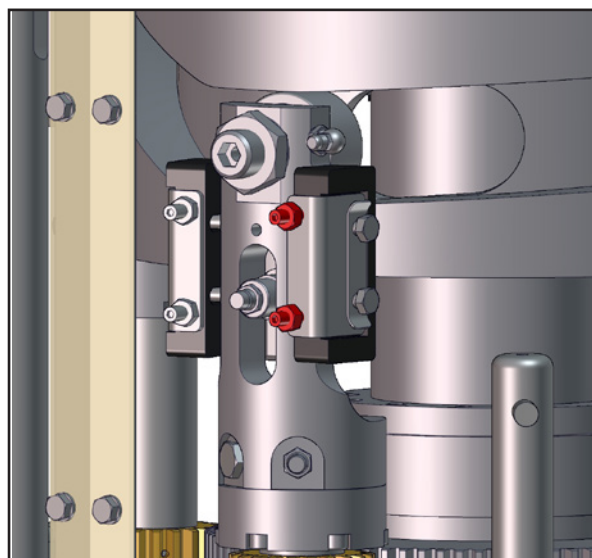


Fig. 146



- Fit a 0.1 mm shim between the two pads and the spindles driving cam.

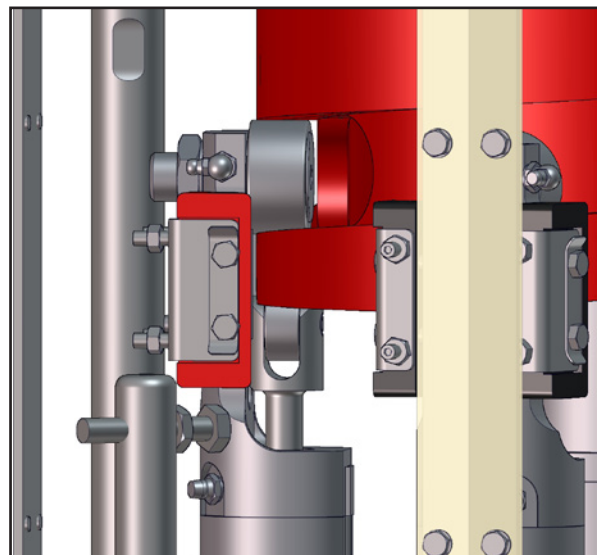


Fig. 147

- Push the pads against the shim.
- Register the position of the head driving piston so that the distance between the pad and the cam is the same for both pads.
- Verify the distances and lock the previously loosened bolts.
- Screw the adjusting screws until they are in contact with the pads.
- Screw all the grub screws locking nuts.

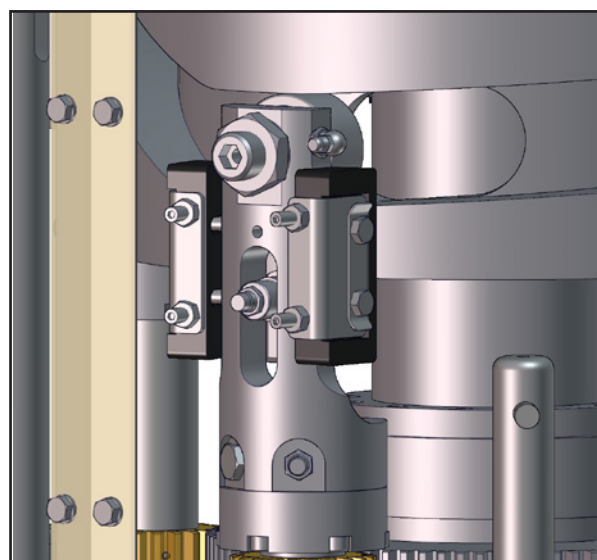


Fig. 148



A clearance between 0.1 and 0.2 mm must be kept all along the race cam.

This condition must be checked on the cam circumference in at least four diametrically opposed points (e.g. rear and front parts, left and right side).

- After adjustment, again bend the ends of the pads fixing safety plates so as to prevent that bolts may loosen.

Repeat the operation for the other possible spindles to be adjusted.



11.3.3 MAINTENANCE OF THE CLOSURE HEAD

11.3.3.1 "SSD" CLOSURE HEAD

The table below shows the necessary tools for maintenance of the closure head.

TOOLS	
	Roller washer mounting wrench
	Hook wrench 34-36 with ball
	Mouth-presser wrench

11.3.3.1.1 REMOVAL OF THE ROLLER HOLDER ARMS

Remove the closure head from the closing machine as described in sub-chapter 10.2.1 REMOVAL AND INSTALLATION OF THE CLOSURE HEADS page 60.

Proceed as follows:

- Insert the the closure head supporting equipment into a jaw.

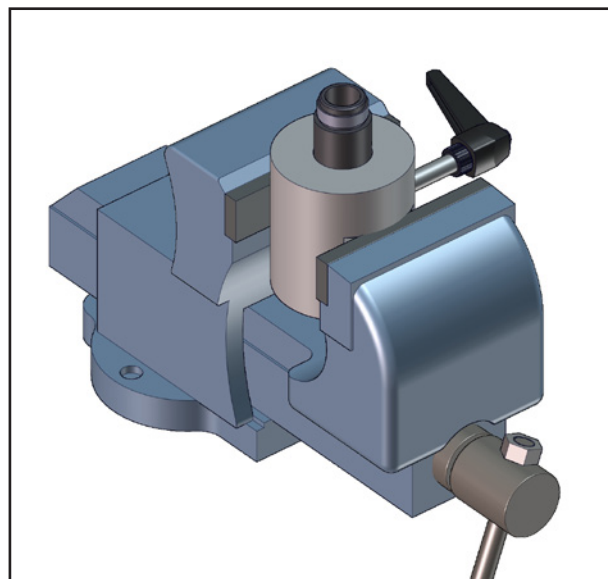


Fig. 149



- Screw the closure head on the sleeve of the supporting equipment.

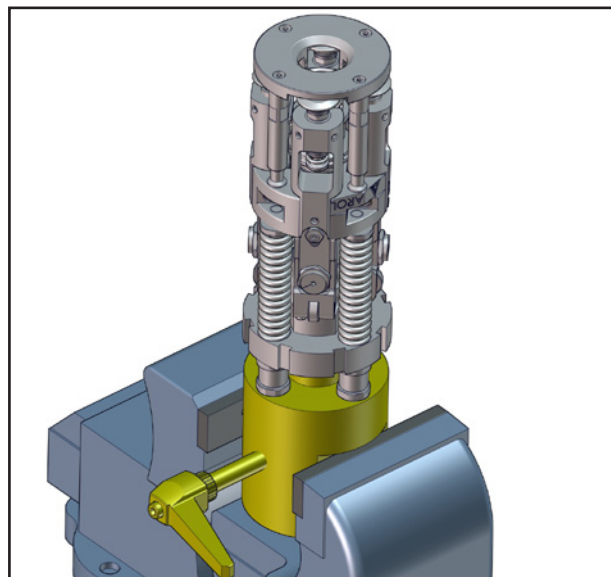


Fig. 150

- Unscrew the centering device fixing screws with a 3 mm Allen wrench.

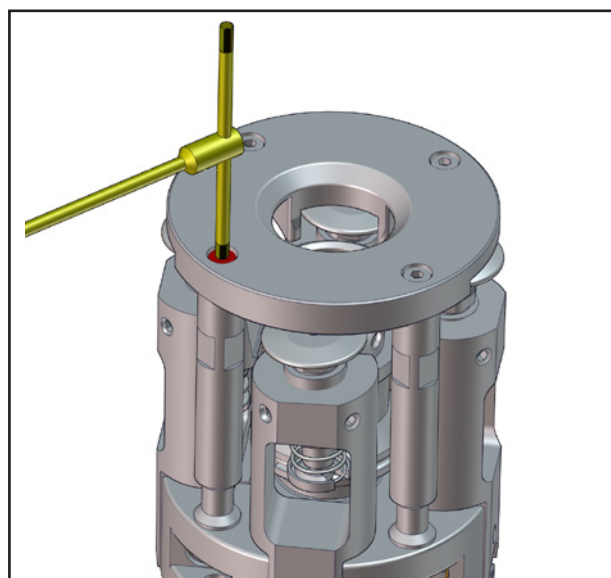


Fig. 151

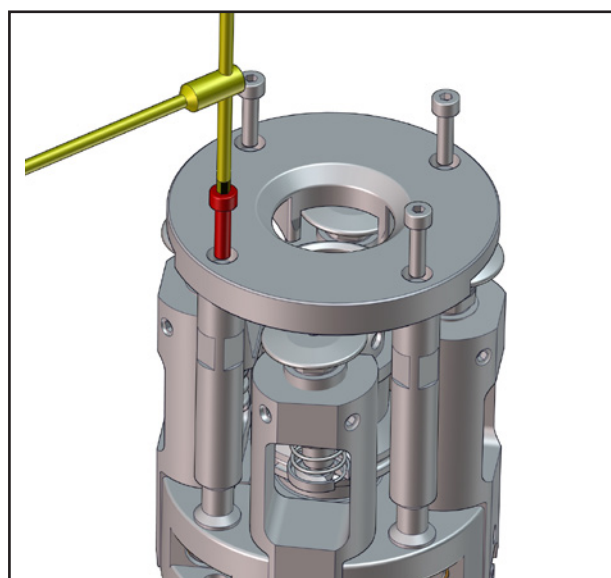


Fig. 152



- Remove the centering device.

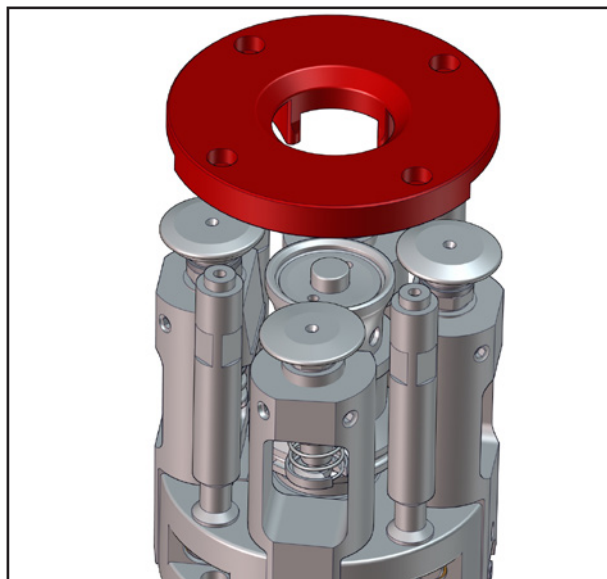


Fig. 153

- Unscrew the centering device's support columns with a 10 mm exagonal wrench, and remove the centering device's support columns.

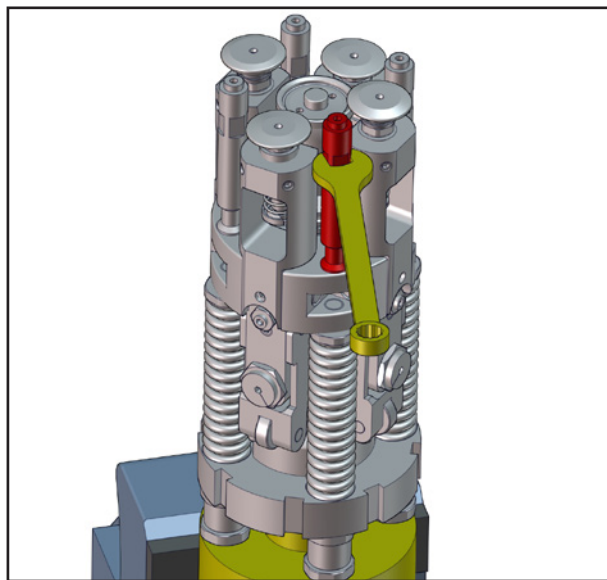


Fig. 154

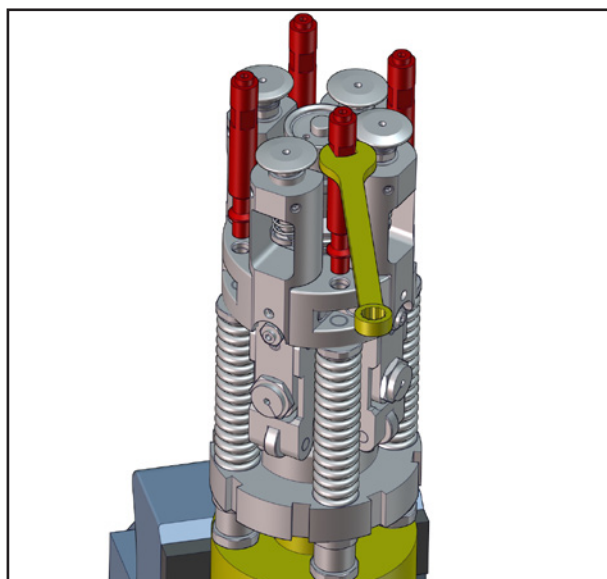


Fig. 155



- Unscrew the dowel fixing grub screw with a 3 mm Allen wrench.

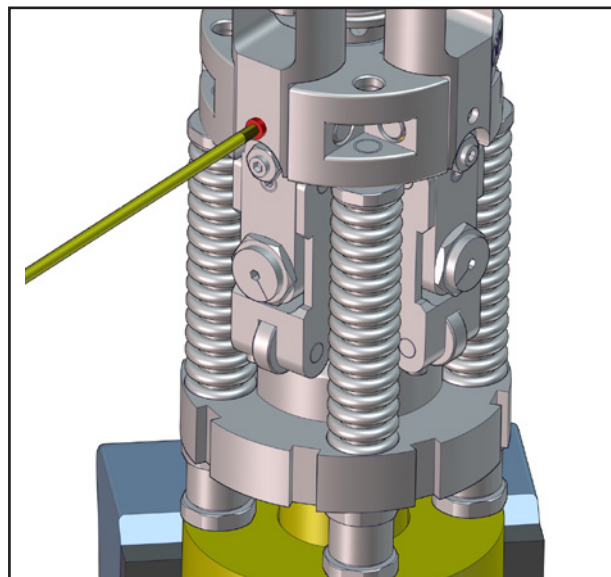


Fig. 156

- Withdraw the arm movement pin.

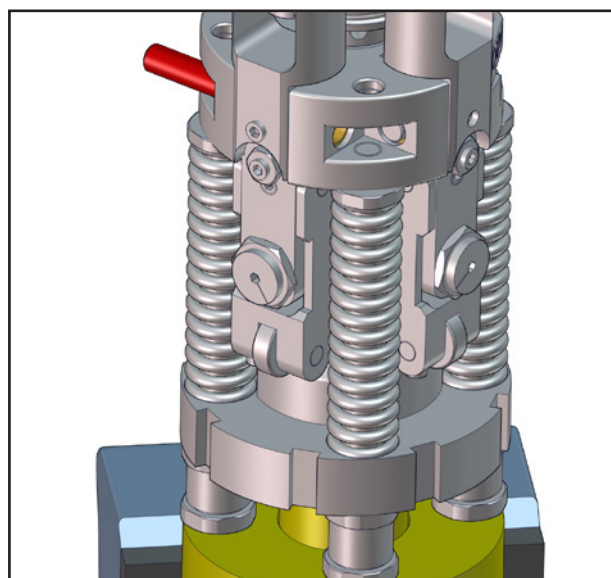


Fig. 157

- Remove the roller support arm.
- Proceeding in the same way to remove the remaining roller holder arms.



Before removing the roller holder arms, it is necessary to mark the arms in such way so to remount them in their original position.

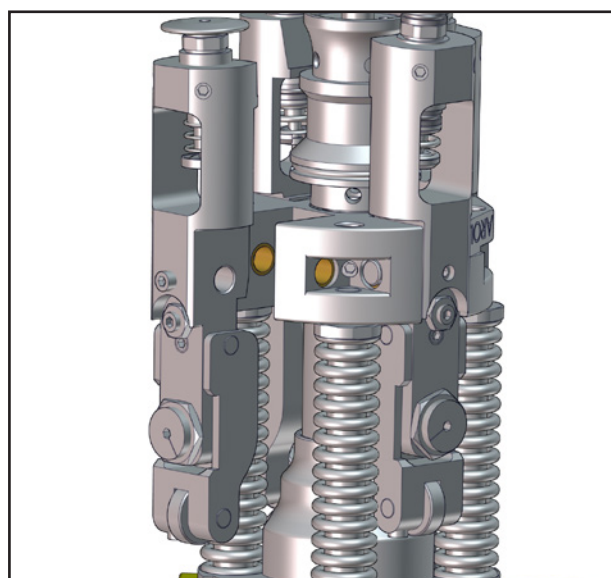


Fig. 158



11.3.3.1.2 BOTTLE MOUTH-HOLDER REPLACEMENT

Proceed as follows:

Remove the roller holder arms operating as described in subchapter 11.3.3.1.1 REMOVAL OF THE ROLLER HOLDER ARMS page 100.

- Fit the mouth-presser dismounting wrench in one of the holes of the central shaft.

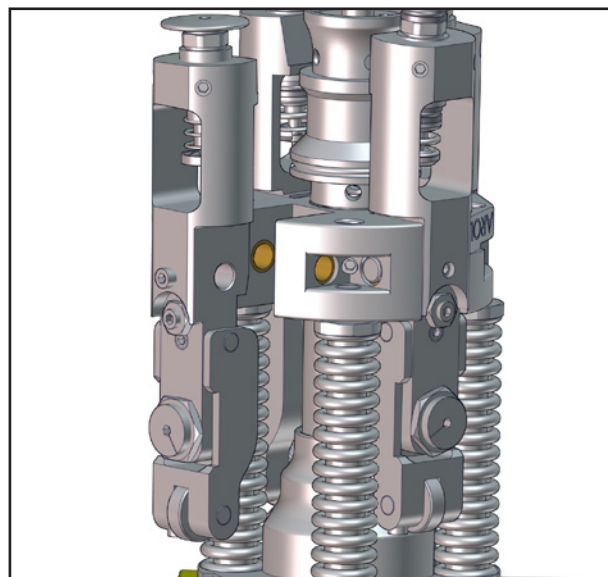


Fig. 159

- Fit the mouth-presser dismounting wrench in one of the holes of the mouth-presser.

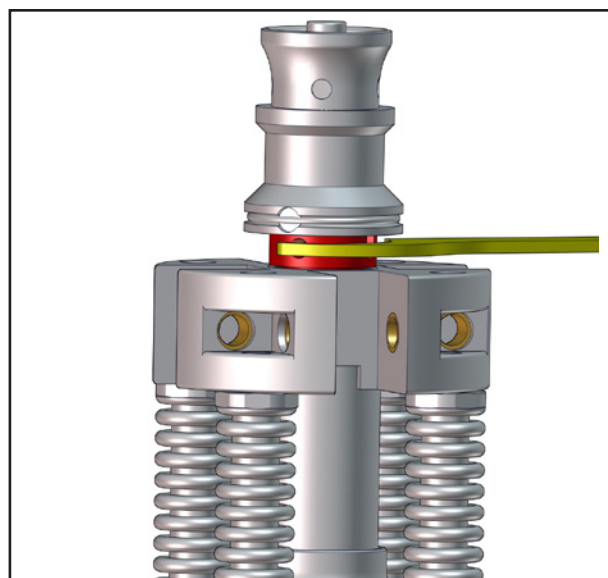


Fig. 160

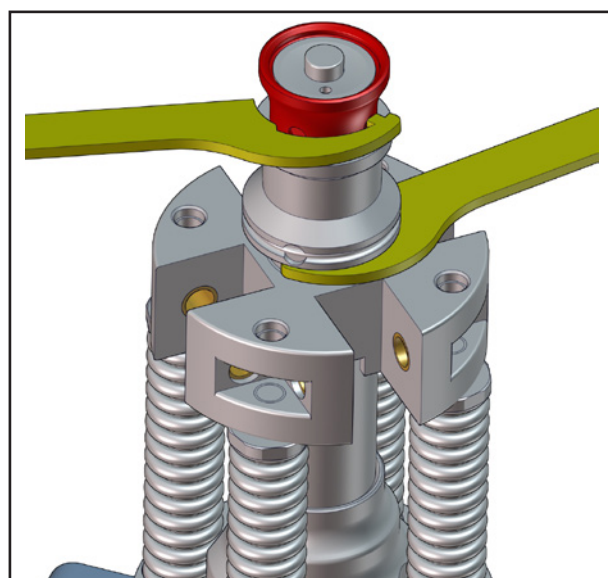


Fig. 161



- Unscrew the bottle mouth-presser with the mouth-presser wrenches, and remove the bottle mouth-presser.

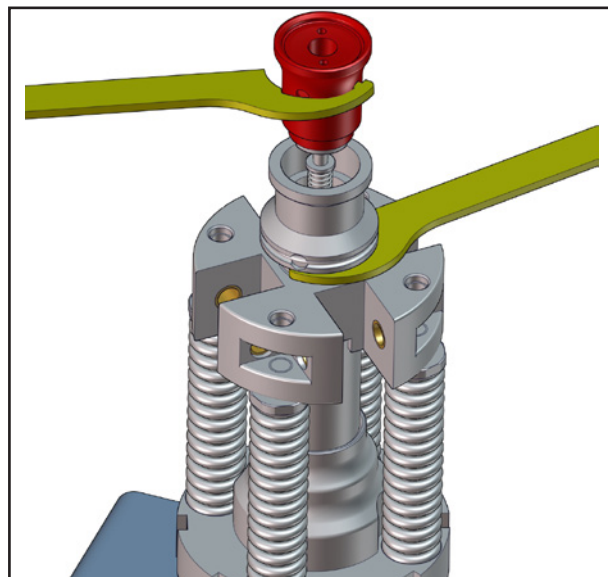


Fig. 162



The mouth-presser threading is left handed.

- Proceed in assembling the new mouth-presser.
- Reassemble the closure head, carrying out what was previously described in the opposite direction.
- Carry out the adjustment as described in sub-chapter 10.3 ADJUSTMENTS OF THE CLOSURE HEAD page 73.



11.3.3.1.3 REPLACING THE SPRINGS

11.3.3.1.3.1 COMPENSATING SPRINGS



Before working, measure and make a note of the distance between the adjustment bush and the head holder flange.

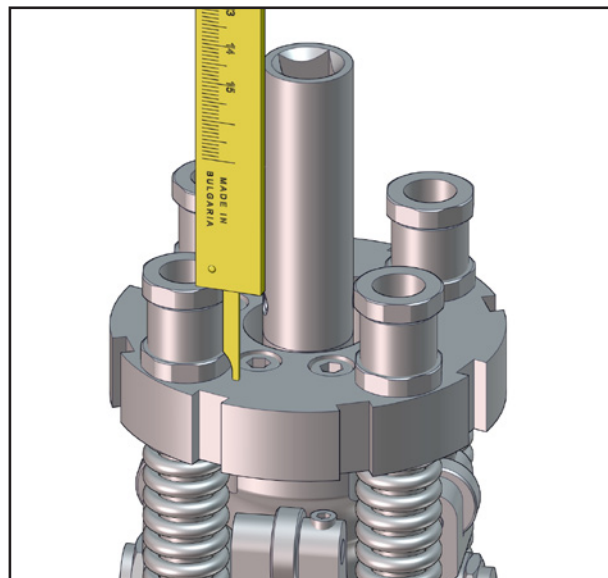


Fig. 163

Proceed as follows:

- Loosen the bush fixing nuts with the spring adjustment hexagonal wrench.

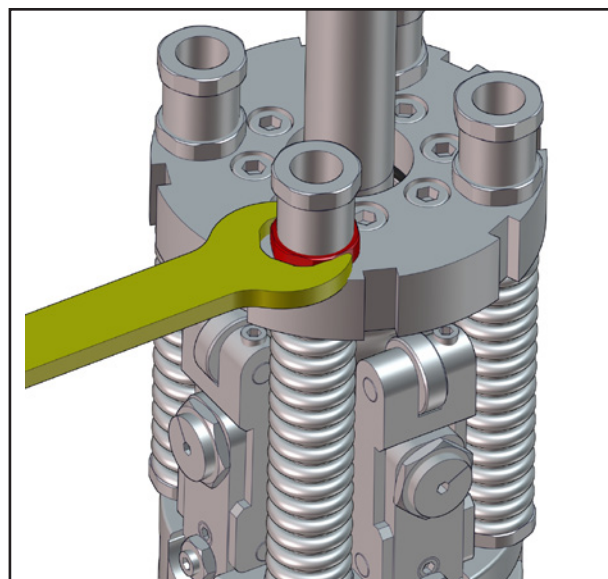


Fig. 164

- Completely unscrew the adjustment bushes with the spring adjustment hexagonal wrench.

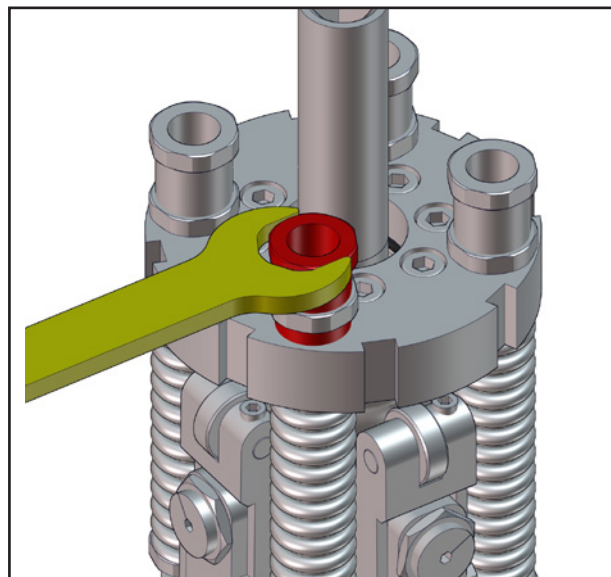


Fig. 165

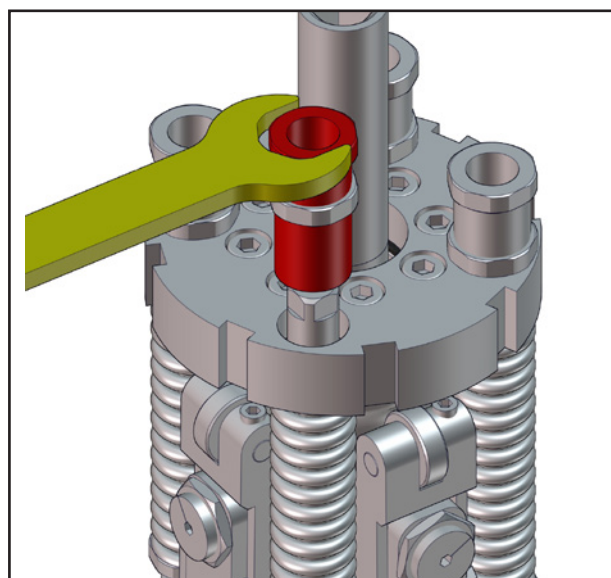


Fig. 166

- Unscrew the tie-rod fastening nuts with the springs adjustment wrench using the 10 mm hexagonal wrench to hold the spring guide tie-rods.

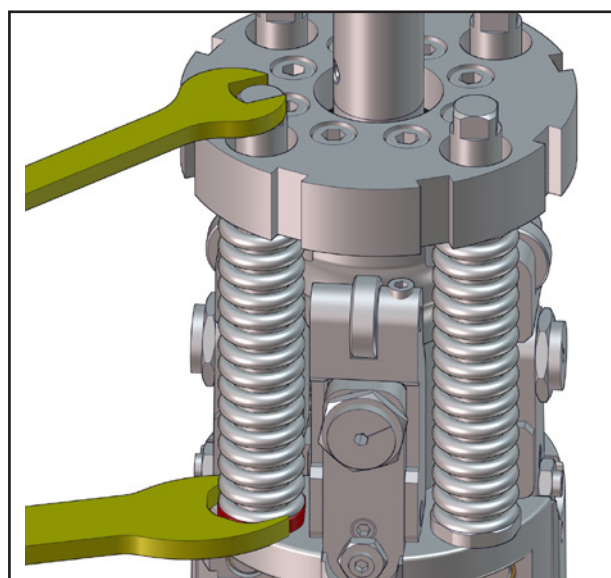


Fig. 167



- Completely unscrew the spring guide tie-rods with a 10 mm hexagonal wrench.

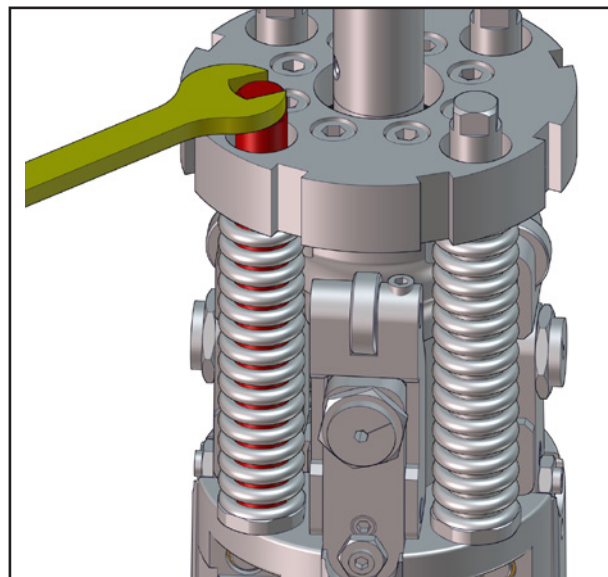


Fig. 168

- Slide out the spring guide tie-rods and tie-rod locking nuts.

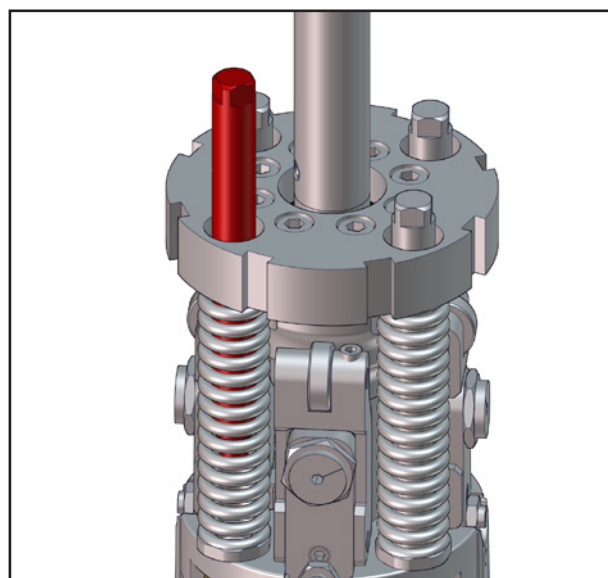


Fig. 169

- Remove the compensating springs, tie rod locking nuts and springs supporting washers.
- Reassemble inversely to that described for disassembly, taking care to restore the previously measured distances.

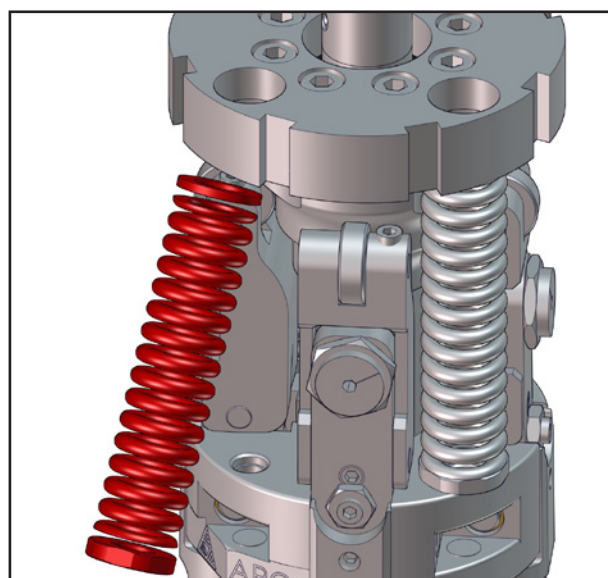


Fig. 170

11.3.3.1.3.2 SIDE LOADING SPRINGS

Proceed as follows:

- Insert a 3 mm Allen screw wrench into the seat of the spring holder bush

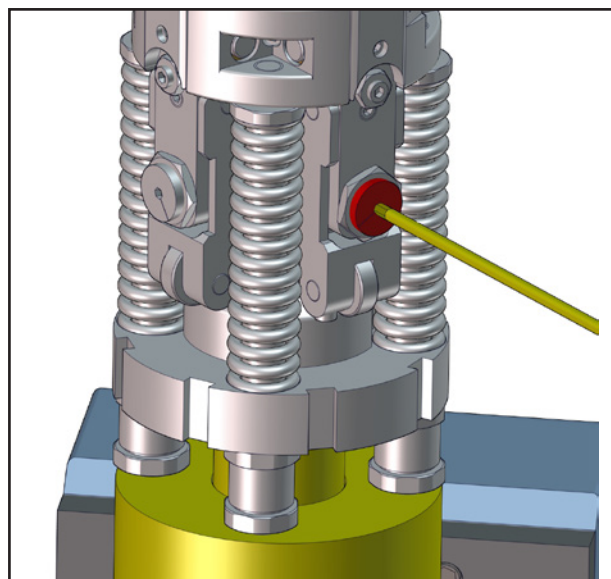


Fig. 171

- Insert the 19 mm hexagonal wrench on the locking nut.
- Holding a 3 mm Allen screw, loosen the locking nut with a 19 mm hexagonal screw.

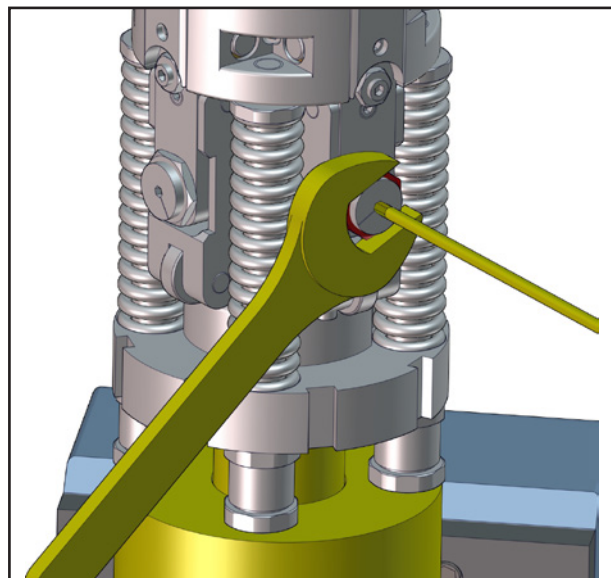


Fig. 172

- Loosen the spring holder bush with a 3 mm Allen wrench, and remove the bush.

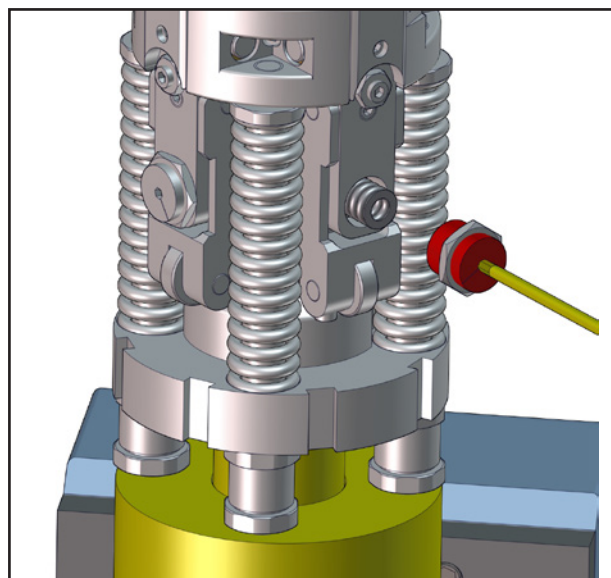


Fig. 173



- Extract the lateral pressure spring and replace it with a new one.



Lubricate the new spring proceeding as described in sub-chapter 11.4.7

LUBRICATION OF THE CLOSURE HEAD
page 131.

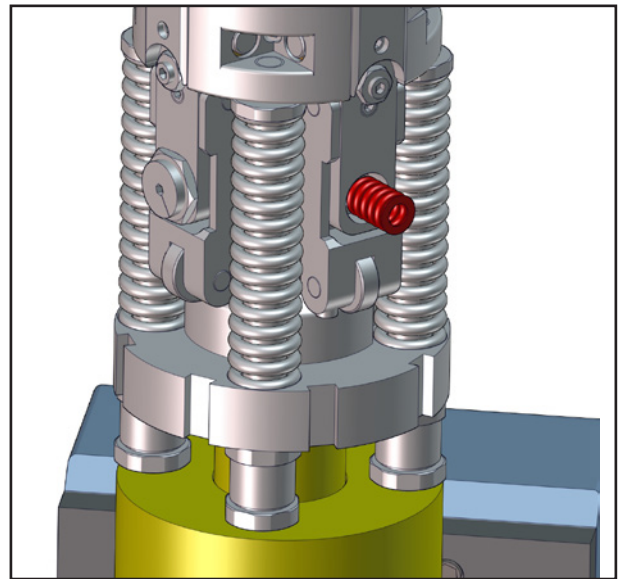


Fig. 174

- Reassemble the spring holder bush.
- Adjust the threading rollers side loading as described in sub-chapter 10.3.1.2
ADJUSTING THE THREADING ROLLER page 80.
- Adjust the skirting rollers side loading as described in sub-chapter 10.3.1.3
ADJUSTING THE SKIRTING ROLLER page 86.
- Keeping a 3 mm Allen wrench still, screw the locking nut with the 19 mm hexagonal wrench.

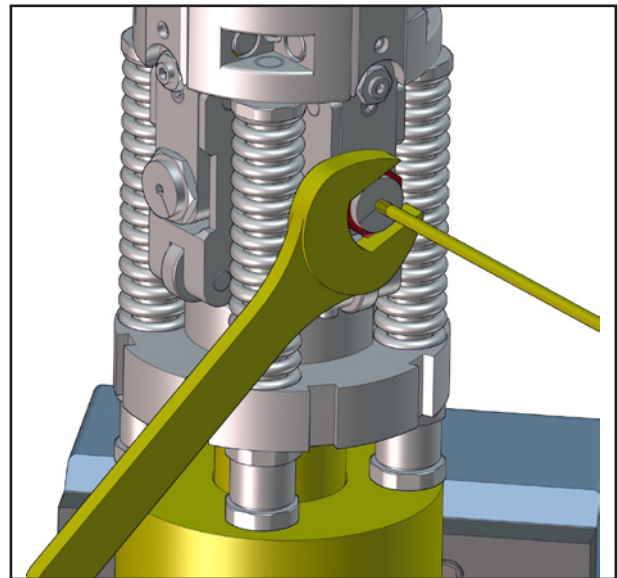


Fig. 175

11.3.3.1.3.3 THREADING ROLLER SPRINGS

Proceed as follows:

- remove the threading arm as described in sub-chapter 11.3.3.1.1 REMOVAL OF THE ROLLER HOLDER ARMS page 100.

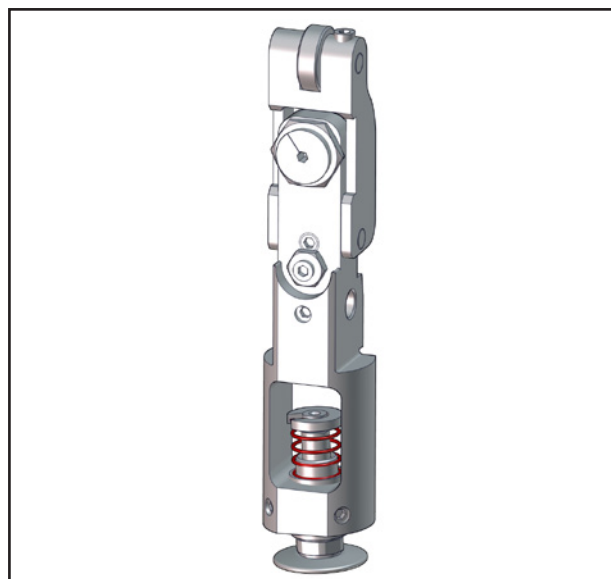


Fig. 176

- compress the threading roller spring.

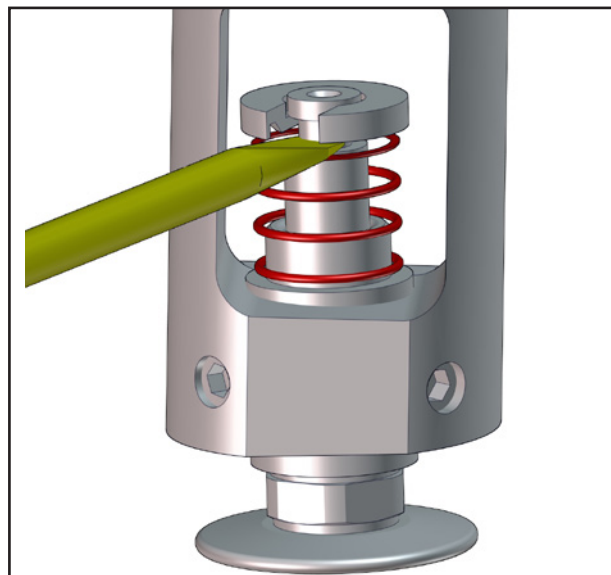


Fig. 177

- Extract from the roller seat the threading roller washer.

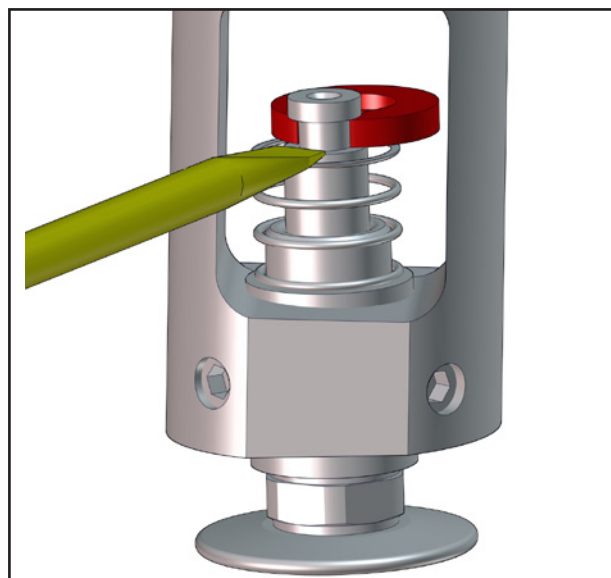


Fig. 178



- Extract the threading roller spring.

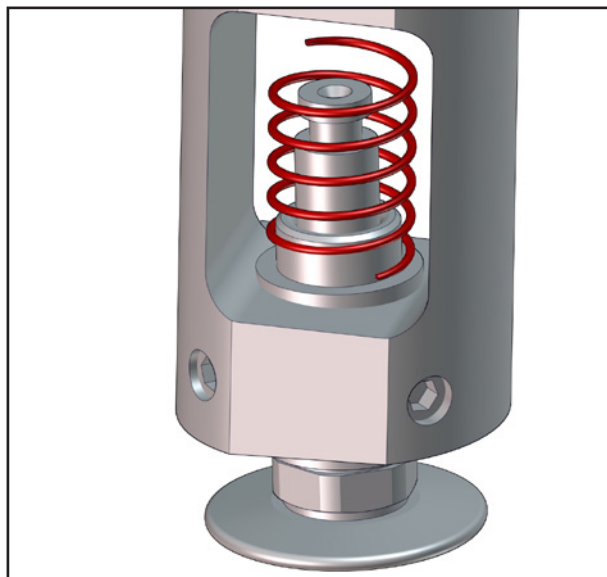


Fig. 179

- Insert the new spring on the threading roller and keep it compressed.
- Reposition the threading roller washer in threading washer seat.



11.3.3.1.3.4 SKIRTING ROLLER SPRINGS

Proceed as follows:

- remove the skirting arm as described in sub-chapter 11.3.3.1.1 REMOVAL OF THE ROLLER HOLDER ARMS page 100.
- Using a suitable tool (for example a screwdriver) compress, pushing down, the skirting roller spring.

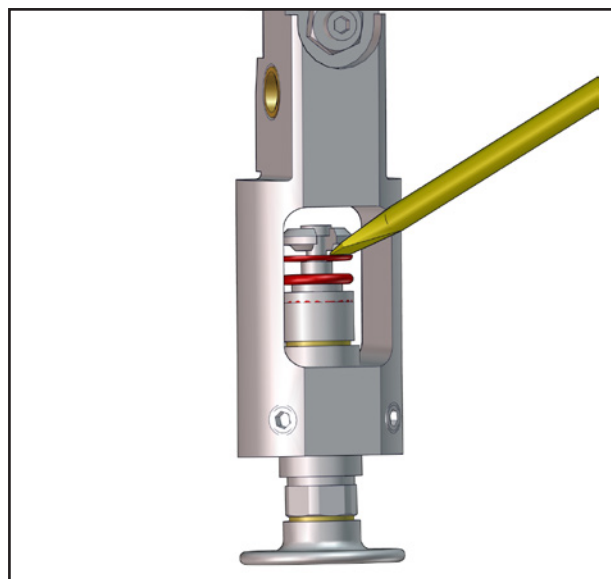


Fig. 180

- Extract from the roller seat the skirting roller washer.

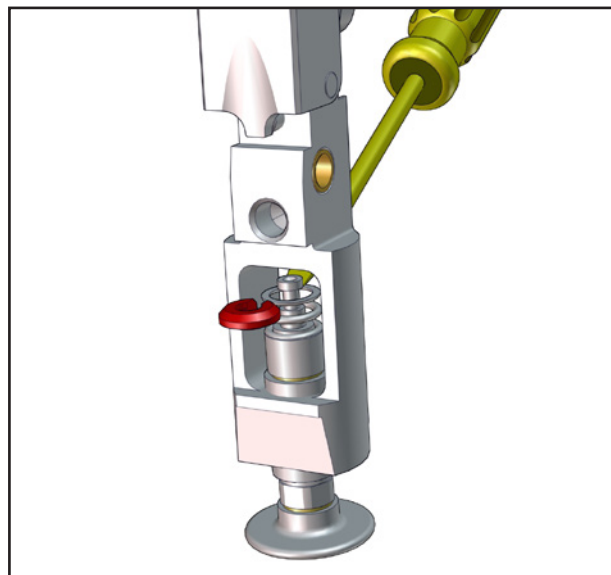


Fig. 181

- Extract the skirting roller spring.

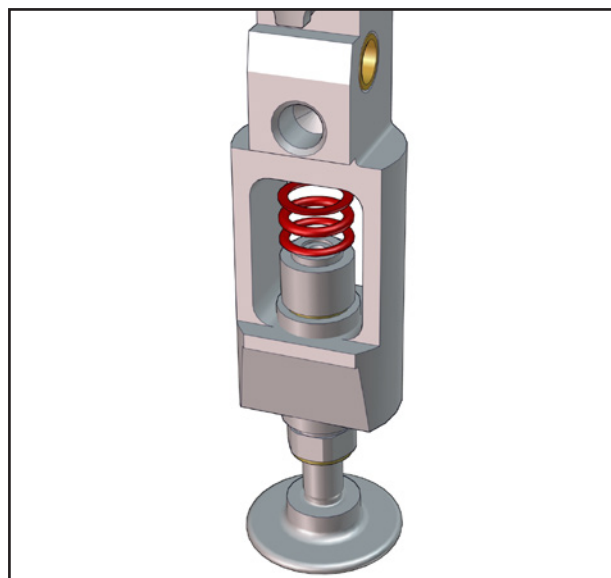


Fig. 182



- Insert the roller support arm complete with the skirting roller into a vice with soft jaws.
- Insert the new spring on the skirting roller.

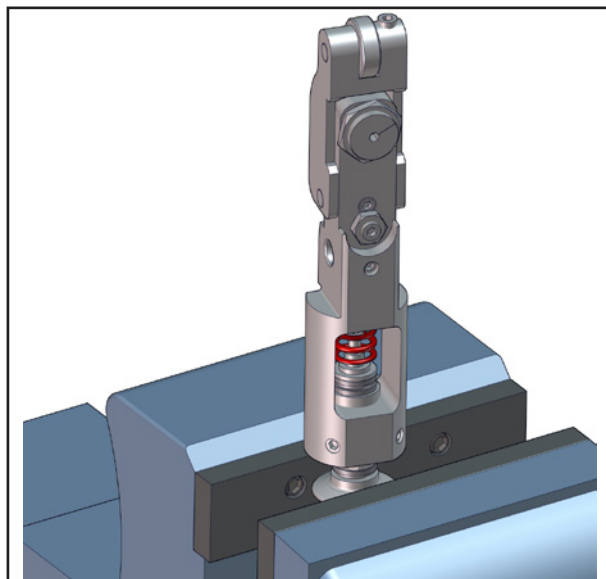


Fig. 183

- Keep compressed the skirting roller spring with the roller washer mounting wrench.

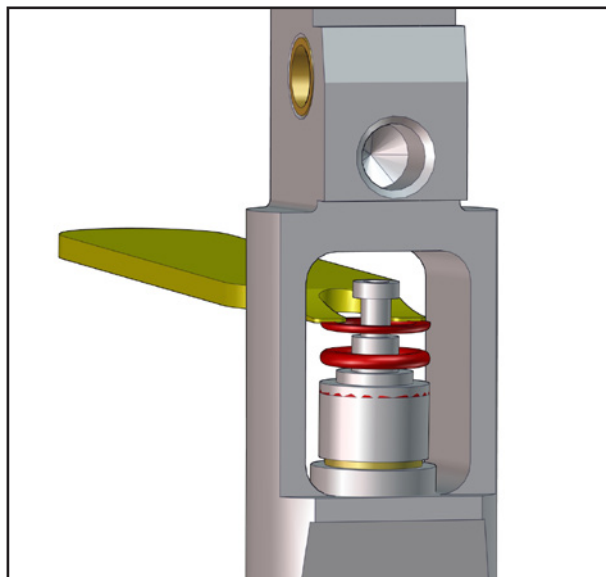


Fig. 184

- Reposition the washer in skirting roller seat.

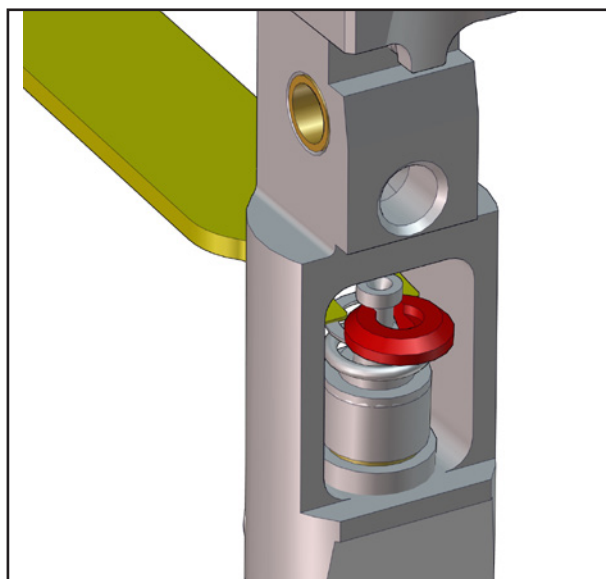


Fig. 185



11.3.3.1.3.5 MOUTH-PRESSER SPRING

Remove the mouth-presser from the closure head as described in sub-chapter 11.3.3.1.2
BOTTLE MOUTH-HOLDER REPLACEMENT page 104.

Proceed as follows:

- Insert the mouth-presser wrench in a vice.

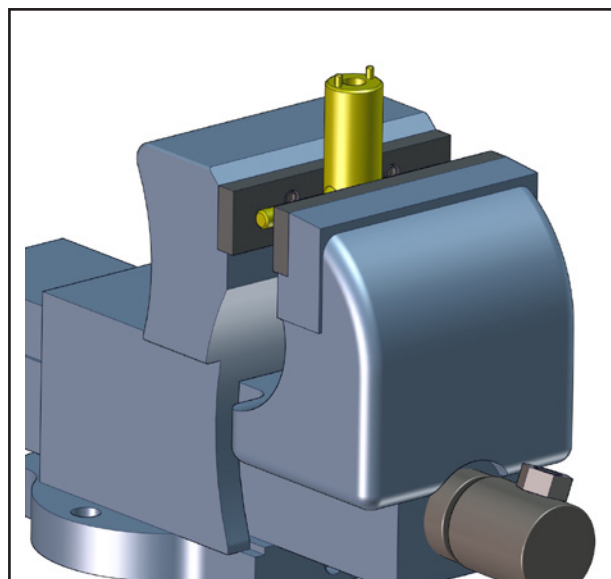


Fig. 186

- Insert the mouth-presser on the wrench.

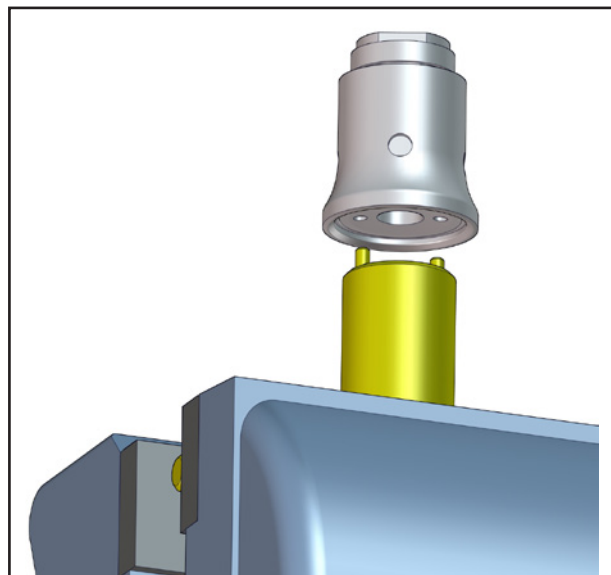


Fig. 187

- Unscrew the GAS feeler pin bush with a 17 mm exagonal wrench.

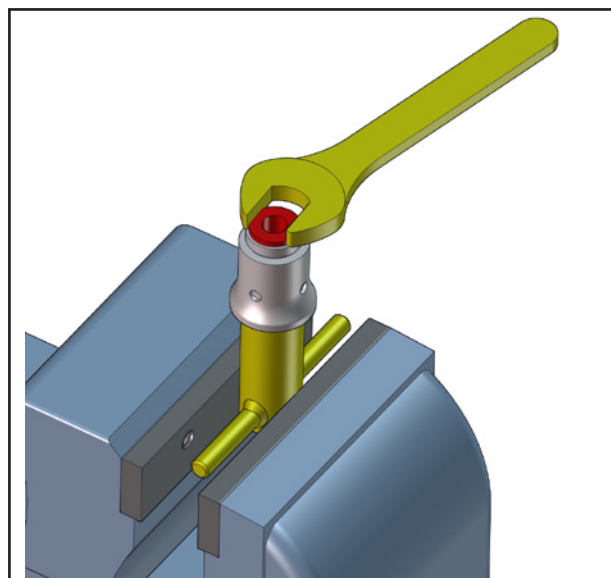


Fig. 188



- Unscrew the GAS feeler pin bush with a 17 mm exagonal wrench.

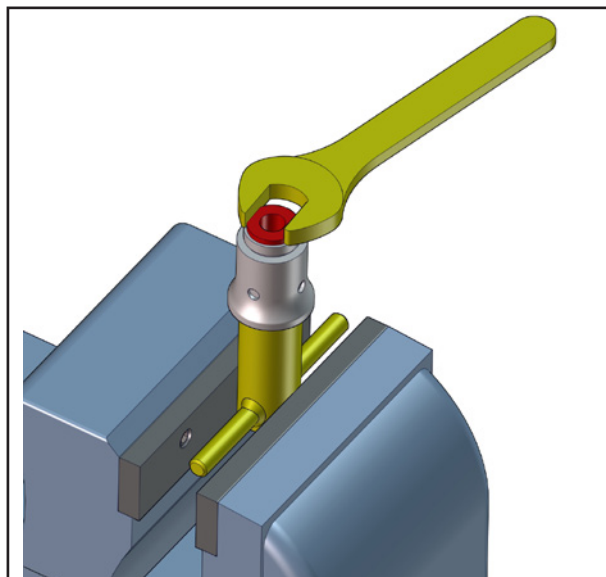


Fig. 189

- Remove the GAS feeler pin bush and the plunger mouth-presser from the mouth-presser

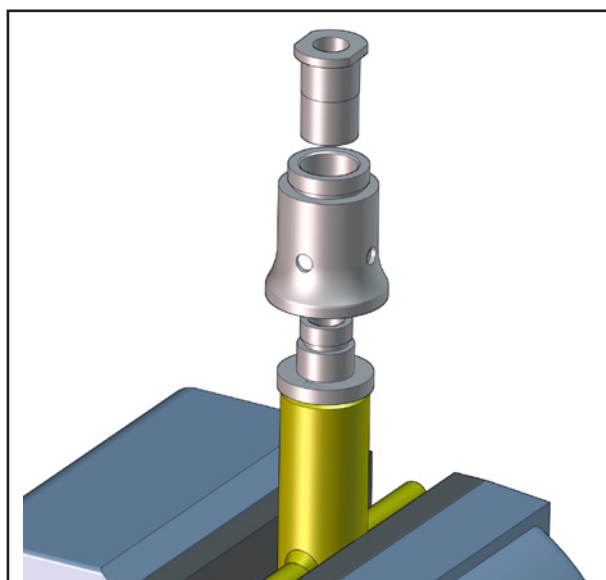


Fig. 190

- If present, remove the spacers from the mouth-presser inside.

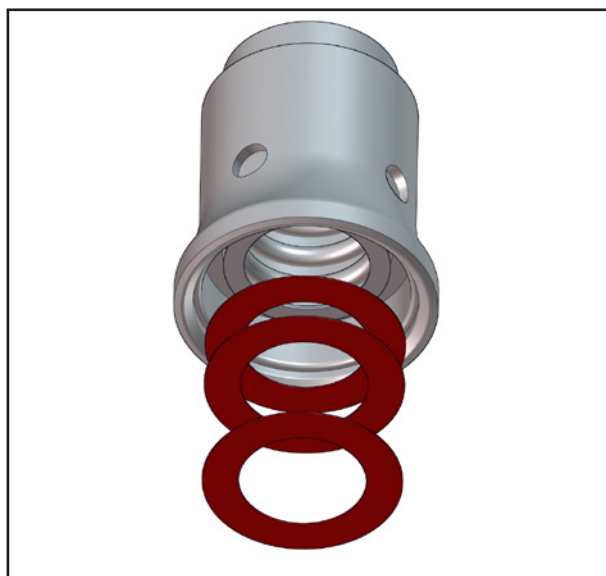


Fig. 191



- Extract the GAS mouth-presser spring and replace it with a new one.



Lubricate the new spring proceeding as described in sub-chapter 11.4.7

LUBRICATION OF THE CLOSURE HEAD
page 131.

- Remount the GAS feeler pin bush.

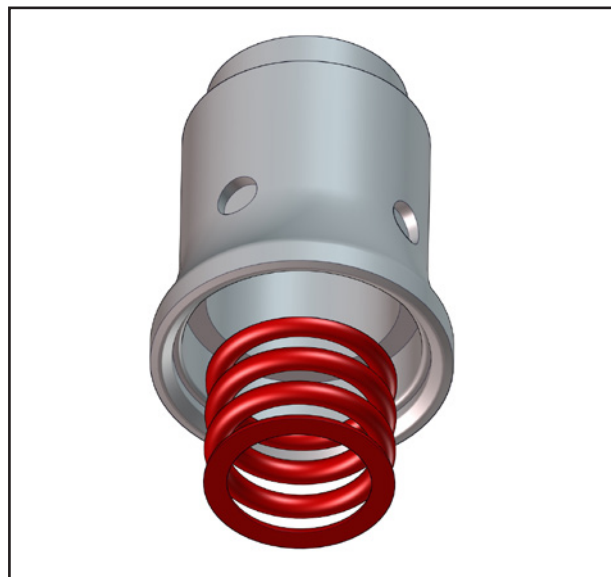


Fig. 192



11.3.4 MAINTENANCE OF THE "FLAT" CENTRIFUGAL CAPS SORTER



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

If there is an anomalous distribution, it is necessary to carry out the following operations:

- check that the nozzles placed on the exit mouth are not obstructed and are in the correct position to push caps, not correctly orientated, into the discharge tube.
- Check that the nozzle or nozzles, if more than one, placed on the circumference is not obstructed.

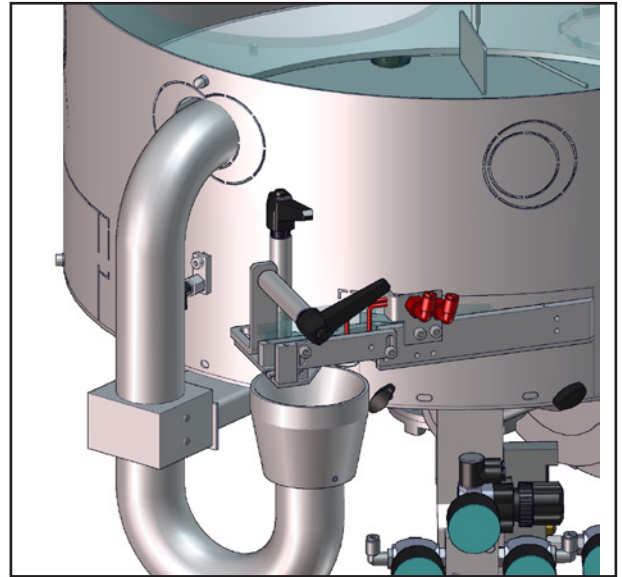


Fig. 193

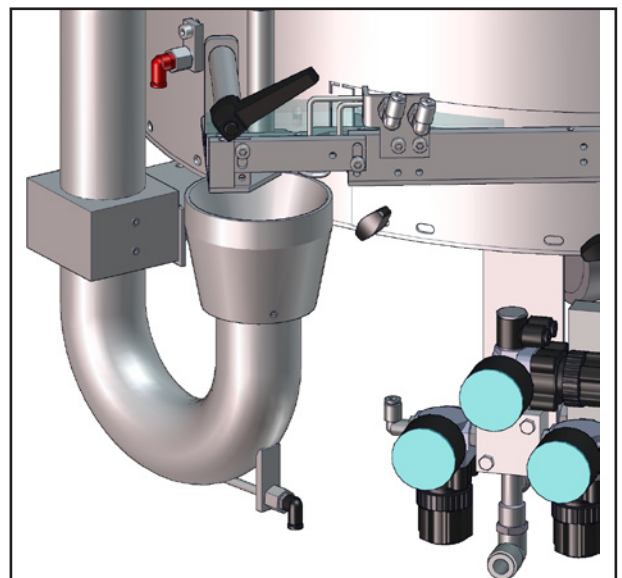


Fig. 194



- Check that the nozzle placed on the discharge tube is not obstructed.
- Check that in the tube there are no caps; if there are caps it is necessary to remove them.

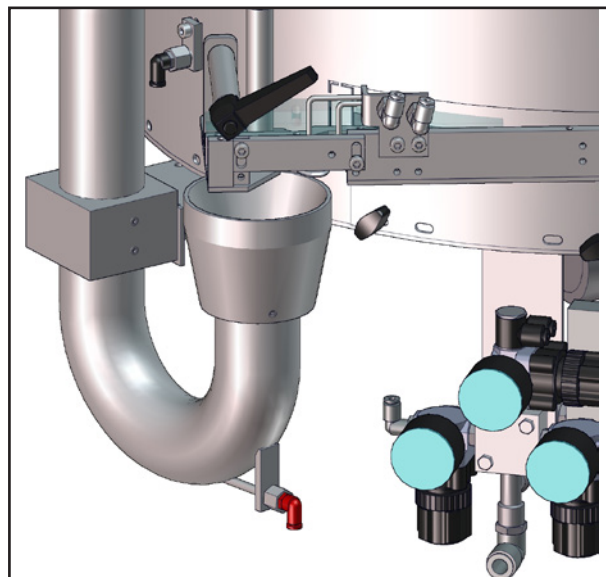


Fig. 195

- For the removal of the caps from the discharge tube it is necessary:
 - unscrew the fixing grub screw placed on the holding clamp;

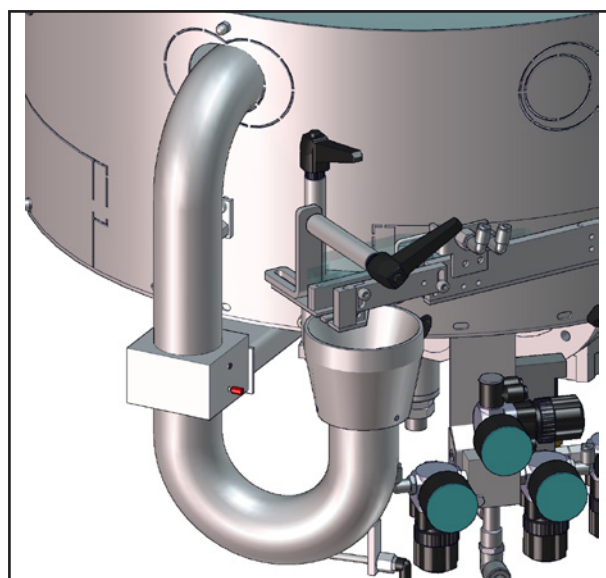


Fig. 196

- Pull down the lower discharge tube.

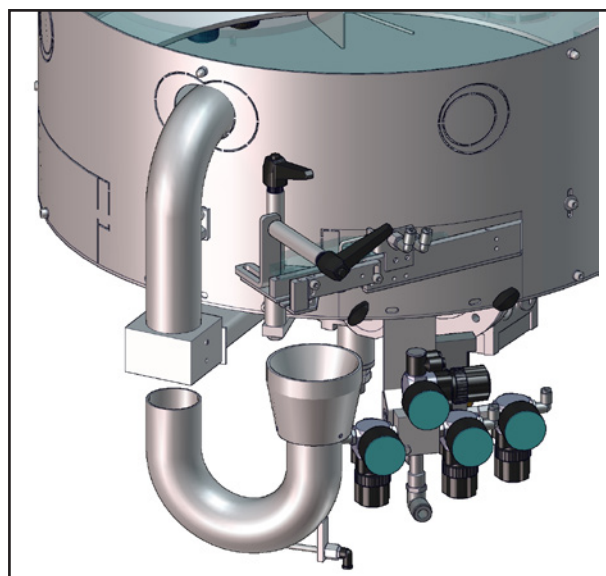


Fig. 197



11.3.5 MAINTENANCE OF THE UV RAY STERILIZER



Before carrying out maintenance operations on the lamp, allow it to cool down. Use specific gloves that comply with current regulations.

The most important maintenance aspect for the device concerns the UV ray emission source (U.V. ray tube).

To access it remove power and remove the safety guard as indicated in sub-chapter 11.3.5.1 U.V. TUBE REPLACEMENT.

After 6000 operation hours, the tube must be replaced. For the replacement of the tube U.V. proceed as described in sub-chapter 11.3.5.1 U.V. TUBE REPLACEMENT.

11.3.5.1 U.V. TUBE REPLACEMENT



Before carrying out maintenance operations on the lamp, allow it to cool down. Use specific gloves that comply with current regulations.

To replace the U.V. tube, proceed as follows:

- disconnect the power supply.
- unscrew and remove the fixing bolts on both sides of the safety guard.

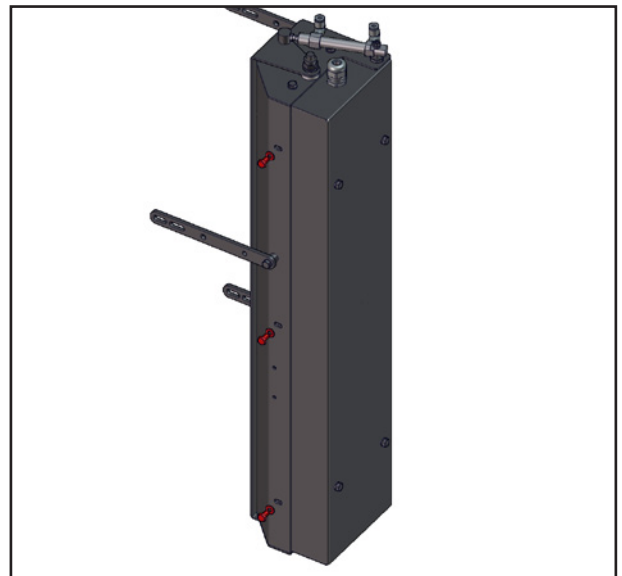


Fig. 198



- Remove the lamp coupling guard with U.V. tube.

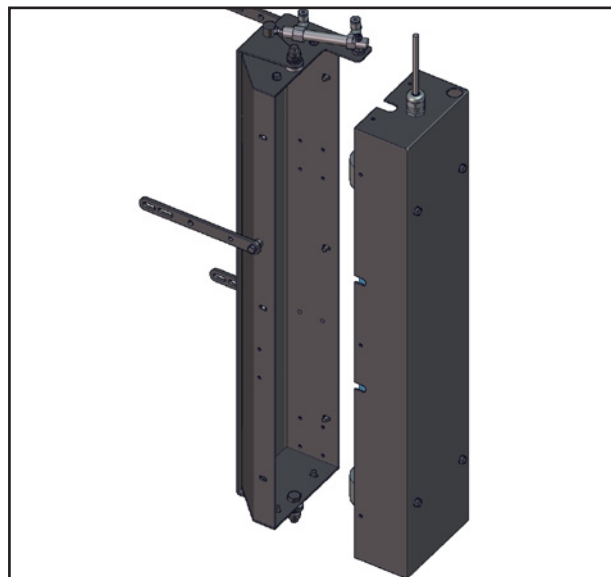


Fig. 199

- Hold the tube with fingers at its ends so it does not rotate.
- Unscrew the threaded rings of the bulb socket and pull out along the tube.



Fig. 200

- Turn the tube itself 90° on the socket (always clasp the two ends).
- First lift one side of the tube and the other in the direction vertical of the socket.
- Remove the U.V. tube from the protection cylinder made of glass.
- To mount the new U.V. tube, proceed inversely to the above described.



When installing the U.V. tube, pay attention and avoid touching your hands on the glass part (pure quartz). Handle the U.V. tube using appropriate gloves.



Fig. 201



11.4 LUBRICATION



11.4.1 GENERAL INFORMATION



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.



Before starting the closing machine, check the lubrication of all components described in the following subchapters.

If the closing machine is not used for a long time, lubricate it again.

Mineral oils, when not used for over six months, lose their properties and must be changed.

11.4.2 LUBRICANT TYPE-APPROVAL ACCORDING TO NSF

Due to the fact that the closing machine is used in food industry, AROL S.p.A. dedicated and special lubricants should be used on the machines, to prevent possible contamination problems of the food/drink in case of occasional or accidental contact between lubricant and food and drink. To this purpose, the selection of the lubricant was made according to the lubricant approval standard, according to NSF, a standard applied all over the world for food industry.

The type-approval according to NSF includes two categories: NSF - H1 and NSF - H2.

- **H1** indicates *Food grade lubricants*, that can thus be used in all friction points of machines and systems for the food and pharmaceutical industries, where occasional, and technically unavoidable, contacts can occur between lubricant and food.
- **H2** indicates lubricants that are recommended for the general use in the food and pharmaceutical industries, supposing that a contact with the food is totally to be excluded.

Due to the fact an accidental contact with the food/drink cannot be totally avoided, AROL S.p.A. decided to use always lubricants of category **H1** (*Food grade lubricants*).

The user company must use category H1 lubricants as described in the following tables.



11.4.3 TYPES OF LUBRICANTS

AROL S.p.A., after careful studies and functional tests, recommends to use only the lubricants indicated in the table below to lubricate its machines:



The lubricants indicated in the table are classified **NSF-H1**.

SIGNS		PRODUCT	NLGI	Base oil	Thickener [Viscosity Index]	Base oil viscosity [cSt]	Worked penetration [0.1mm]	T. Min [°C]	T. Max [°C]
A		Klubersynth UH1 64-62	2	Synth-PAO	Gel silica	60	265-295	-40	150
		G4501	1	Synth-PAO	AIK	100	310-340	-40	150
B		Kluberlub NH1-11-222	2	MO	AIK	220	265-295	-15	110
		Klubersynth UH1-14-222	2	Synth-PAO	AIK	220	265-295	-2	120
		Kluberfood NH1 94-402	1/2	Synth	Complex calcium	400	280-300	-30	160
C		Paraliq P150		Synth+MO		150		-15	100
D		Klubersummit FG46		Synth-PAO	[130]	46		-45	135
		L0346FG		MO	[100]	46		-21	
		Purity AW 46		HT/MO	[108]	46		-15	
E		Purity FG Synth	2	Synth-PAO	Complex sulfonate calcium	46	294	-45	200
F		Kluberpaste UH1 84-201	1	Synth-PAO	PTFE	200			
		P1900	1	MO	AIK	85	290-340	-30	300
G		Barrierta L55 / 1	1	Synth-PFPE	PTFE	400		-40	260
		Krytox GPL 226	2	Synth-PFPE	PTFE [155]	240		-30	288
H		Klubersynth UH1 64-62	2	Synth-PAO	Gel silica	60	265-295		
		G4501	1	Synth-PAO	AIK	100	310-340	-40	150
I		Purity MF Spray		HT/MO	[150]	151		-9	150



SIGNS		PRODUCT	NLGI	Base oil	Thickener [Viscosity Index]	Base oil viscosity [cSt]	Worked penetration [0.1mm]	T. Min [°C]	T. Max [°C]
I		Purity MF Spray		HT/MO	[150]	151		-9	150
J		Klubersynth UH1 64-62	2	Synth - PAO	Gel silica	60	265-295	-40	150
		G4501	1	Synth - PAO	AIK	100	310-340	-40	150
K		Paraliq GB 363	3	Synth+MO	Bentonite	2400	215-245	-30	140
L		Paraliq GB 363	3	Synth+MO	Bentonite	2400	215-245	-30	140
M		Kluberlub NH1-11-222	2	MO	AIK	220	265-295	-15	110
		Klubersynth UH1-14-222	2	Synth - PAO	AIK	220	265-295	-25	120
		Kluberfood NH1 94-402	1/2	Synth	Complex calcium	400	280-300	-30	160
N		Kluberoil 4 UH1-220N		Synth - PAO	[150]	220		-25	120
		L1122FG		Synth - PAO / MO	[142]	197		-33	
		Purity FG Gear Oil 220		HT/MO	[92]	206		-18	
O		G4501	1	Synth - PAO	AIK	100	310-340	-40	150
P		Klubersynth UH1 64 1302	2	Synth	Silicate	1300	265-295	-10	150
Q		Purity FG EP 150		MO	[95]	144		-18	247
R		Kluberfood NH1 4-220N Spray		Synth - PAO	[150]	220		-30	120
S		Klubersynth UH1 64 1302	2	Synth	Silicate	1300	265-295	-10	150
		Kluberfood NH1 94-402	1/2	Synth	Complex calcium	400	280-300	-30	160
T		Cassida Grease GTS 2	2	Synth - PAO	Complex sulfonate calcium	46	265-295	-45	170
U		Kluberfood NH1 1-17		MO					



11.4.3.1 LUBRICANT RE-ORDERING

The lubricants used for the machine can be directly ordered to *AROL S.p.A.* or to the local distributors. See the table here below for the international addresses.

	http://www.klueber.com
	http://www.molykote.com
	http://www.petro-canada.ca
	http://www.krytox.com
	http://www.fuchs-oil.com

11.4.4 LEGEND FOR LUBRICATION SCHEMES

SYMBOL	DESCRIPTION
	Lubricate with manual pump for grease
	Lubricate with brush for grease
	Lubrication glass with disposable cartridge
	Lubricate with spray
	Lubricate with manual pump for oil
	Lubrication glass



11.4.5 POINTS TO BE LUBRICATED



To protect the closing machine from wear, seizure or other severe damage to the mechanisms, it is necessary to lubricate and grease periodically all indicated points.



AROL S.p.A. guarantees the maintenance intervals indicated hereby and the correct operation of the mechanisms exclusively with the recommended lubricants.

11.4.5.1 INSTRUCTIONS FOR READING THE "LUBRICATION POINT LIST" TABLE

The following pages show:

- the description table (*Lubrication point table*) relevant to time intervals, methods and used lubricants for the maintenance interventions on the closing machine herein.
- the position of the lubricating points (*Lubrication point scheme*) described herein.

The table "*Lubrication point list*" is designed to find immediately the operations to be carried out for the closing machine lubrication.

The "*Lubrication point scheme*" indicates the position of the points described in the table on the closing machine.

The points marked by a double frame indicate that the lubrication is carried out in a centralized way. Thus, the quantity of lubricant indicated in the table must be filled; then, the system will distribute the correct quantities to the different components.

POINT CODE	The column indicates the reference of the point to be lubricated, indicated on the closing machine next to the greaser and in the " <i>Lubrication point scheme</i> "
POINT DESCRIPTION	This column briefly describes the greasing point.
LUBRICANT TYPE	The reference letter for the lubricant to be used is indicated in this column. See the table for the lubricant type to be used.
METHOD	This column shows the tool by which the relevant greasing point must be greased.
INLET NUMBER [NO.]	It indicates the number of greasing points on the closing machine.
QUANTITY/POINT [gr.]	It indicates the lubricant quantity to be fed in each greasing point.
TOTAL QUANTITY [gr.]	It indicates the total lubricant quantity to be used.
INTERVAL [h]	It indicates every how many hours the different lubrication points must be greased.



After the lubrication of each points by grease, check that there are not leaks from seal gaskets or from the ditribution pipes.



11.4.5.2 LUBRICATION POINT LIST AND SCHEMES

LUBRICATION POINTS LIST

Order Ack.	S/N	Version	Reference	Typ	Model	P.D.	H.N.
	17203	0		T	EURO VA IES LIGHT	288	6
Point	Description	Lubr. Type	Lubr. Method	Number of points	Qty per point [g]	Total qty [g]	Lubr. Int. [h]
6	Central bearings or central slewing bearing	M		1	50	50	500
8A	Piston upper roller	A		6	6	36	500
8C	Piston fork bearing	A		6	6	36	500
8D	Piston liner	A		6	15	90	500
16A	Lower central bearing or column slewing bearing.	B		1	30	30	500
16B	Upper central bearing or column slewing bearing.	B		1	30	30	500
4	Pistons controlling cam	M					1000
7	Sliding bush	M		1	5	5	1000
11	Heads rotation gearing	M					1000
1	Lifting screw	N		1	120	120	3000 *
17	Central column inner bearing	N		1	120	120	3000 *
18	Lifting axial bearing	N		1	120	120	3000 *

Rotating point

* Value that depends on environmental conditions

Fig. 202 Lubrication point table

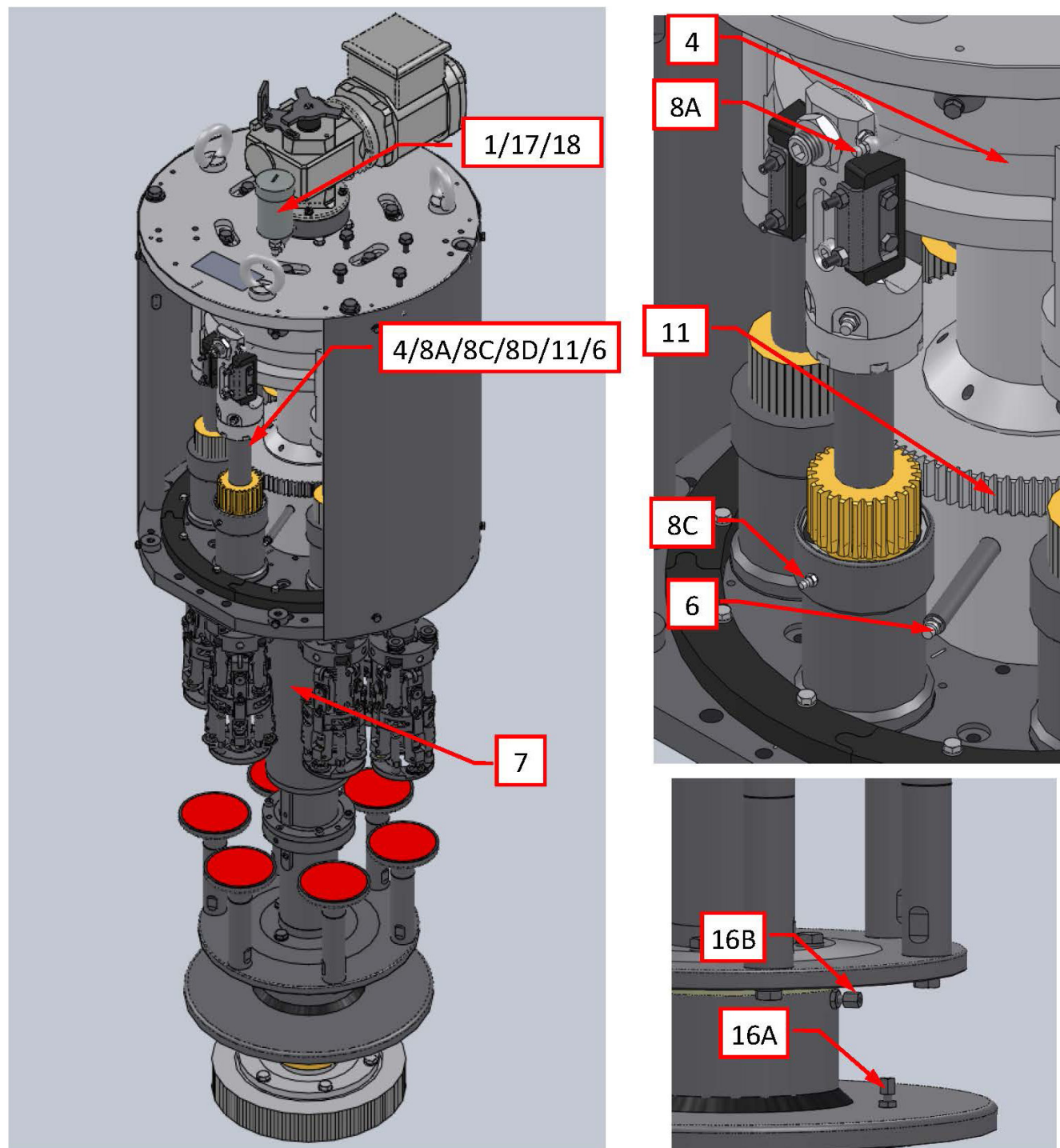


Fig. 203 Lubrication point scheme



11.4.6 LUBRICANT CARTRIDGE



At the closing machine delivery, the lubricant cartridge is not fit, thus it is not active. Before starting the closing machine, install and activate the cartridge.

11.4.6.1 LUBRICATOR DESCRIPTION

The cartridge of the lubricant is composed of:

- an adjustment screw (1).
- A body containing the lubricant (2).



The lubricant cartridge may provide lubrication intervals from 1 to 12 months. The interval is defined according to the alignment of the screw with the number on the ring nut of the cartridge.

AROL S.p.A. suggests for the lubrication of its machine to rotate the adjustment screw on number 6 of the ring nut on the cartridge.



Fig. 204

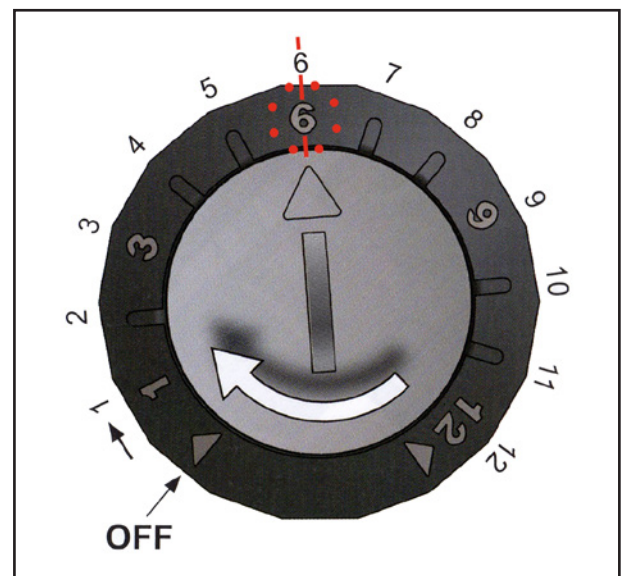


Fig. 205



During the supply period, the lubricator must not be removed to maintain the pressure inside it.



11.4.6.2 CARTRIDGE REPLACEMENT

When the cartridge must be replaced, operate as follows:

- Remove the plug from the discharge opening.
- Screw by hand the lubricating cartridge on the suitable coupling.

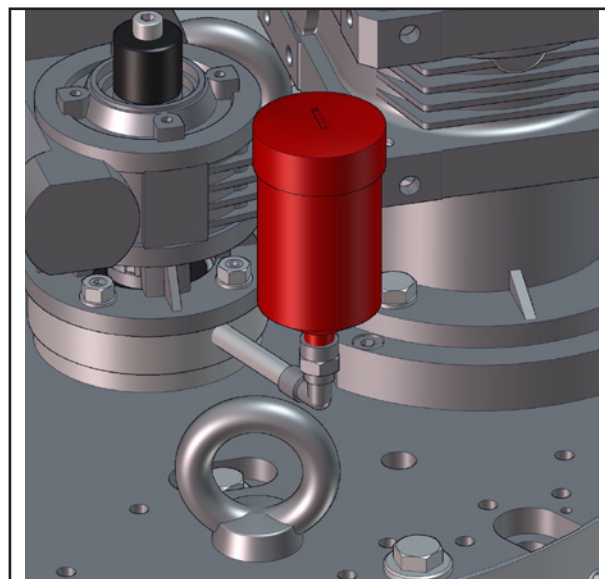


Fig. 206

- Rotate the adjustment screw on number 6 of the ring nut on the cartridge.



Fig. 207

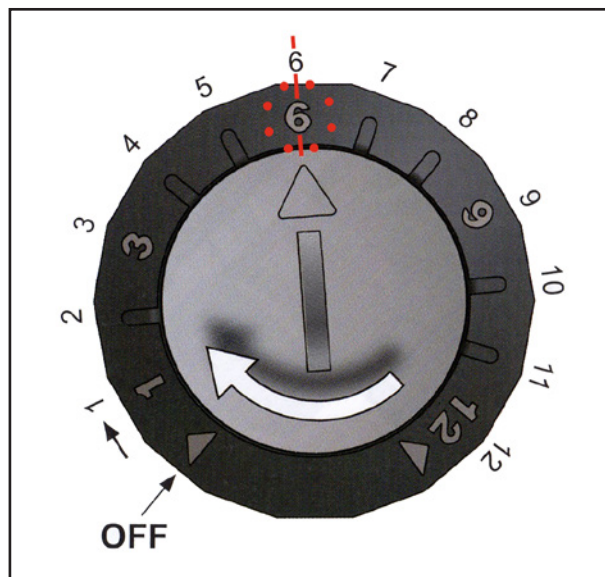


Fig. 208

11.4.7 LUBRICATION OF THE CLOSURE HEAD

11.4.7.1 "SSD" CLOSURE HEAD

Use lubricant "I" type.

- Lubricate the rollers of arms forks with a aerosol spray oil. Lubricate every 8 hours.

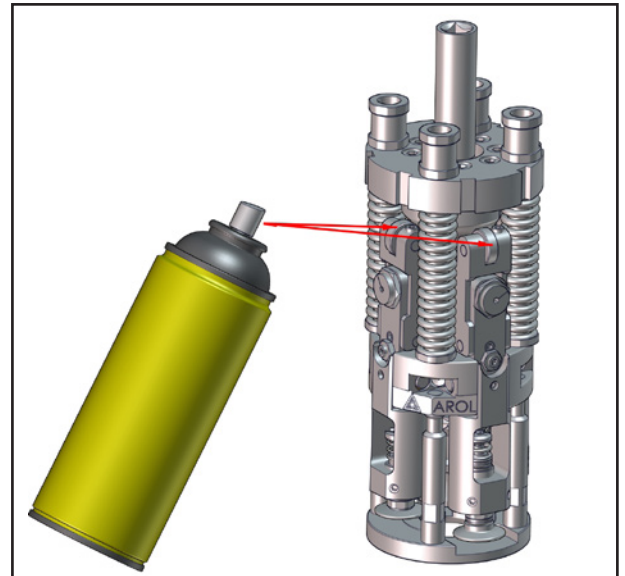


Fig. 209

- Lubricate the springs and threading rollers with a aerosol spray oil. Lubricate every 8 hours.



Fig. 210

- Lubricate the springs and skirting rollers with a aerosol spray oil. Lubricate every 8 hours.

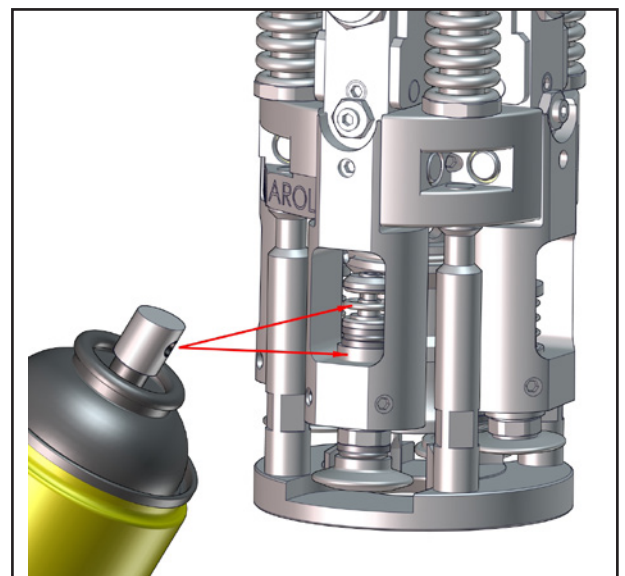


Fig. 211



Use lubricant "J" type.

- Lubricate the head central block with a grease pressure syringe. Lubricate every 120 hours, introducing 10 gr of lubricant.



Fig. 212

- In case you must lubricate the side load springs and/or the GAS mouth-presser spring, after their replacement, using a brush apply a layer of lubricant on all the spring coils. Use a brush.



Since it is objectively difficult to express an amount of lubricant to be laid with a brush, with "stroke" refers to the amount of grease that is retained, plunging up to half the bristles of a brush number 8 off about 10 mm high 18 mm bristle.

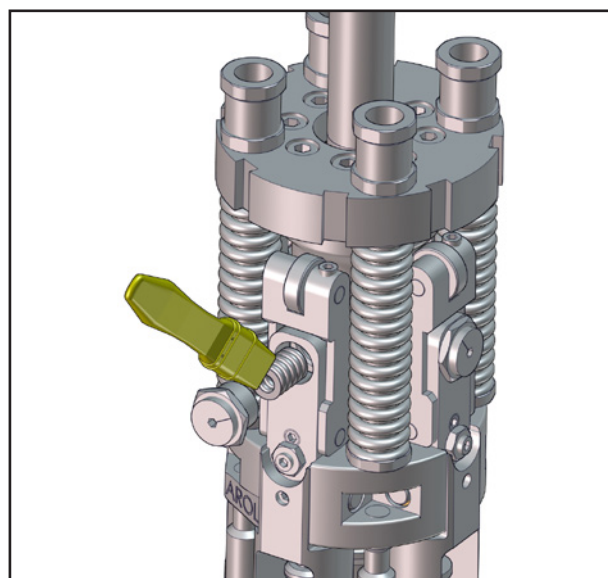


Fig. 213

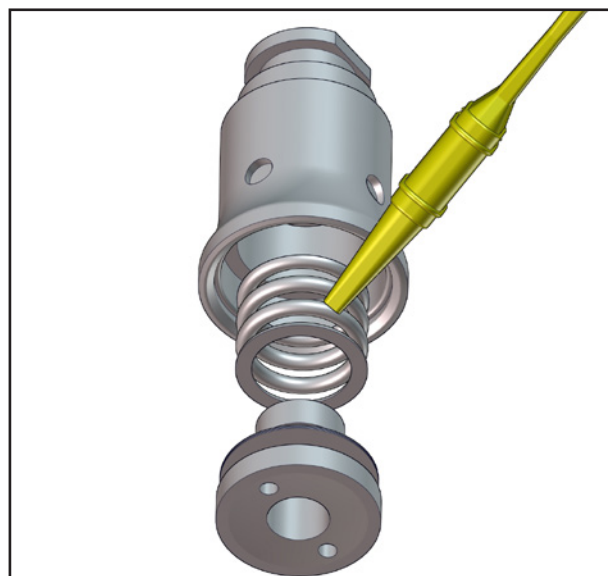


Fig. 214



11.4.8 BONFIGLIOLI REDUCER LUBRICATION

11.4.8.1 INTRODUCTION

On the reducer is fitted with the nameplate on the side shown. On this plate there are all the references and all the indispensable indications for its correct identification.

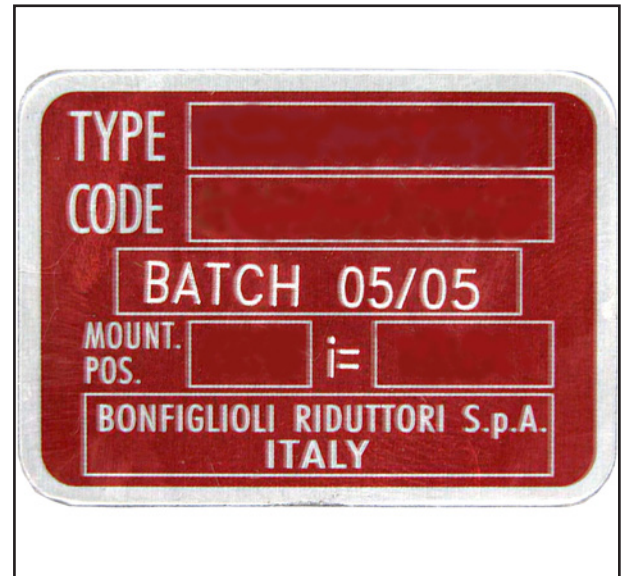


Fig. 215

11.4.8.2 LUBRICATION

VF30....VF49 AND W63....W86 REDUCERS

For reducers of this group, that are on a average at high power, the oil lubrication is adopted. These reducers have not charge, discharge and level oil plugs and so do not need any maintenance having already the right oil quantity.

VF130....VF150 AND W110 REDUCERS

For reducers of this group, that are on a average at low power, the permanent lubrication with synthetic oil is adopted. Reducers have charge, discharge and level oil plugs



Reducers are supplied with a lubrication oil load **TIVELA OIL S320** type.



It is recommended, if the lubricants is not of the recommended ones, that it has an equivalent composition on synthetic nature and viscosity. Moreover it must have the suitable antifoam additives.

OIL QUANTITY TABLE (LITRES)

		B3	B6	B7	B8	V5
VF130	HS	3,90	2,50	2,50	2,30	3,30
	P(IEC)	3,00	2,50	2,50	2,30	3,30
VFR130		0,40	0,50	0,50	0,70	0,40
W110	P80...P132	1,50	1,70	1,70	1,90	1,70
	M2-M3	1,50	1,70	1,70	1,90	1,70
	$7 < i < 15$	1,50	1,70	1,70	1,90	1,70
	$20 < i < 100$	2,70	1,70	1,70	1,90	1,70



11.4.8.2.1 OIL REPLACEMENT



Do not mix oil with different trade marks and characteristics and check that the oil in use has high anti-foam characteristics and EP.

If an identical lubricant is not available, completely empty the reducer and clean it inside with a light solvent, before the next filling.

- Place a container of adequate capacity under the discharge plug.
- Remove the charge, discharge and level plug and let the oil freely flow.



To make easier the discharge operation, it is recommended to operate with hot oil.

- Wait some minutes so that all the oil is went out, then screw again the discharge plug, after the gasket has been replaced.
- Pour new oil only after the reducer is its definitive position, until the half of the level plug is reached.
- Screw the charge plug, after the gasket has been replaced.



The exhausted oil must be disposed according to the environment protection norms.

If there are leaks, before to reset the lubricant quantity, it is necessary to find the cause of the problem before starting up again the reducer.



11.5 EXTRAORDINARY MAINTENANCE

11.5.1 SPINDLES CONTROL CAM WEAR CHECK



To check the piston control cam race wear proceed as follows:



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

- Remove the safety guard.
- Check the conditions of the spindles control cam in the cap application point. The worn part is usually the roller sliding race; if the wear is such to form a difference over 1 mm the cam must be replaced.
- Check the wear along the whole sliding race and above all in the area where the piston reaches its bottom dead center.

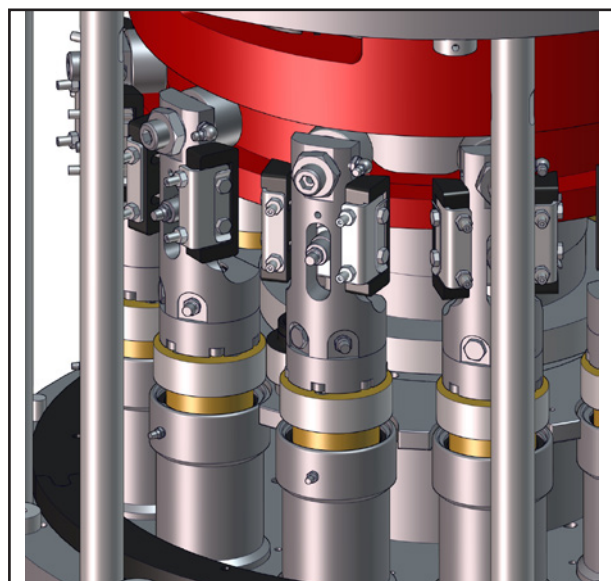


Fig. 216

To replace the cam, contact the *Technical Service*.



11.5.2 REPLACEMENT OF THE BOTTLE SUPPORTING PLATES

To replace the bottle supporting plates, proceed as follows:

- insert a flat head screwdriver or a similar tool between the plate and its support.

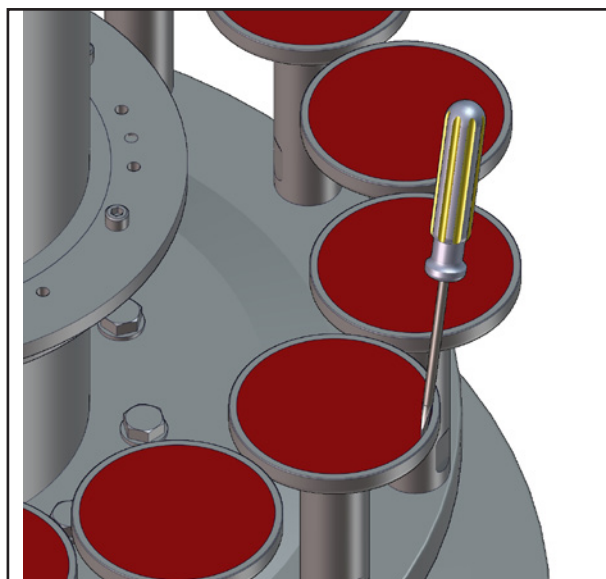


Fig. 217

- Prise the plate out of the support using flat head screwdriver or similar tool.
- Complete the plate removal from the support.

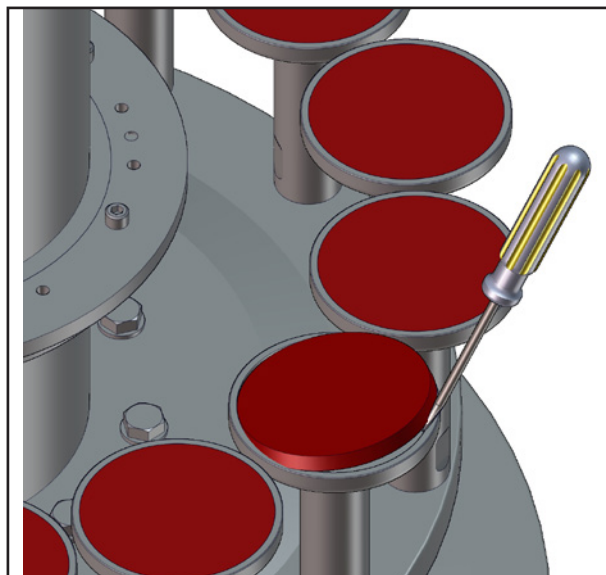


Fig. 218

- Position the new plate in the support seat.

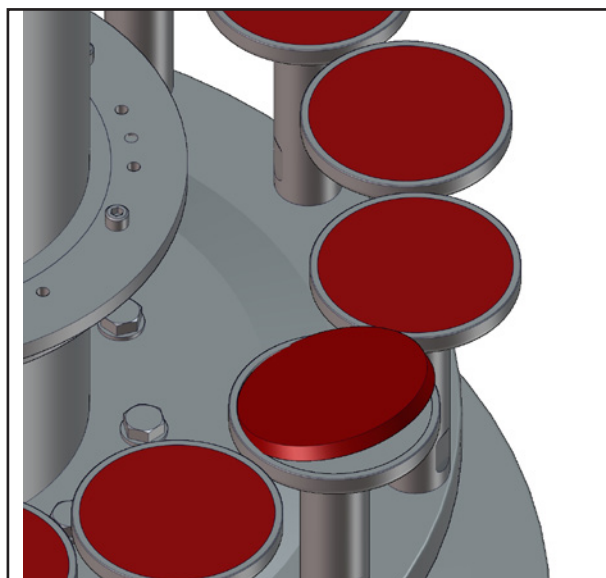


Fig. 219



- Complete the plate insertion inside the support, using a rubber hammer and beating on the plate side

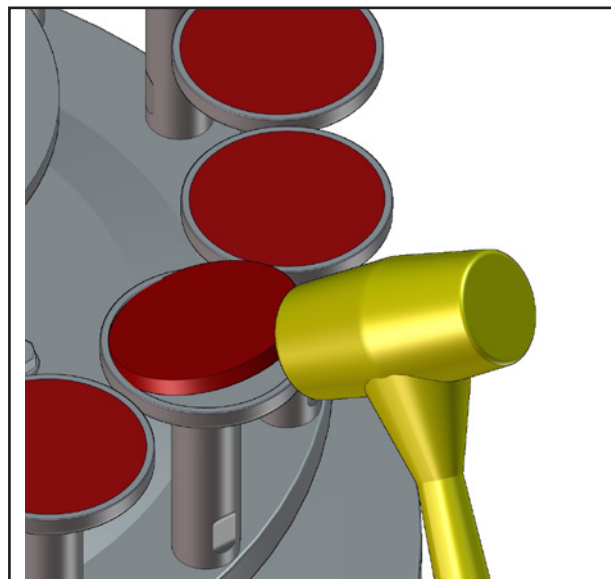


Fig. 220



11.5.3 TRANSPORT EQUIPMENT TIMING

11.5.3.1 ALIGNMENT BETWEEN BOTTLE CENTERING DEVICE AND CENTRAL STAR-WHEEL

The closure head must always be aligned with a cavity of the central star-wheel

If the closure head must be axially aligned with the star-wheel cavity, operate as follows:

- Fit a bottle in the central star-wheel cavity.
- Check the vertical alignment between capping head axis and star-wheel axis.

- If the alignment is not correct:



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

- loosen the fixing screws.
- Manually rotate the star-wheel support complete with star-wheel, till getting a correct alignment between a closure head and a star-wheel cavity.
- At the end, tighten the fixing screws strongly again.
- Remove the bottle.

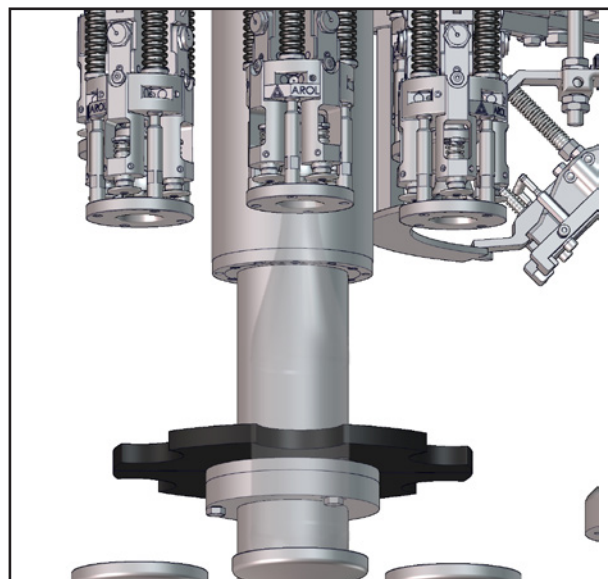


Fig. 221

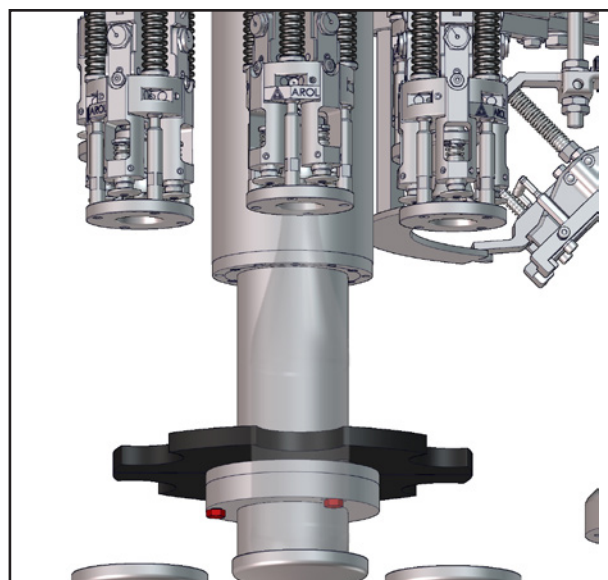


Fig. 222

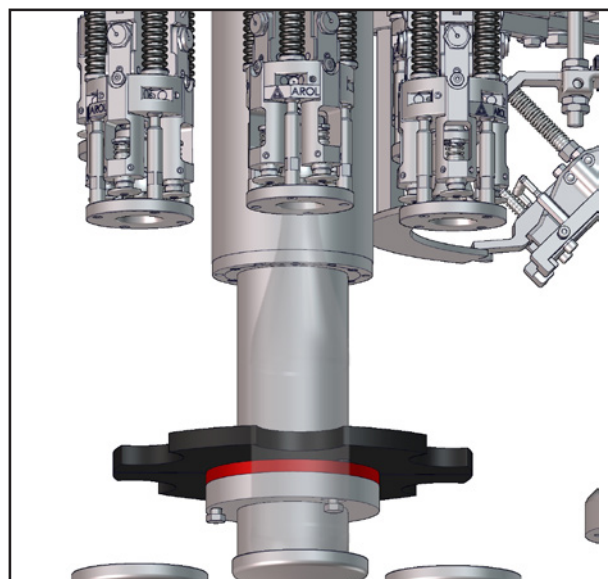


Fig. 223



12 FAULTS AND REMEDIES

The list of the faults in the following pages is useful to detect as fast as possible their reasons and possible solutions.

The research and the interventions must be carried out methodically (from the most probable to the less probable reason) and in logical sequence.

Proceed by steps: a first intervention, a test on the obtained result, then passage to the next intervention.

The main closing machine units are:

- Closure unit
- Caps/corks distribution.
- Caps/corks feeding.
- UV-rays sterilizer.

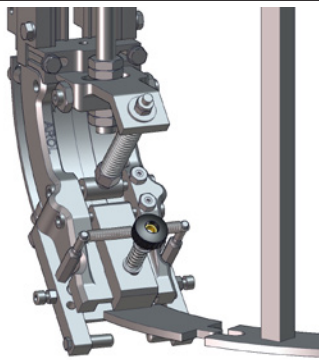


Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.



FAULT	CAUSE	REMEDY
Lifting or lowering the closure carousel, the same does not move vertically.	Interference with the container conveying equipment, foreign matters in the lifting mechanism, other.	Identify the reason for the overload (interference with the container conveying equipment, foreign matters in the lifting mechanism, other).
	The fastening screw of the sliding bush has not been loosened.	Loosen the fastening screw.
	Min. or max. limit of closing carousel vertical range reached.	Make sure the closing machine has been preset for the container size to be processed. If so contact <i>Technical Assistance Service</i> .
The containers break in the closure step.	The closure carousel is too low.	Position the closure carousel at the correct height for the container to be processed.
	Container transfer equipment not suitable.	Replace the equipment with the one suitable for the size to be processed.
Caps are not descending from the sorter.	The sorter exit-mouth is clogged.	Remove possible caps, cap parts or foreign matters. Check on the pressure gauges if the pressure of the air jets is correct.
	Caps are not correctly sized or are strained.	Use caps with the suitable size and even features.
The cap is too much little or not at all threaded.	Insufficient pressure of the threading rollers.	Adjust the pressure of the threading rollers.
	Threading rollers are locked (seized up).	Manually check the rotation and the sliding of threading rollers. If they are locked, dismount them, clean them and lubricate them.
	Spring of the threading roller is broken.	Replace the spring of the threading rollers.
	Springs of the closure head are broken.	Replace the springs with those parts. For the replacement see the relative sub-chapter of the closure head.
	Incorrect closure carousel height.	Place the closure carousel at the correct height for the bottle to handle.
The cap is too much little or not at all skirted.	Incorrect height of the skirting rollers.	Adjust the height of the rollers
	Insufficient pressure of the skirting rollers.	Adjust the pressure of the rollers
	Spring of the skirting roller is broken.	Replace the spring of the skirting roller.



FAULT	CAUSE	REMEDY
The cap is cut on the thread.	Too much pressure on the threading rollers.	Adjust the pressure of the threading rollers.
	Threading rollers are locked (seized up).	Manually check the rotation and the sliding of threading rollers. If they are locked, dismount them, clean them and lubricate them.
Problems in the cap distribution.	Incorrect cap gripping.	Verify the height of the closure carousel.
	Check the operating of springs of the caps positioning device.	Check that the caps positioning device is properly operating. If it does not contact the <i>Technical Service</i> . 



13 SCRAPPING AND DISPOSING



As there are in force different norms of observation, it is necessary to follow the prescriptions fixed by laws and authorities of the states where the closing machine will be installed.

Proper precautions must be taken for out-of-order and scrapping of the closing machine in order to avoid possible dangerous situations.

In particular, follow these general instructions:

- disconnect the closing machine from the electric network and check that no residual voltage exists.
- Empty all the lubrication points of oil paying attention in not dispersing and dispose it according the laws in forces in the country where the closing machine has been used.
- Lock all the moving parts of the closing machine, making sure that there are no possibilities of movement, also due to accidental collision during the transfer operation.
- With the help of qualified and authorized personnel proceed to the dismantling of components and the differentiated collection of materials (steel, plastic, electrical components, wires, etc.) in order to dispose everything according to the indications foreseen by the laws in force.



Always use security clothes during the operations that require direct intervention on the closing machine.